结构力学

Structure Mechanics



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第九章

渐　近　法

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§ 9-1　渐近法概述

§ 9-2　力矩分配法基本原理

§ 9-3　用力矩分配法计算连续梁和无侧移刚架

§ 9-4　无剪力分配法

§ 9-5　力矩分配法与位移法的联合应用

§ 9–1 渐近法概述

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用力法、位移法分析超静定结构，都需要求解多元联立方程组，求出基本未知量。当未知量较多时，计算颇为繁重。

渐近法—采用逐步地逼近真实解的方法。

渐近法主要有：

力矩分配法

无剪力分配法

渐近法特点

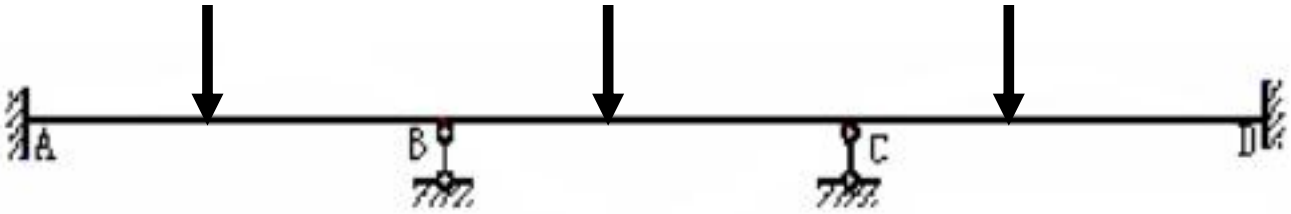
不必求解联立方程，不需求出基本未知量，而是直接以杆端弯矩为计算对象，采用逐步修正并逼近精确结果的算法，因此称为渐近法。

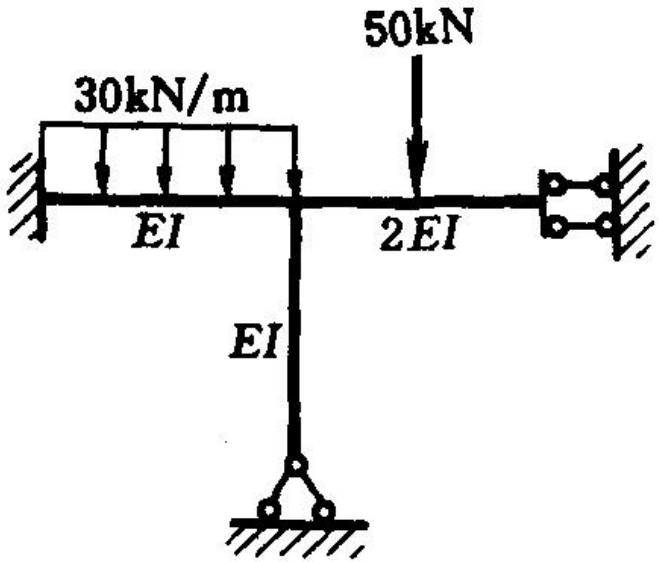
力矩分配法和无剪力分配法就其本质来说，都属位移法的范畴，计算原理及符号规则均与位移法相同，只是计算过程不相同。

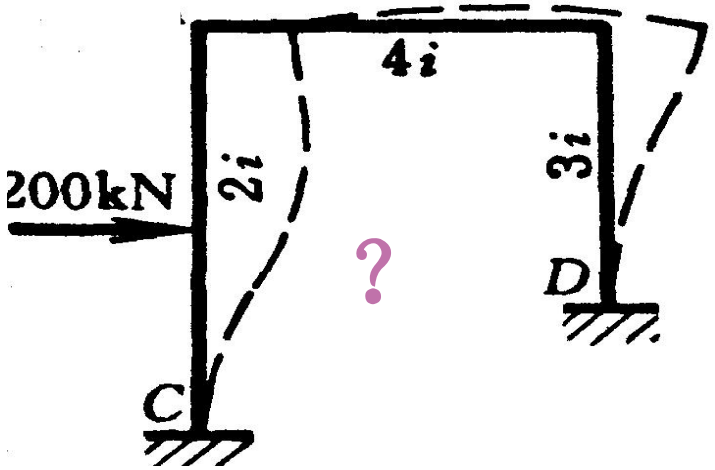
力矩分配法的适用范围：

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只有结点角位移而无结点线位移的梁和刚架。







位移法与力矩分配法的对比

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对于结点无线位移的超静定结构，用位移法求解是为了消除基本结构各个刚结点上的附加约束反力矩，表达为联立方程的形式，通过解方程而一次完成的。

在力矩分配法中，为消除附加约束反力矩，是对每个附加约束逐次松弛、反复多次进行的，从结点被固定的状态出发，将各个结点逐次恢复转角位移的过程，直接表达为各杆端弯矩的逐次修正的过程；

当松弛结束时，变形和内力趋于实际的最终状态。

此法计算过程的数学实质是松弛法求解联立代数方程的过程。

力矩分配法和无剪力分配法适合于手算的方法。

§9–2力矩分配法基本原理

§9-2 力矩分配法基本原理

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力矩分配法的理论基础是位移法，杆端弯矩的正负规定与位移法相同。

下面先解释力矩分配法中使用的几个名词。

1. 名词解释

（1）劲度系数（也称为转动刚度）

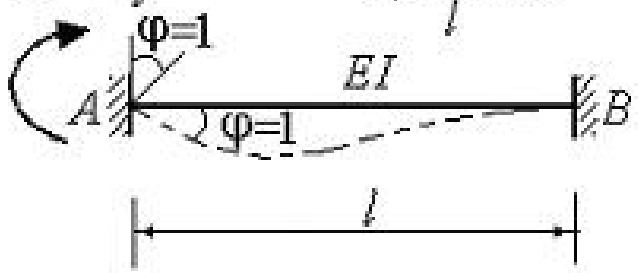
转动刚度表示杆端对转动的抵抗能力。杆端转动刚度以 S_{AB} 表示，它在数值上等于使杆端产生单位转角时所需要施加的杆端弯矩。

S_{AB}

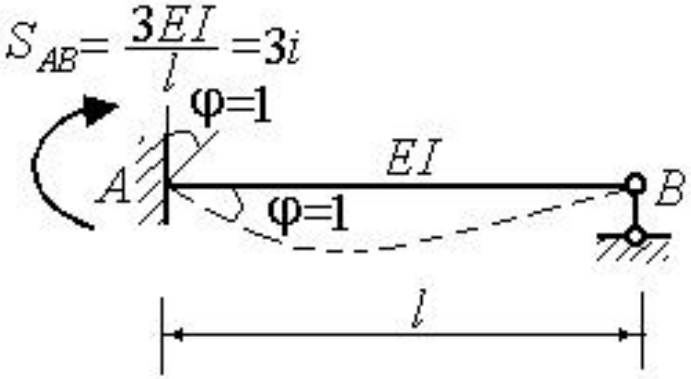
近端、施力端 ← → 远端

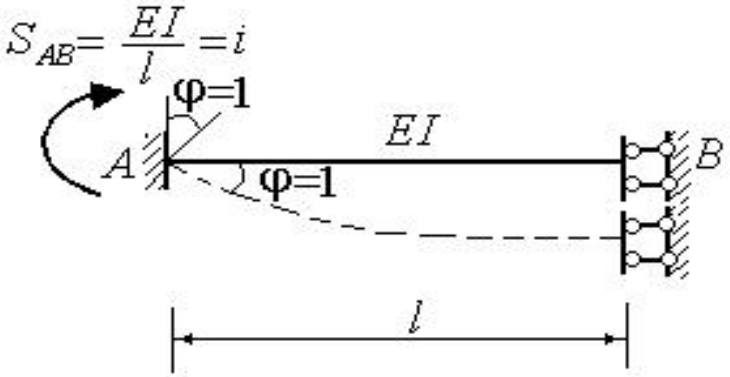
等截面直杆在A端的转动刚度

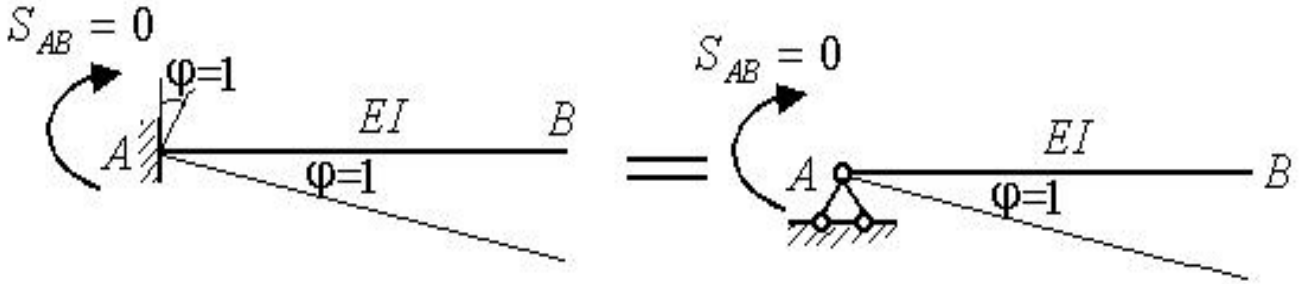
$S_{AB}=\dfrac{4EI}{l}=4i$　　令 $\dfrac{EI}{l}=i$

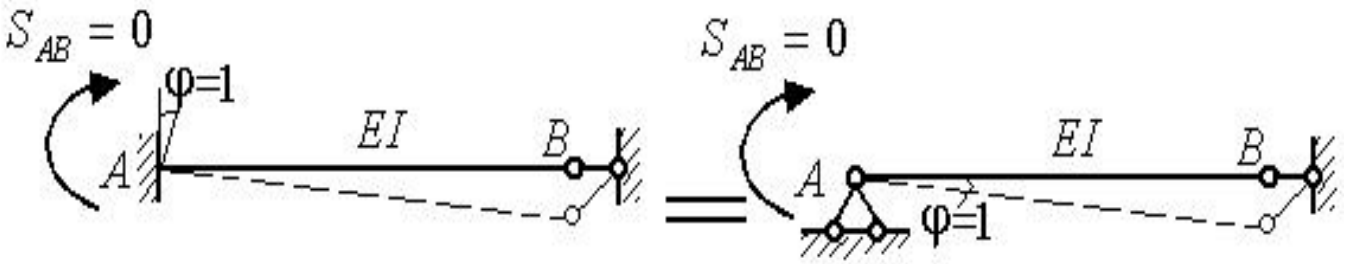


$S_{AB}=\dfrac{3EI}{l}=3i$

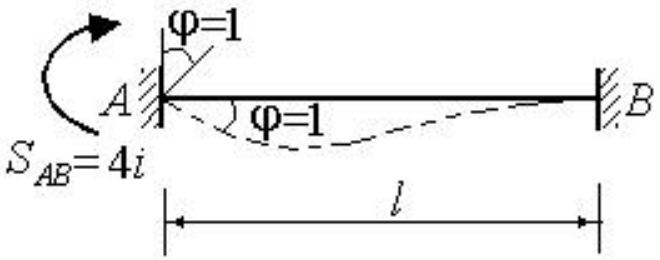


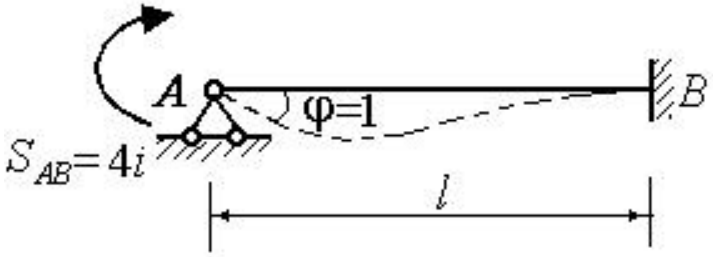

$S_{AB}=\frac{EI}{l}=i$
φ=1
A
EI
B
φ=1
l


$S_{AB}=0$
φ=1
A
EI
B
φ=1
=
$S_{AB}=0$
A
EI
B
φ=1


$S_{AB}=0$
φ=1
A
EI
B
$S_{AB}=0$
A
EI
B
φ=1

- S_{AB} 数值只与远端支承有关,与近端无关。
- S_{AB} 是指施力端A在没有线位移的条件下的转动刚度。

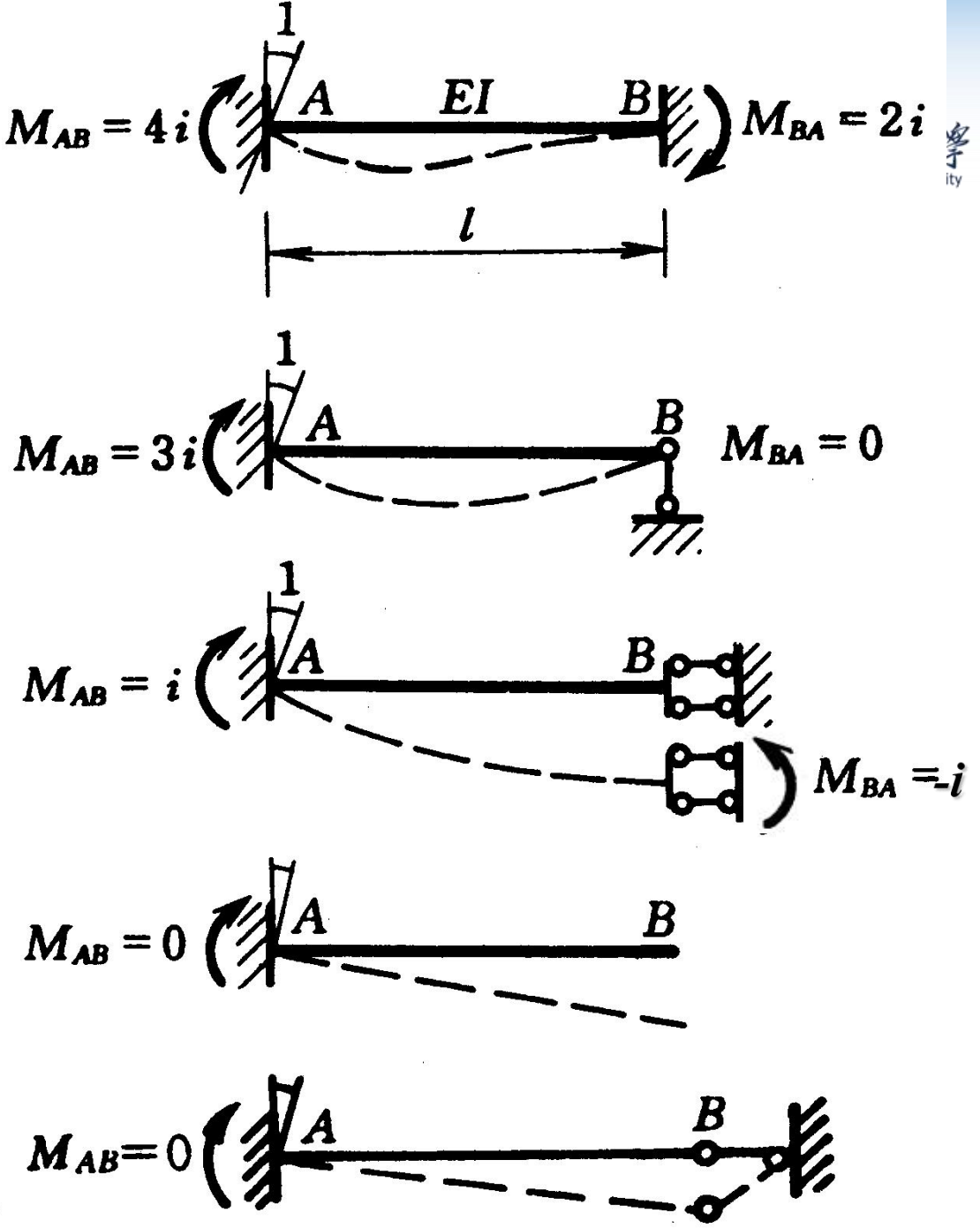


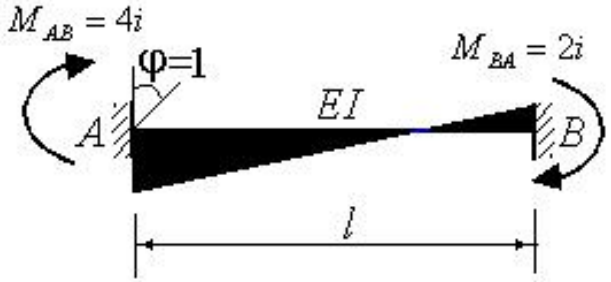


（2）传递系数

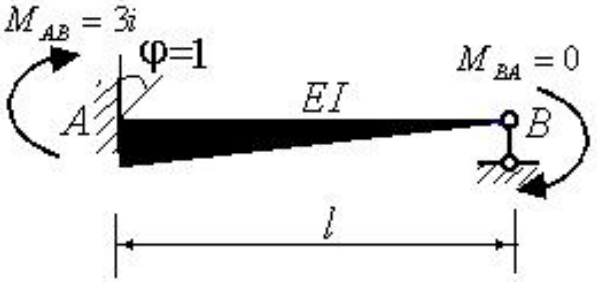
当A端转动时，使各杆近端产生弯矩，同时也使远端产生弯矩，这好比是近端的弯矩按一定的比例传到了远端一样，故将B端的弯矩与A端的弯矩之比称为由A端向B端的传递系数，用 C_{AB} 表示，即

$$C_{AB}=\frac{\text{远端弯矩}}{\text{近端弯矩}}=\frac{M_{BA}}{M_{AB}}$$

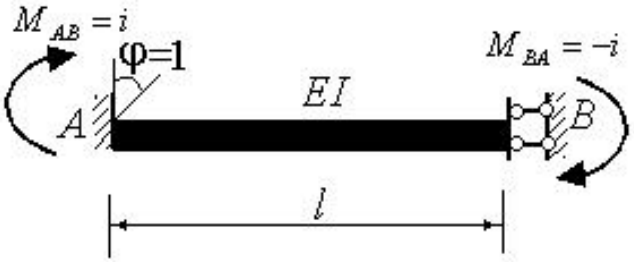




远端固定　$C_{AB}=\frac{1}{2}$



远端铰支　$C_{AB}=0$

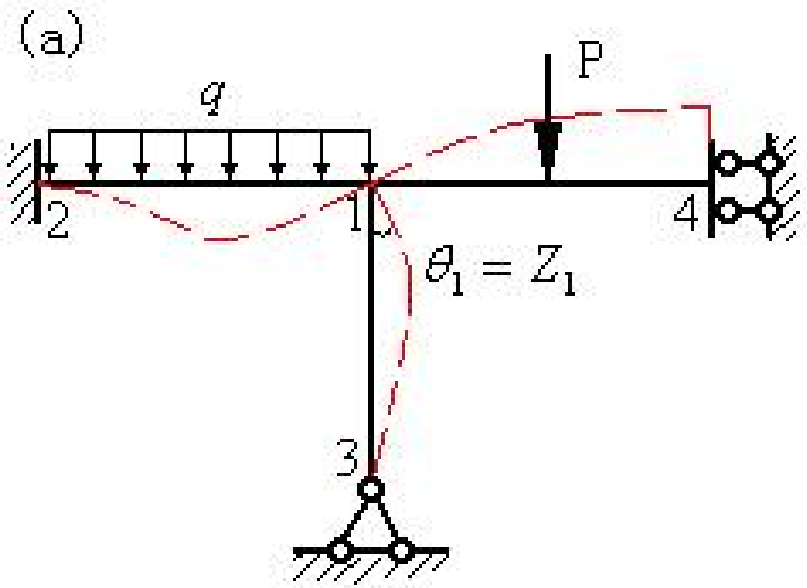


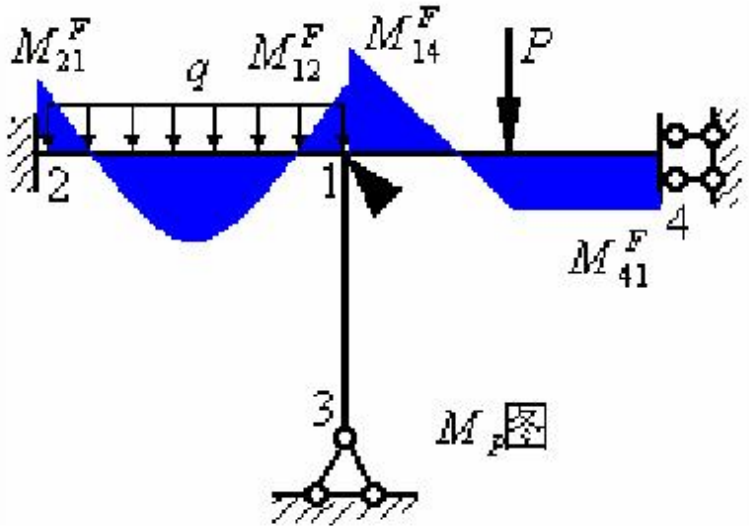
远端滑动　$C_{AB}=-1$

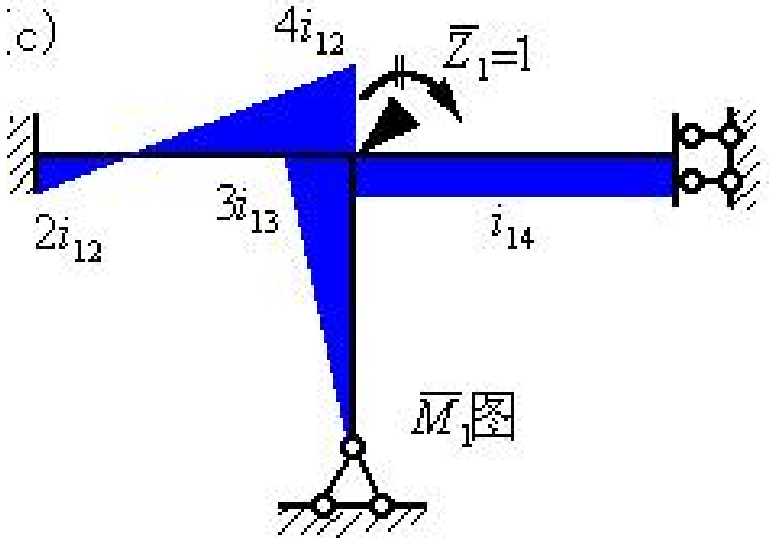
（3）固端弯矩及结点上的不平衡力矩

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下面以位移法计算图示刚架为例来说明固端弯矩及不平衡力矩的概念。



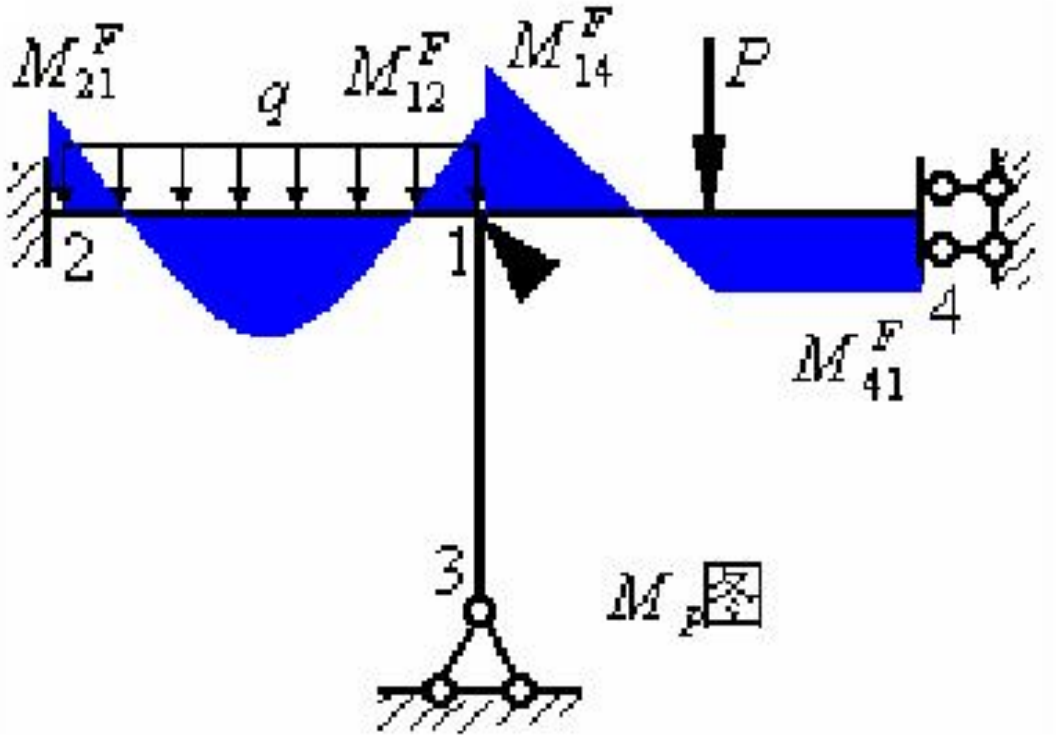




$$r_{11}Z_1 + R_{1P} = 0$$

固端弯矩

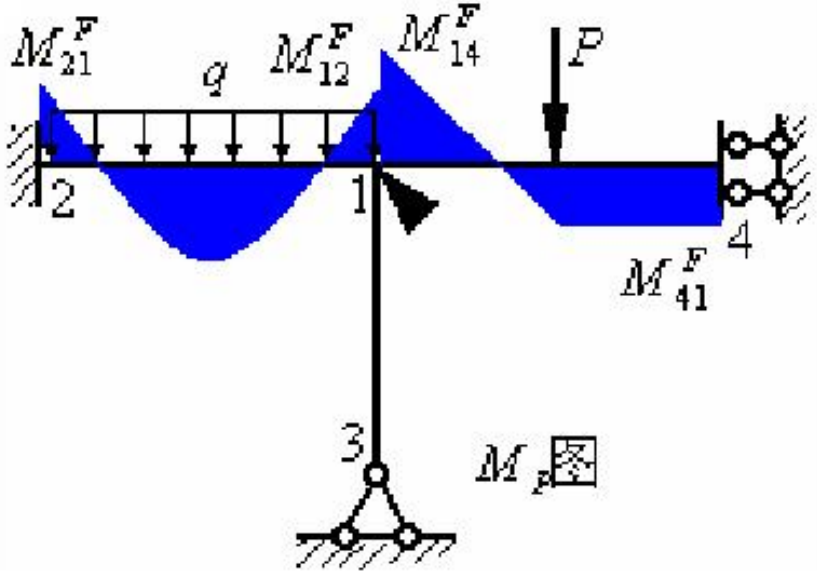
基本结构在外荷载或温度作用下，引起的杆端弯矩M_{ij}^{F}称为固端弯矩。



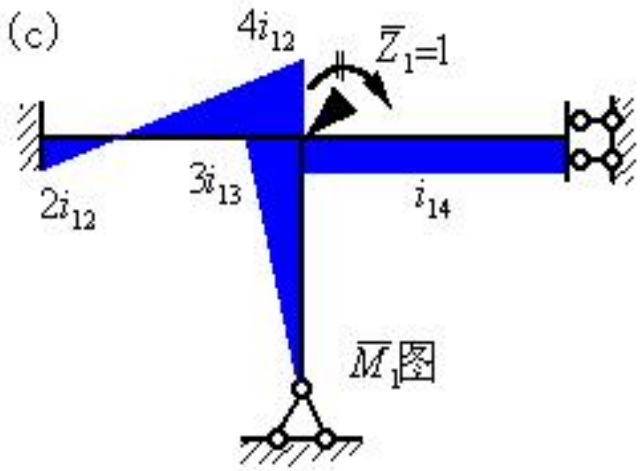
结点上的不平衡力矩

自由项为

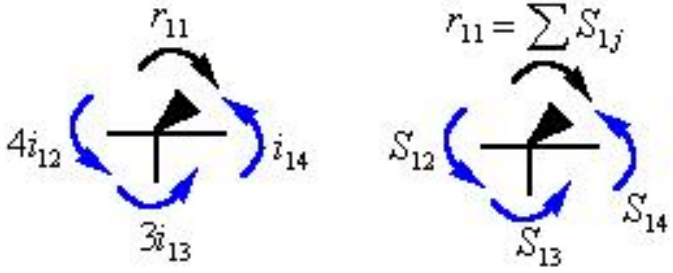
$$R_{1P} = M_{12}^{F} + M_{13}^{F} + M_{14}^{F} = \sum M_{1j}^{F} = M_1$$



自由项是等于交于结点1的各杆端固端弯矩（绕杆端）的代数和 $\sum M_{1j}^{F}$，也就是各固端弯矩所不能平衡的差额，故又称为结点上的不平衡力矩（顺时针方向为正）。



由单位弯矩图，得



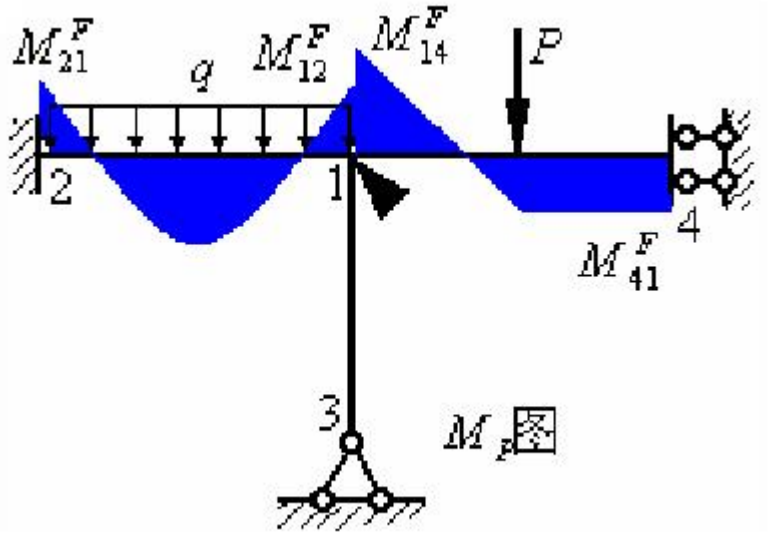
$$r_{11} = 4i_{12} + 3i_{13} + i_{14} = S_{12} + S_{13} + S_{14} = \sum S_{1j}$$

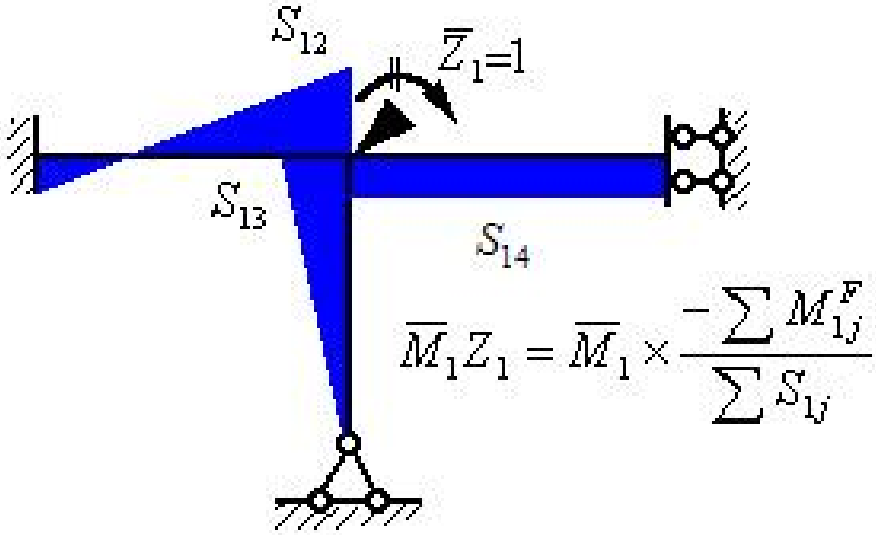
$\sum S_{1j}$代表汇交于结点1的各杆端刚度系数之和。

解典型方程 $$Z_1 = -\frac{R_{1P}}{r_{11}} = \frac{-\sum M_{1j}^{F}}{\sum S_{1j}}$$

按叠加法计算各杆端的最后弯矩，即

$$M = M_P + \overline{M}_1 Z_1$$





各近端弯矩

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$$M_{12}=M_{12}^{F}+\frac{S_{12}}{\sum S_{1j}}\left(-\sum M_{1j}^{F}\right)=M_{12}^{F}+\mu_{12}\left(-\sum M_{1j}^{F}\right)$$

$$M_{13}=M_{13}^{F}+\frac{S_{13}}{\sum S_{1j}}\left(-\sum M_{1j}^{F}\right)=M_{13}^{F}+\mu_{13}\left(-\sum M_{1j}^{F}\right)$$

$$M_{14}=M_{14}^{F}+\frac{S_{14}}{\sum S_{1j}}\left(-\sum M_{1j}^{F}\right)=M_{14}^{F}+\mu_{14}\left(-\sum M_{1j}^{F}\right)$$

以上各式右边第一项为荷载产生的弯矩，即固端弯矩。第二项为结构转动Z角时所产生的弯矩。

第二项相当于把不平衡力矩反号后按刚度系数大小的比例分给各近端，因此称为分配弯矩。

（4）分配系数

μ_{12}、μ_{13}、μ_{14}等称为分配系数，其计算公式为：

$$\mu_{1j}=\frac{S_{1j}}{\sum S_{1j}}$$

显然，同一结点各杆端的分配比例系数之和应等于1，即

$$\sum \mu_{1j}=\mu_{12}+\mu_{13}+\mu_{14}=\frac{S_{12}}{\sum S_{1j}}+\frac{S_{13}}{\sum S_{1j}}+\frac{S_{14}}{\sum S_{1j}}$$

$$=\frac{S_{12}+S_{13}+S_{14}}{\sum S_{1j}}=\frac{\sum S_{1j}}{\sum S_{1j}}=1$$

各远端弯矩

$$M_{21}=M_{21}^{F}+\frac{C_{12}S_{12}}{\sum S_{1j}}\left(-\sum M_{1j}^{F}\right)=M_{21}^{F}+C_{12}\left[\mu_{12}\left(-\sum M_{1j}^{F}\right)\right]$$

$$M_{31}=M_{31}^{F}+\frac{C_{13}S_{13}}{\sum S_{1j}}\left(-\sum M_{1j}^{F}\right)=M_{31}^{F}+C_{13}\left[\mu_{13}\left(-\sum M_{1j}^{F}\right)\right]$$

$$M_{41}=M_{41}^{F}+\frac{C_{14}S_{14}}{\sum S_{1j}}\left(-\sum M_{1j}^{F}\right)=M_{41}^{F}+C_{14}\left[\mu_{14}\left(-\sum M_{1j}^{F}\right)\right]$$

以上各式右边第一项为荷载产生的弯矩，即固端弯矩。

第二项为结构转动Z角时所产生的弯矩，它好比是将各近端弯矩的分配弯矩以传递系数的比例传到各远端一样，故称为传递弯矩。

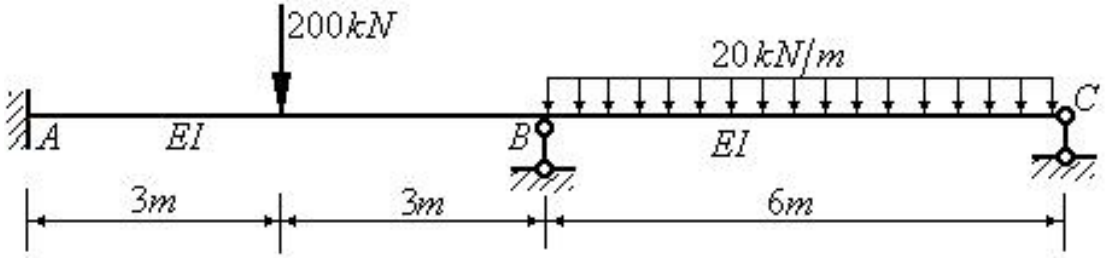
按以上结论计算各杆端弯矩步骤为：

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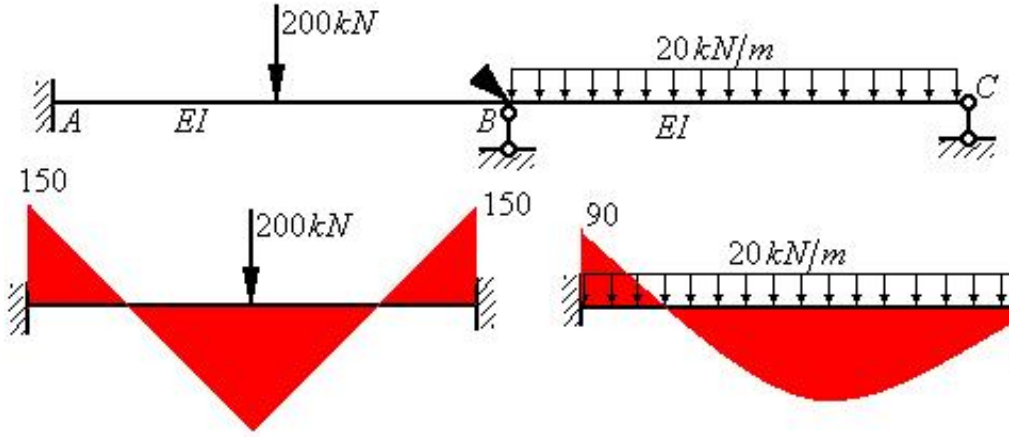
（1） 固定结点，即加入刚臂。此时各杆端有固端弯矩，而结点上有不平衡力矩，它暂时由刚臂承担。

（2）放松结点，即取消刚臂，让结点转动。这相当于在结点上加入一个反号的不平衡力矩，于是不平衡力矩被取消而结点获得平衡。此反号的不平衡力矩按劲度系数大小的比例分配给近端，于是各近端得到分配弯矩，同时各自向其远端进行传递，各远端得到传递弯矩。

例1 用力矩分配法作图示连续梁的弯矩图

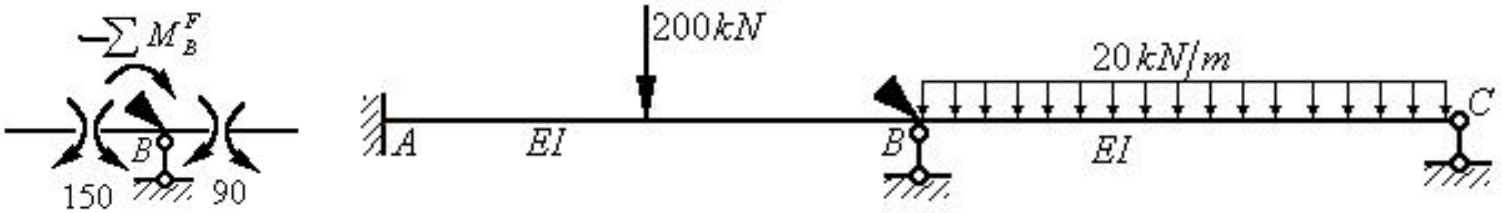


（1）固定结点B



$$M_{AB}^{F}=\frac{Pl}{8}=-\frac{200kN\times 6m}{8}=-150kN\cdot m,\qquad M_{BA}^{F}=150kN\cdot m$$

$$M_{BC}^{F}=-\frac{ql^{2}}{8}=-\frac{20kN/m\times(6m)^{2}}{8}=-90kN\cdot .m$$

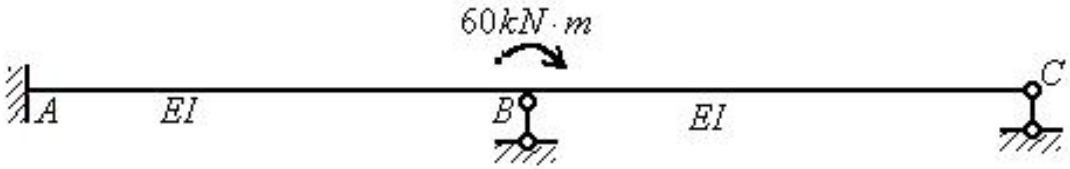


结点B的不平衡力矩为：

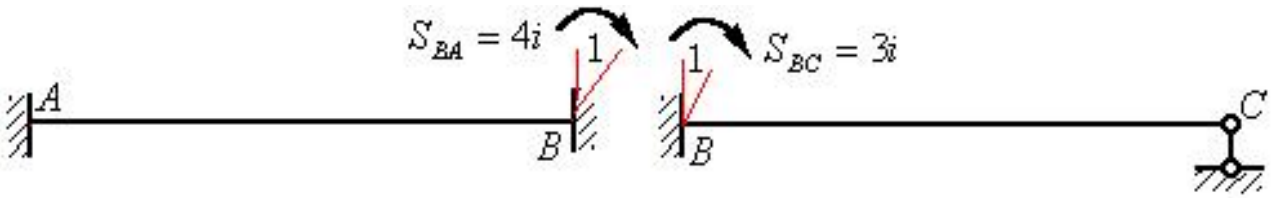
$$\sum M_{B}^{F}=150kN\cdot m-90kN\cdot m=60kN\cdot m,$$

（2）放松结点*B*

这等于在结点*B*新加一个外力偶 –60kN.m。此力偶按分配系数分配于两杆的*B*端，并使*A*端产生传递弯矩。



*AB*和*BC*杆的线刚度相等，转动刚度：



分配系数、分配力矩为

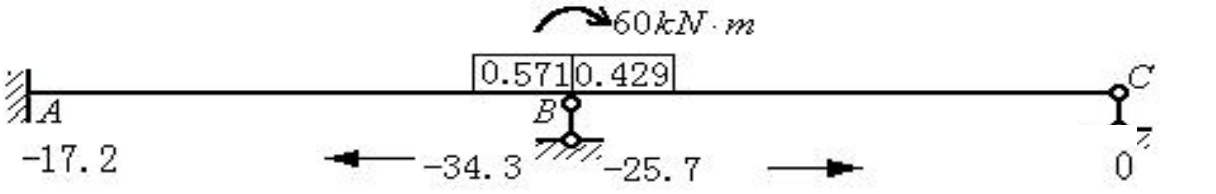
$$\mu_{BA}=\frac{S_{BA}}{S_{BA}+S_{BC}}=\frac{4i}{4i+3i}=\frac{4}{7}=0.571,$$

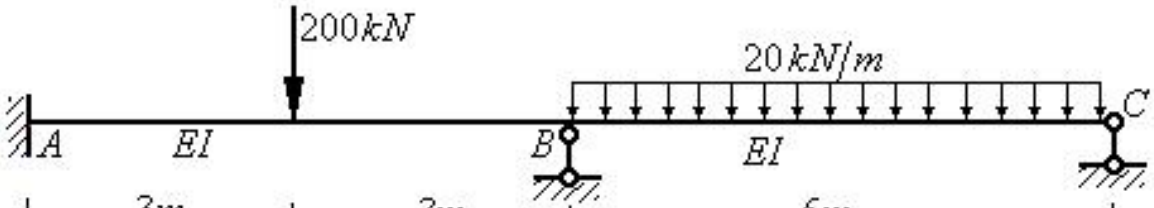
$$\mu_{BC}=\frac{S_{BC}}{S_{BA}+S_{BC}}=\frac{3i}{4i+3i}=\frac{3}{7}=0.429$$

$$\mu_{BA}+\mu_{BC}=1$$

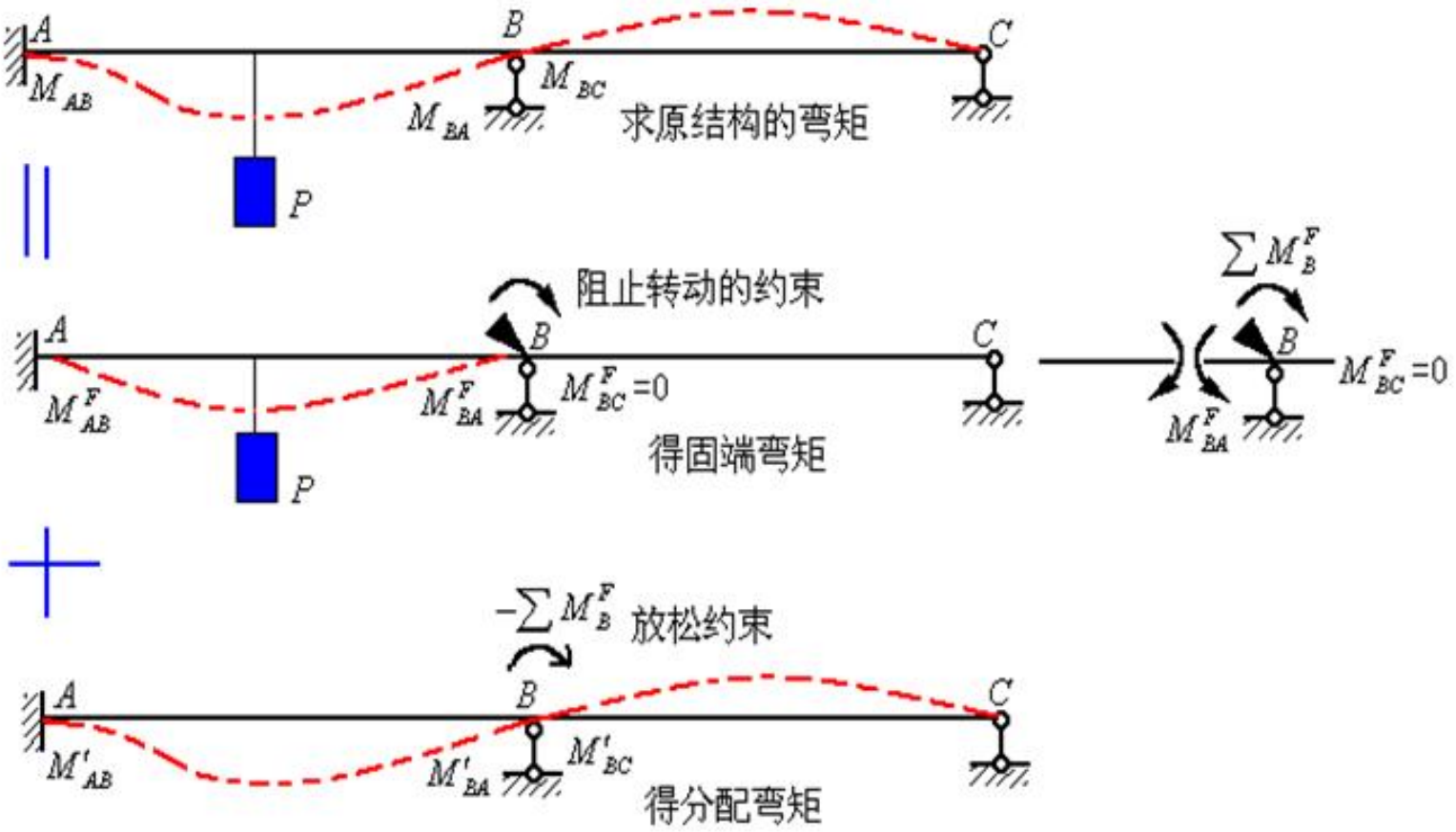
$$M'_{BA}=0.571\times(-60kN\cdot m)=-34.3kN\cdot m$$

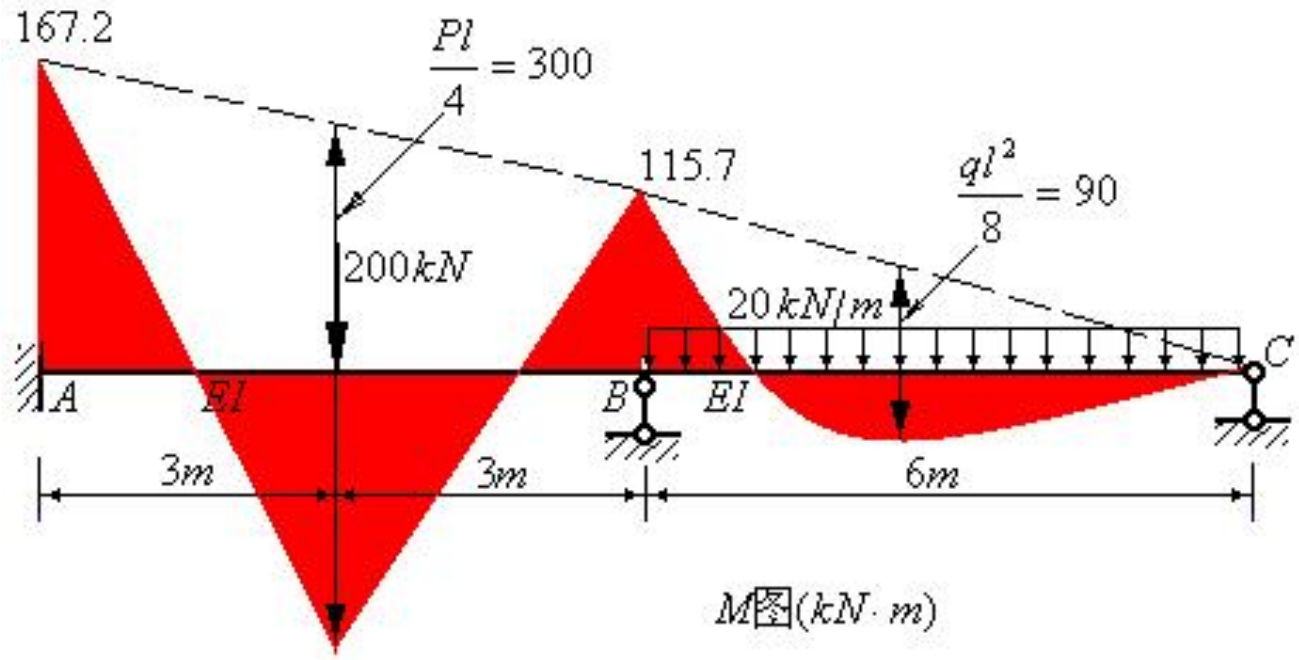
$$M'_{BC}=0.429\times(-60kN\cdot m)=-25.7kN\cdot m$$





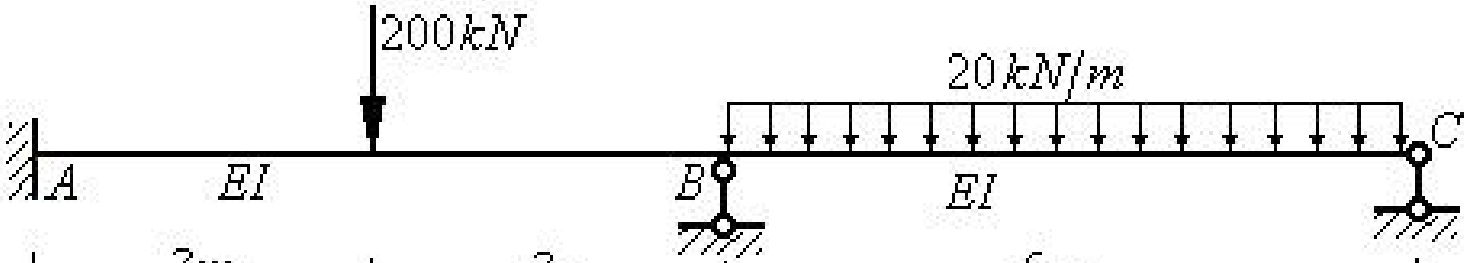
分配系数		0.571	0.429	
固端弯矩M^F	-150	150	-90	0
分配弯矩及传递弯矩	-17.2	-34.3	-25.7	0
杆端弯矩M	-167.2	115.7	-115.7	0


A
B
C
M_{AB}
M_{BA}
M_{BC}
P
求原结构的弯矩
阻止转动的约束
M^F_{AB}
M^F_{BA}
$M^F_{BC}=0$
得固端弯矩
$\sum M^F_B$
$-\sum M^F_B$
放松约束
M'_{AB}
M'_{BA}
M'_{BC}
得分配弯矩

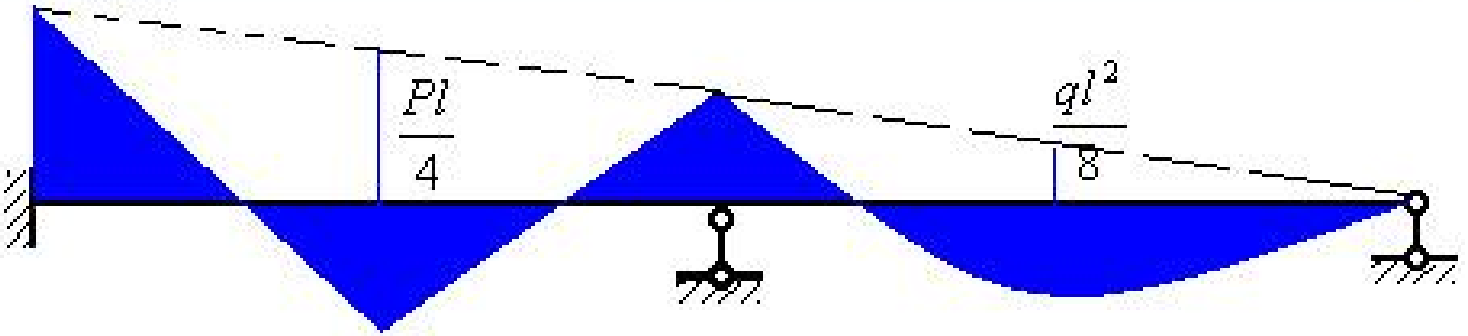

167.2
$\frac{Pl}{4}=300$
115.7
$\frac{ql^2}{8}=90$
200kN
20kN/m
A
EI
B
EI
C
3m
3m
6m

M图(kN·m)

例2



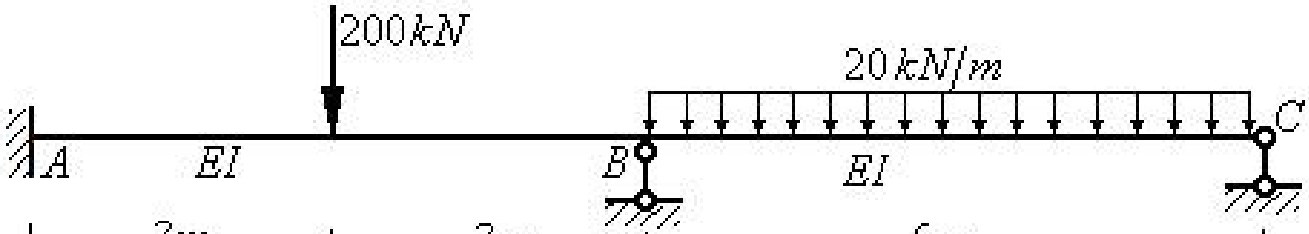
分配系数		0.571	0.429	
固端弯矩M^F	-150	150	-90	0
分配弯矩及传递弯矩	-17.2	-34.3	-25.7	0
杆端弯矩M	-167.2	115.7	-115.7	0



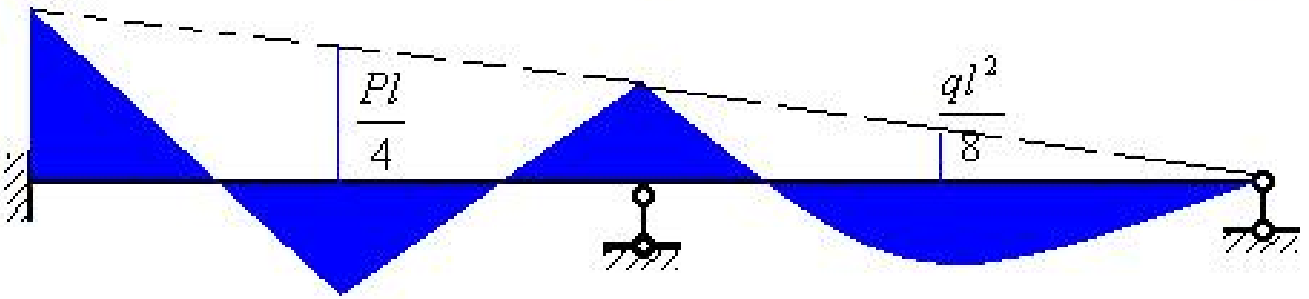
AB杆弯曲刚度为2EI，BC杆弯曲刚度为EI

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令 $i=\dfrac{EI}{6}$，$\mu_{BA}=\dfrac{4\times(2i)}{4\times(2i)+3i}=\dfrac{8}{11}=0.727$，$\mu_{BC}=0.273$



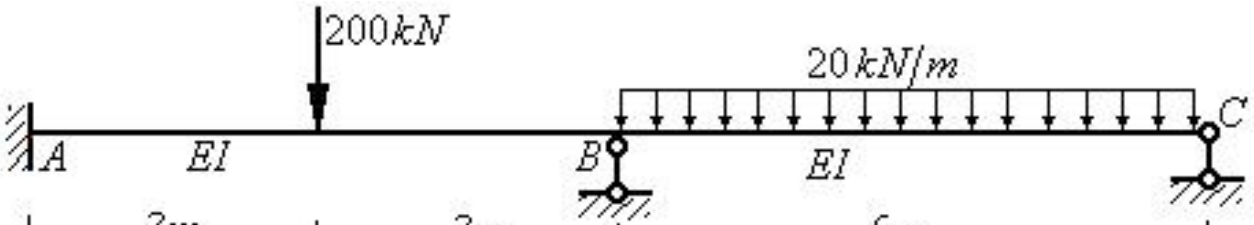
分配系数		0.727	0.273	
固端弯矩M^F	-150	150	-90	0
分配弯矩及传递弯矩	-21.81 ←	-43.62	-16.38	→ 0
杆端弯矩M	-171.81	106.38	-106.38	0



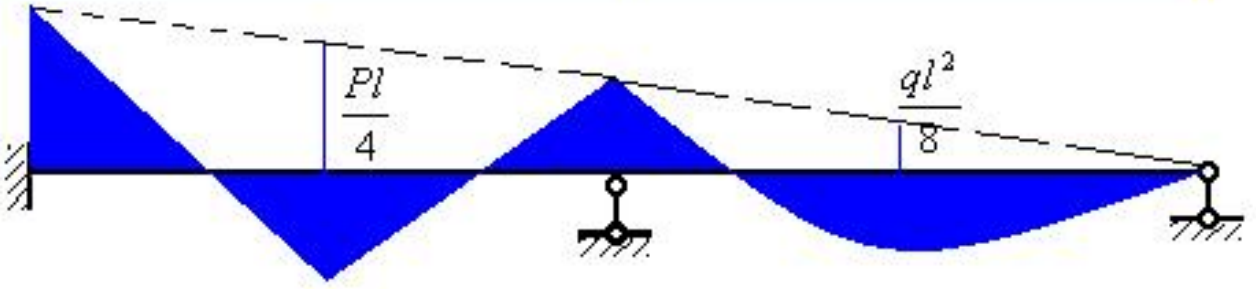
AB杆弯曲刚度为4EI，BC杆弯曲刚度为EI

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令 $i=\dfrac{EI}{6}$，$\mu_{BA}=\dfrac{4\times(4i)}{4\times(4i)+3i}=\dfrac{16}{19}=0.842$，$\mu_{BC}=0.158$



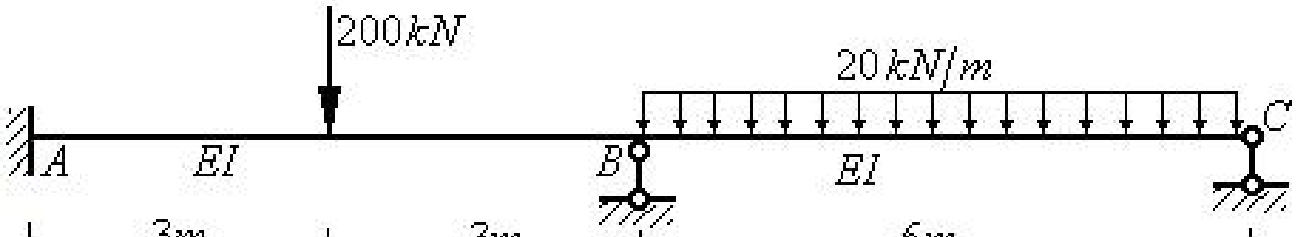
分配系数		0.842	0.158	
固端弯矩M^F	-150	150	-90	0
分配弯矩及传递弯矩	-25.26 ←	-50.52	-9.42	→ 0
杆端弯矩M	-175.26	99.48	-99.48	0



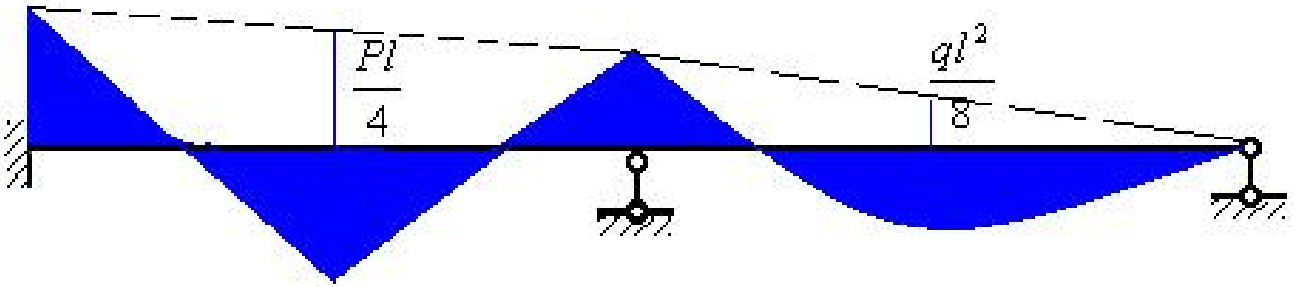
AB杆弯曲刚度为EI，BC杆弯曲刚度为3EI

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令 $i=\dfrac{EI}{6}$，$\mu_{BA}=\dfrac{4\times i}{4\times i+3\times(3i)}=\dfrac{4}{13}=0.308$，$\mu_{BC}=0.692$

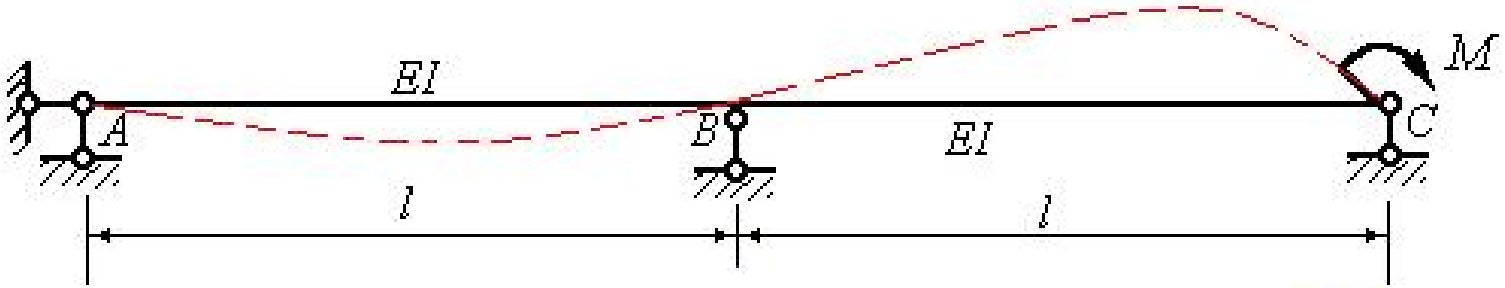


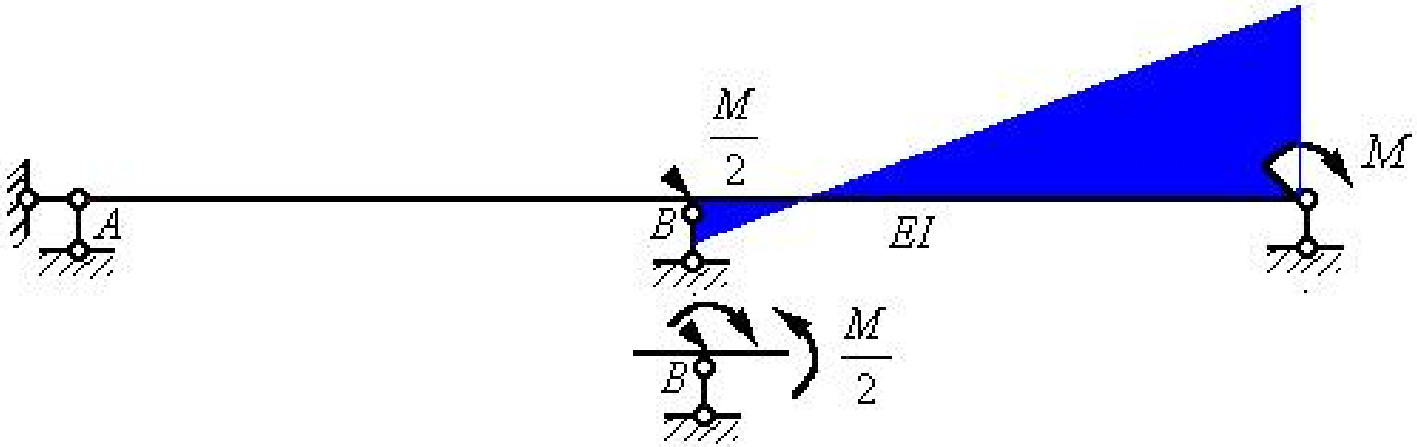
分配系数		0.308	0.692	
固端弯矩M^F	-150	150	-90	0
分配弯矩及传递弯矩	-9.24 ←	-18.48	-41.52	→ 0
杆端弯矩M	-156.24	131.52	-131.52	0

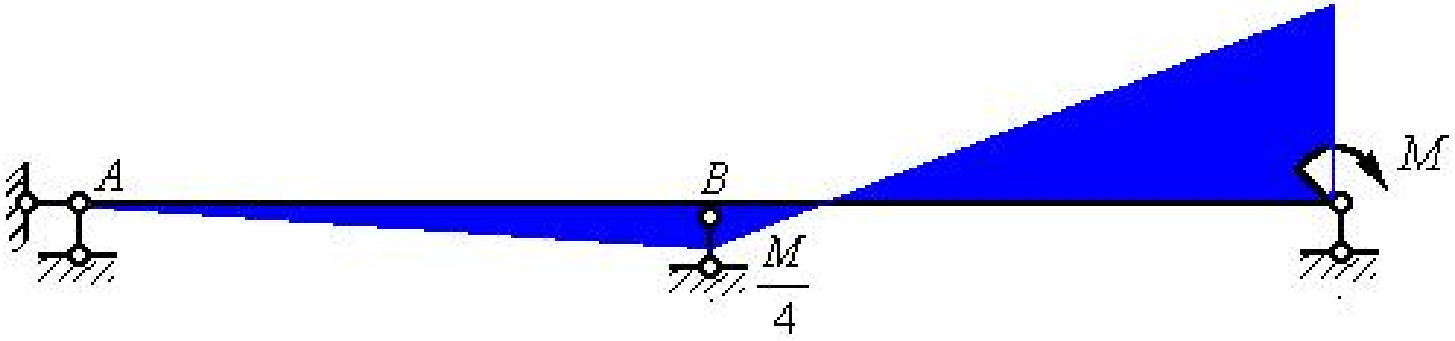


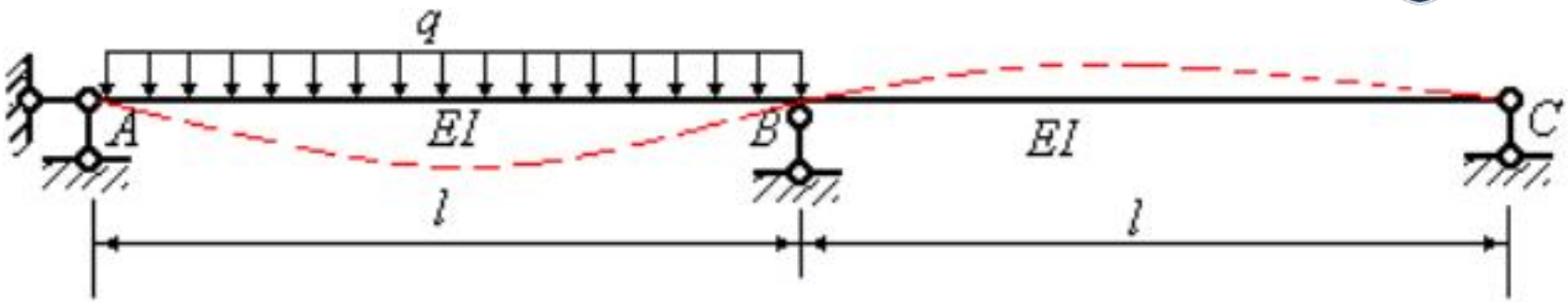
单结点力矩分配法练习题

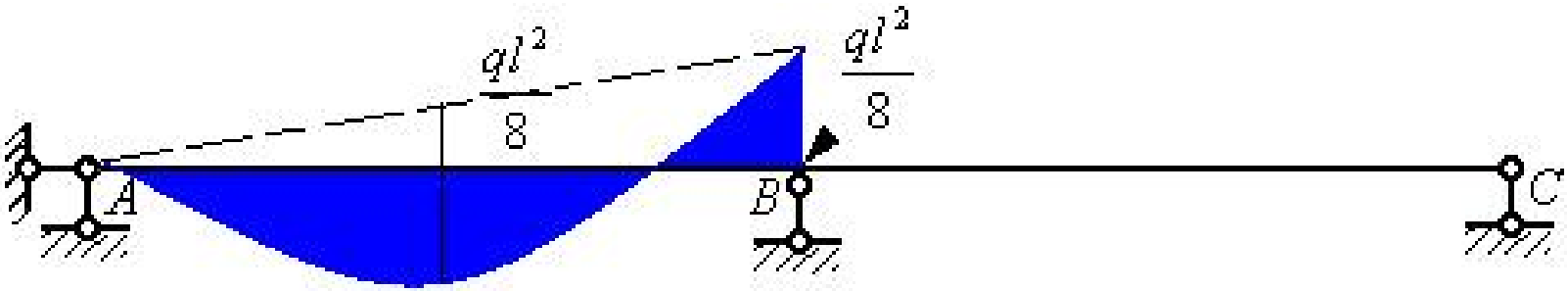
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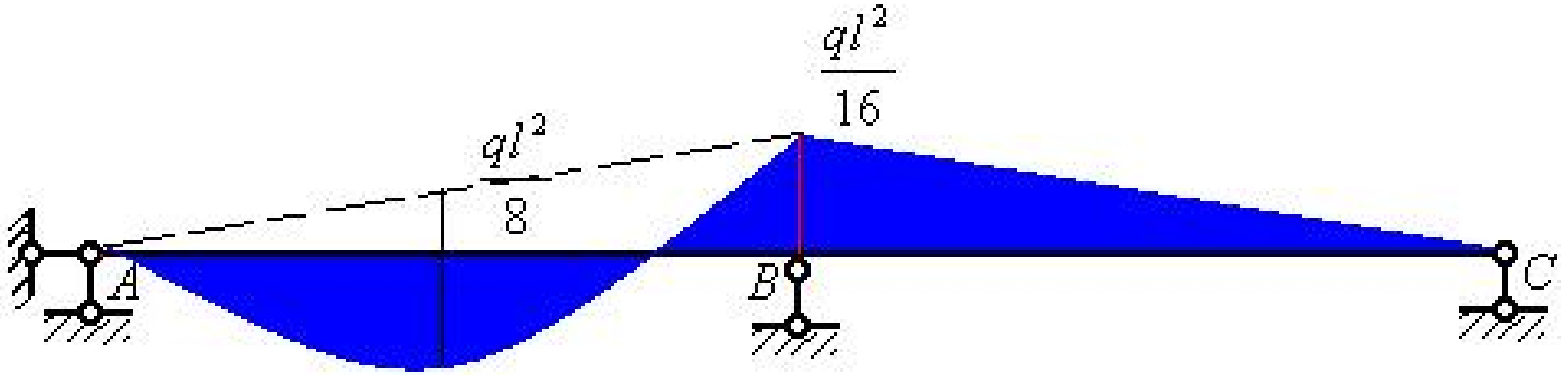


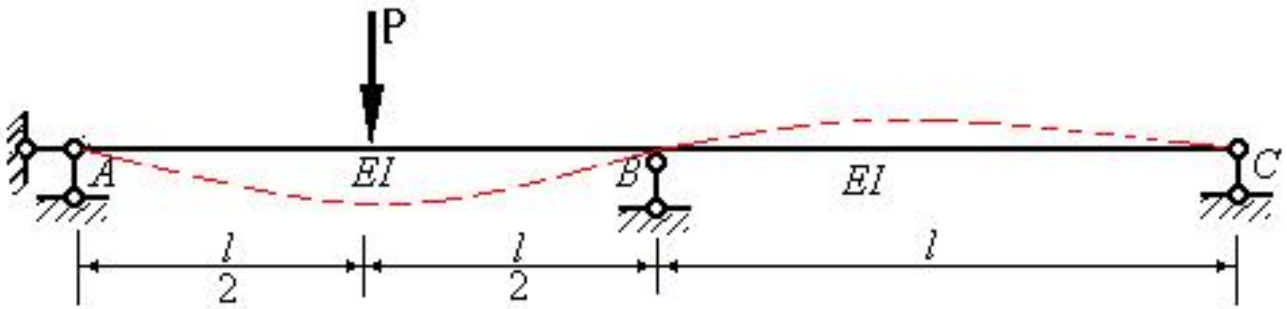


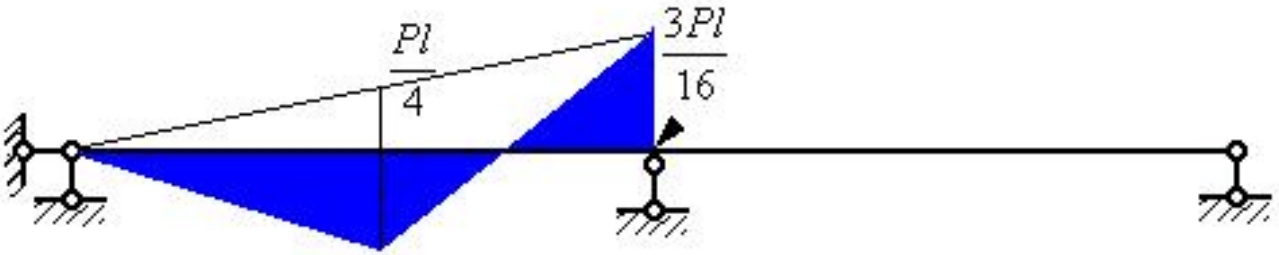


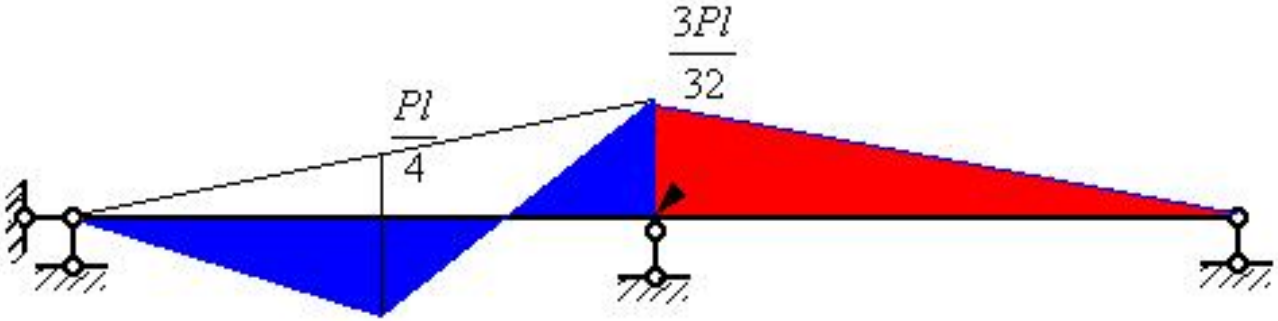

q
A
EI
B
EI
C
l
l

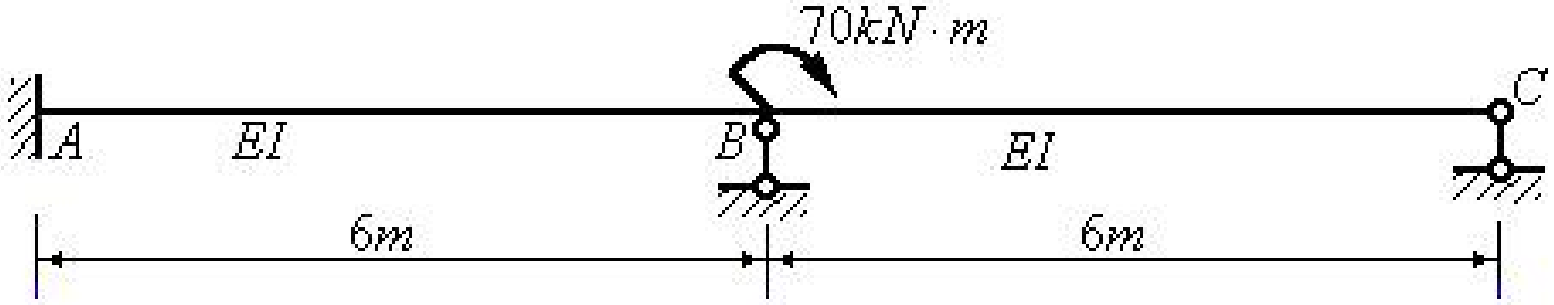

$\frac{ql^2}{8}$
$\frac{ql^2}{8}$
A
B
C

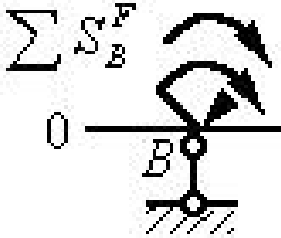

$\frac{ql^2}{16}$
$\frac{ql^2}{8}$
A
B
C


P
A
EI
B
EI
C
l/2
l/2
l


Pl/4
3Pl/16

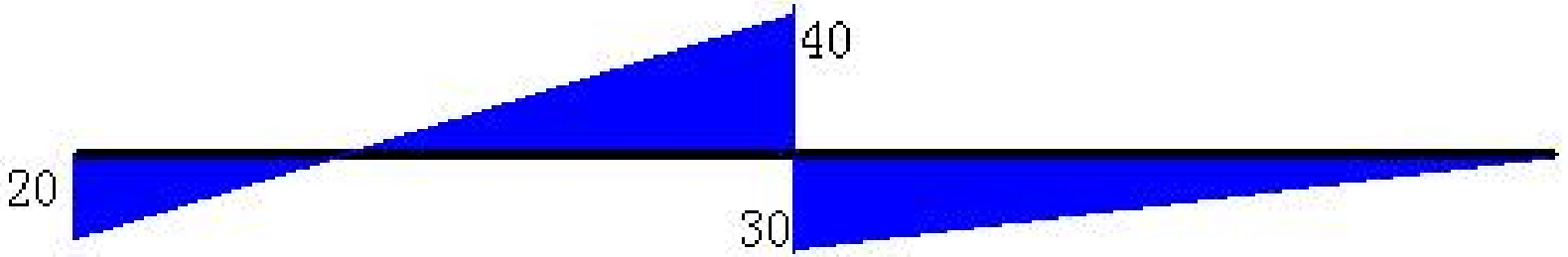

3Pl/32
Pl/4

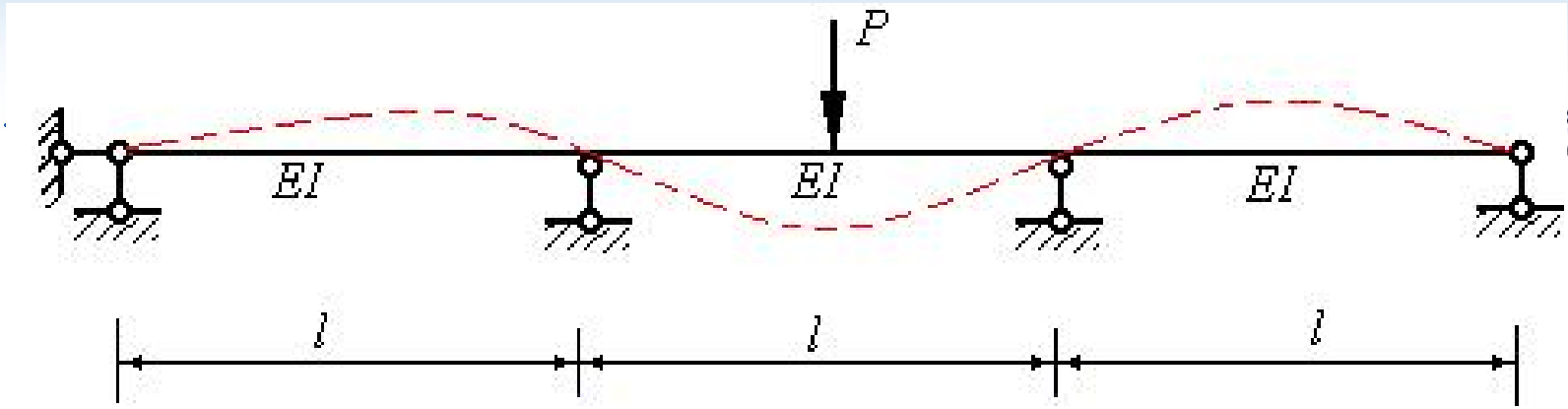


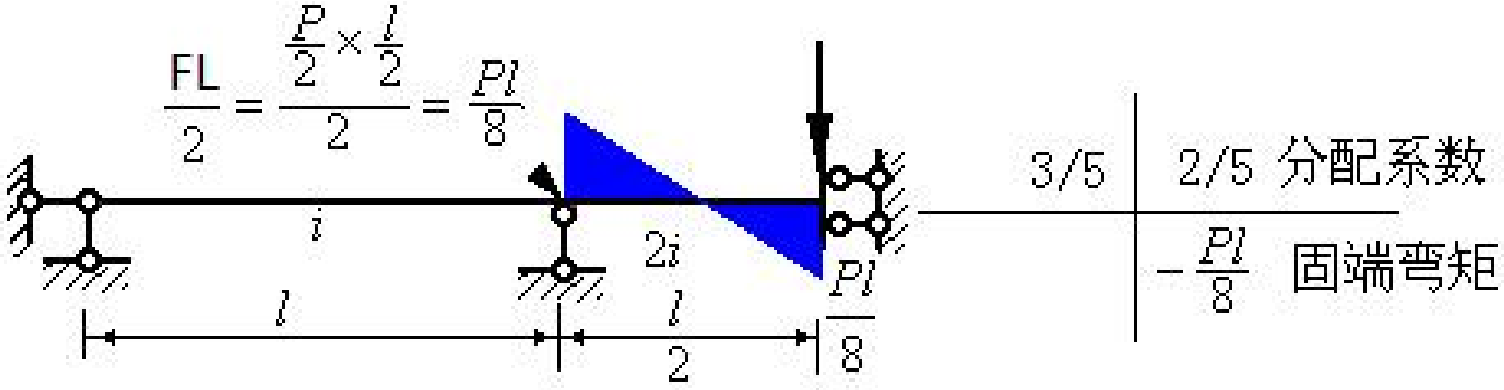


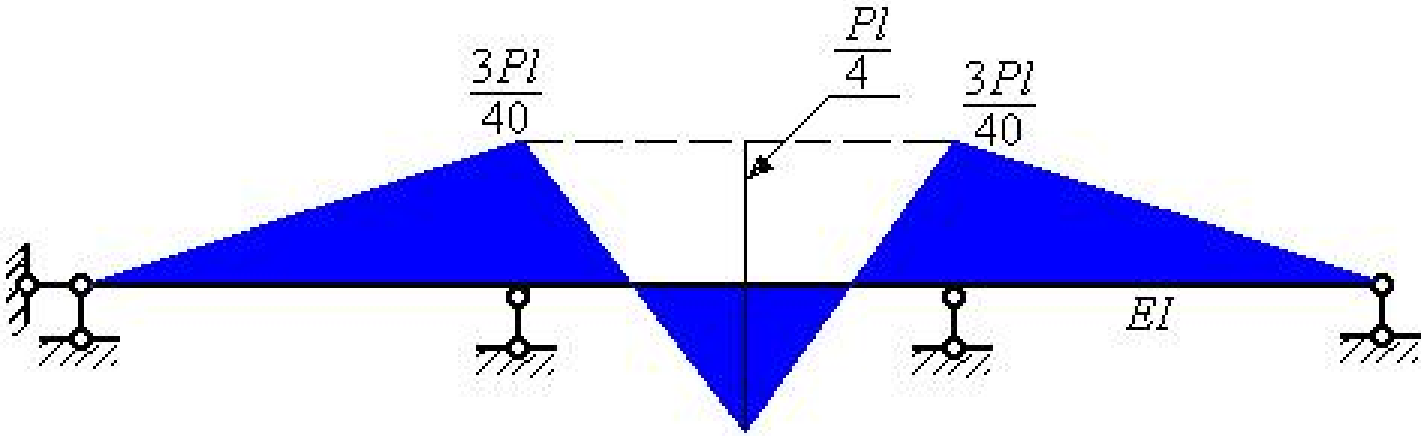
$$\sum S_B^F = R_{1P} = -70kN \cdot m$$

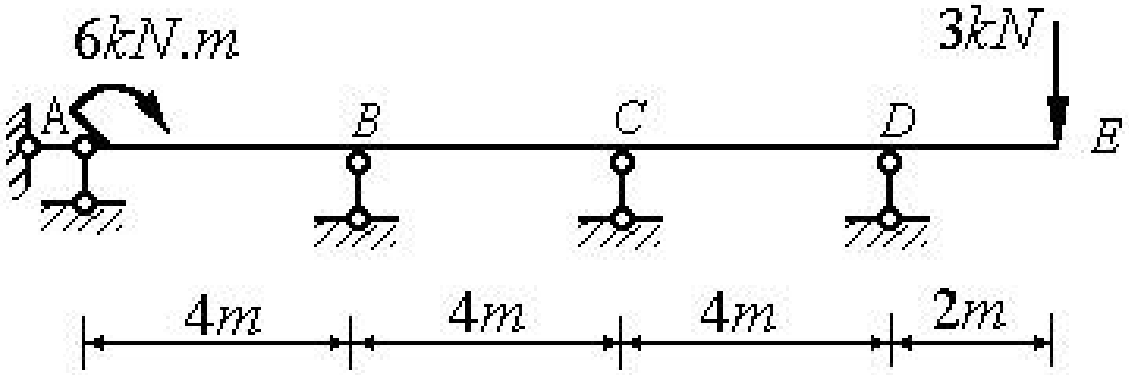
分配系数		$\frac{4}{7}$	$\frac{3}{7}$	
固端弯矩	0	0	0	0
			70	
分配弯矩	20	← 40	30 →	0
杆端弯矩	20	40	30	0

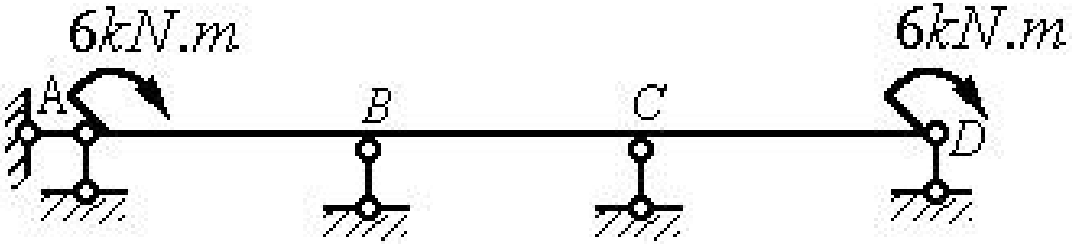


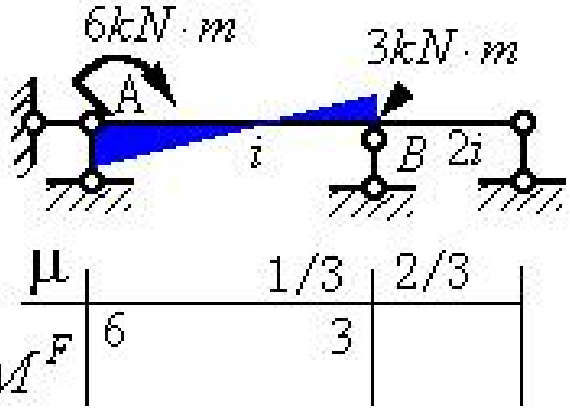

P
EI
EI
EI
l
l
l

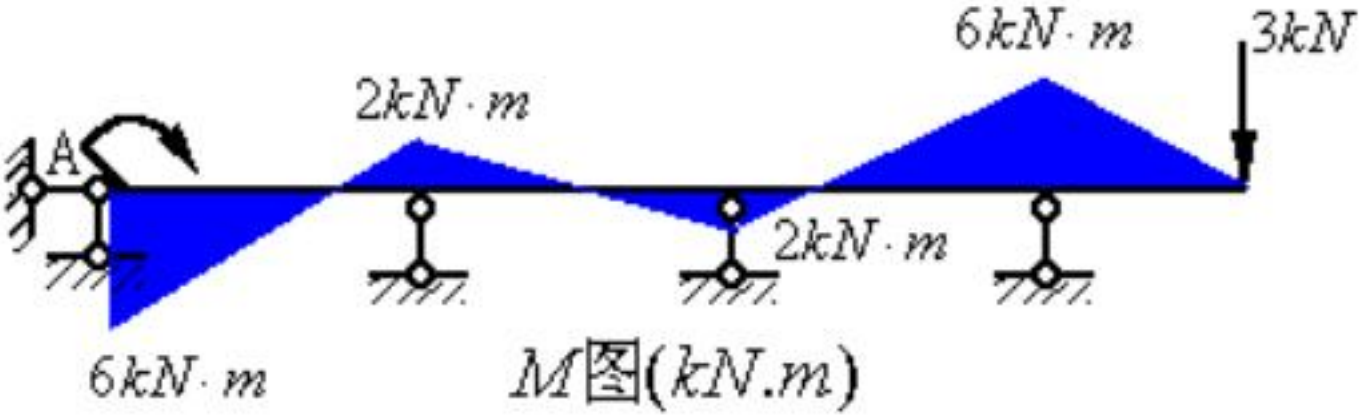

FL/2 = (P/2 × l/2)/2 = Pl/8
i
2i
l
l/2
Pl/8
3/5
2/5 分配系数
−Pl/8 固端弯矩

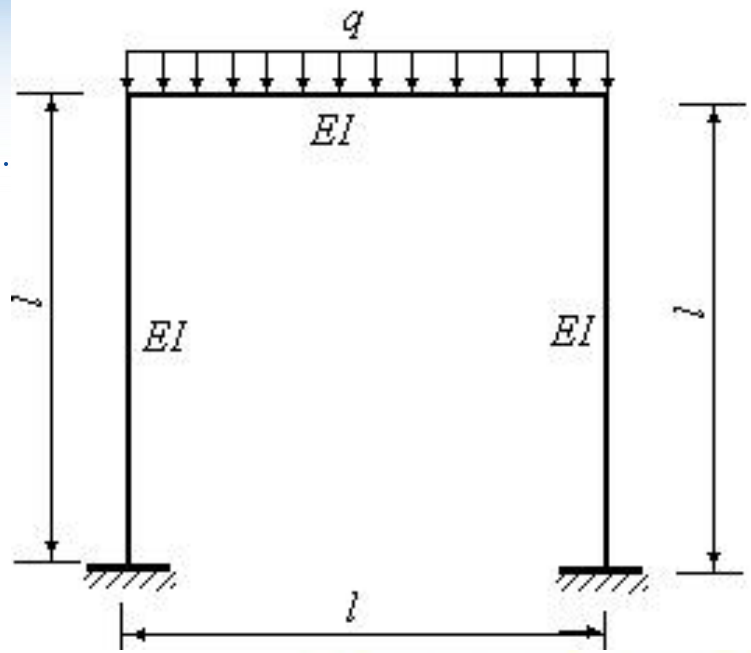

3Pl/40
Pl/4
3Pl/40
EI

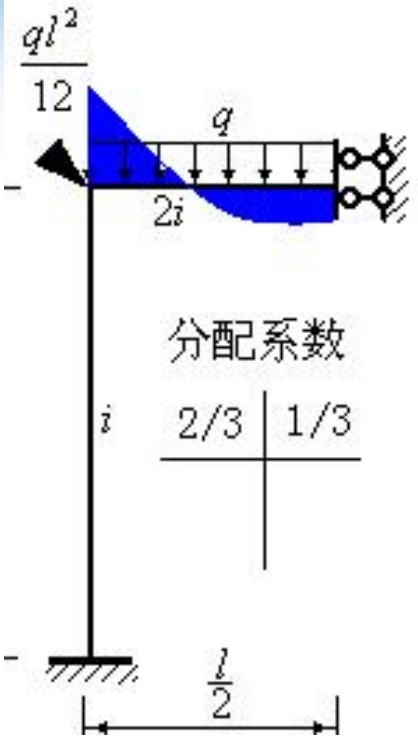

6kN.m
3kN
A
B
C
D
E
4m
4m
4m
2m

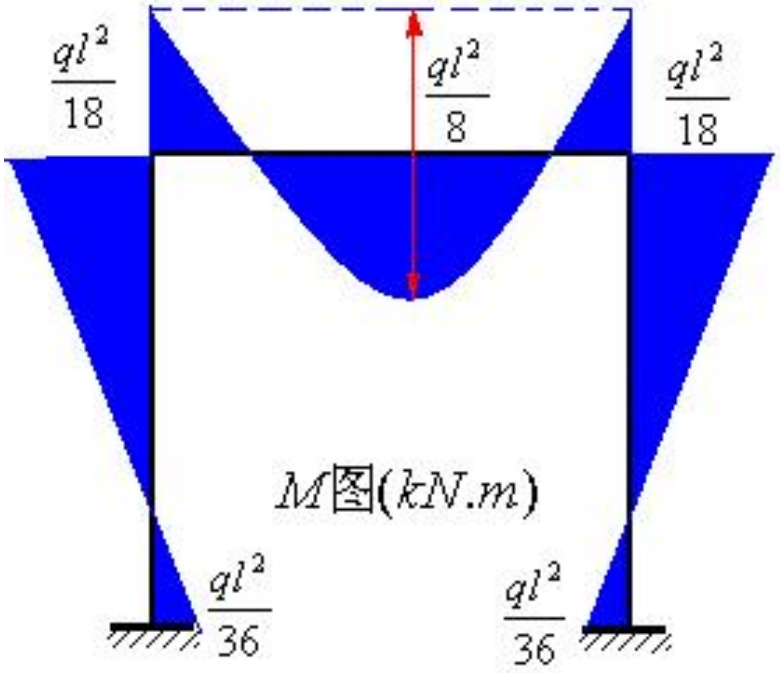

6kN.m
6kN.m
A
B
C
D


6kN·m
3kN·m
A
i
B
2i
μ
1/3
2/3
M^F
6
3


6kN·m
3kN
2kN·m
A
2kN·m
6kN·m
M图(kN.m)

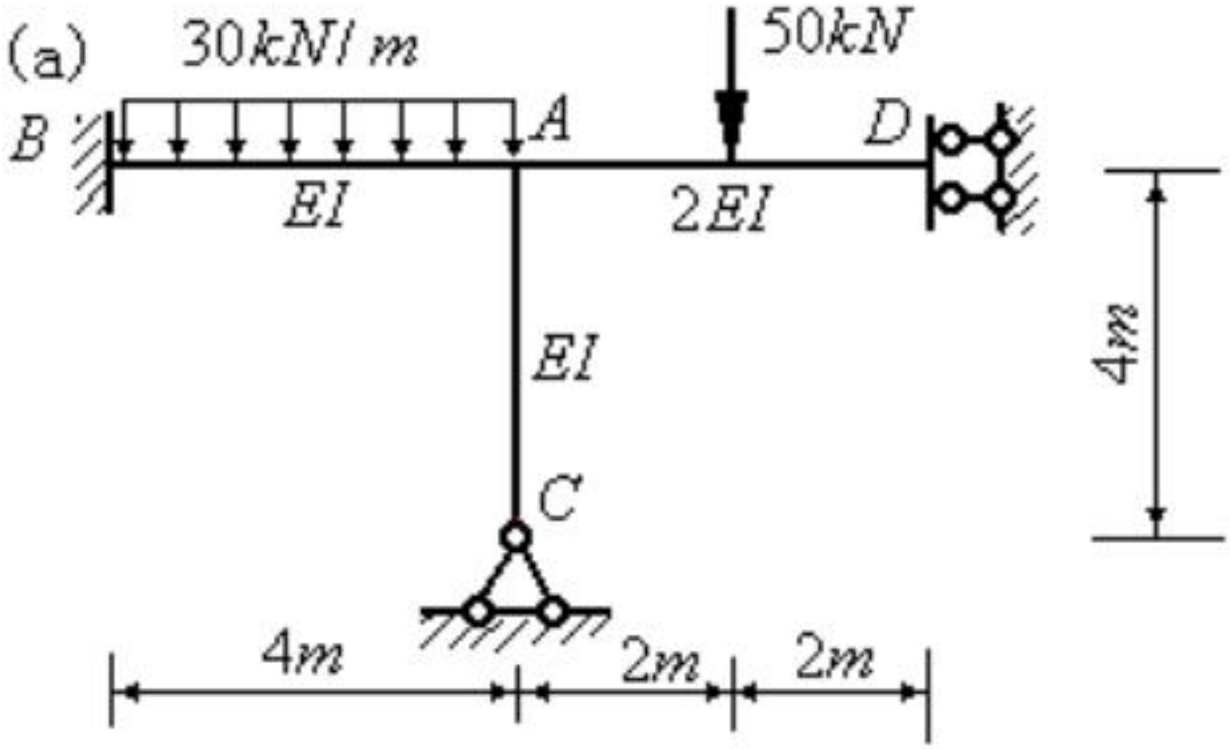




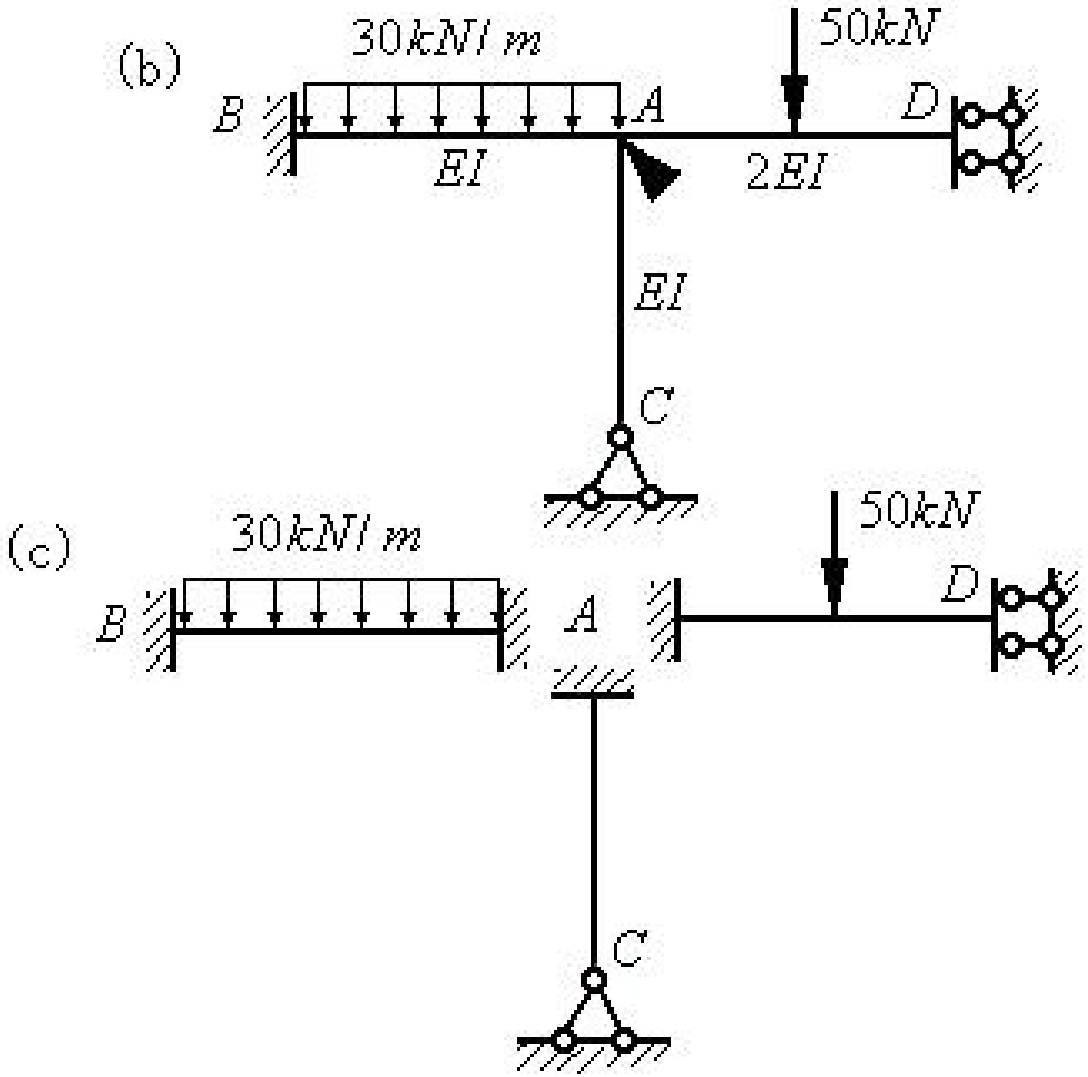


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用力矩分配法作图示刚架的弯矩图



固定结点A



固端弯矩、分配系数

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在结点A加约束。计算固端弯矩及分配系数，令$i=\dfrac{EI}{4}=1.$

$$M_{BA}^{F}=-\frac{ql^2}{12}=-40kN\cdot m,\qquad M_{AB}^{F}=\frac{ql^2}{12}=40kN\cdot m$$

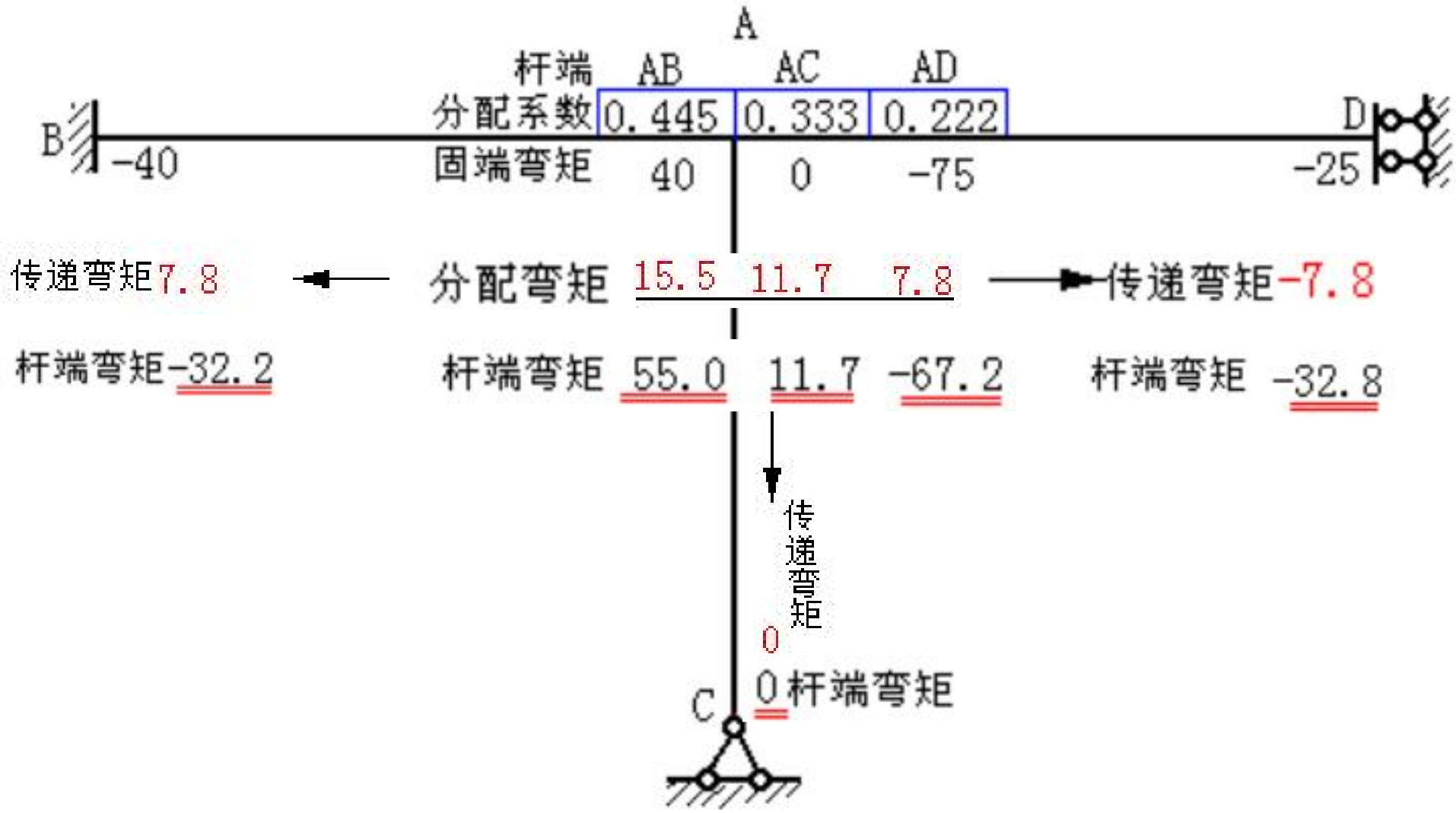
$$M_{AD}^{F}=-\frac{3Pl}{8}=-75kN\cdot m,\qquad M_{DA}^{F}=-\frac{Pl}{8}=-25kN\cdot m$$

计算分配系数：

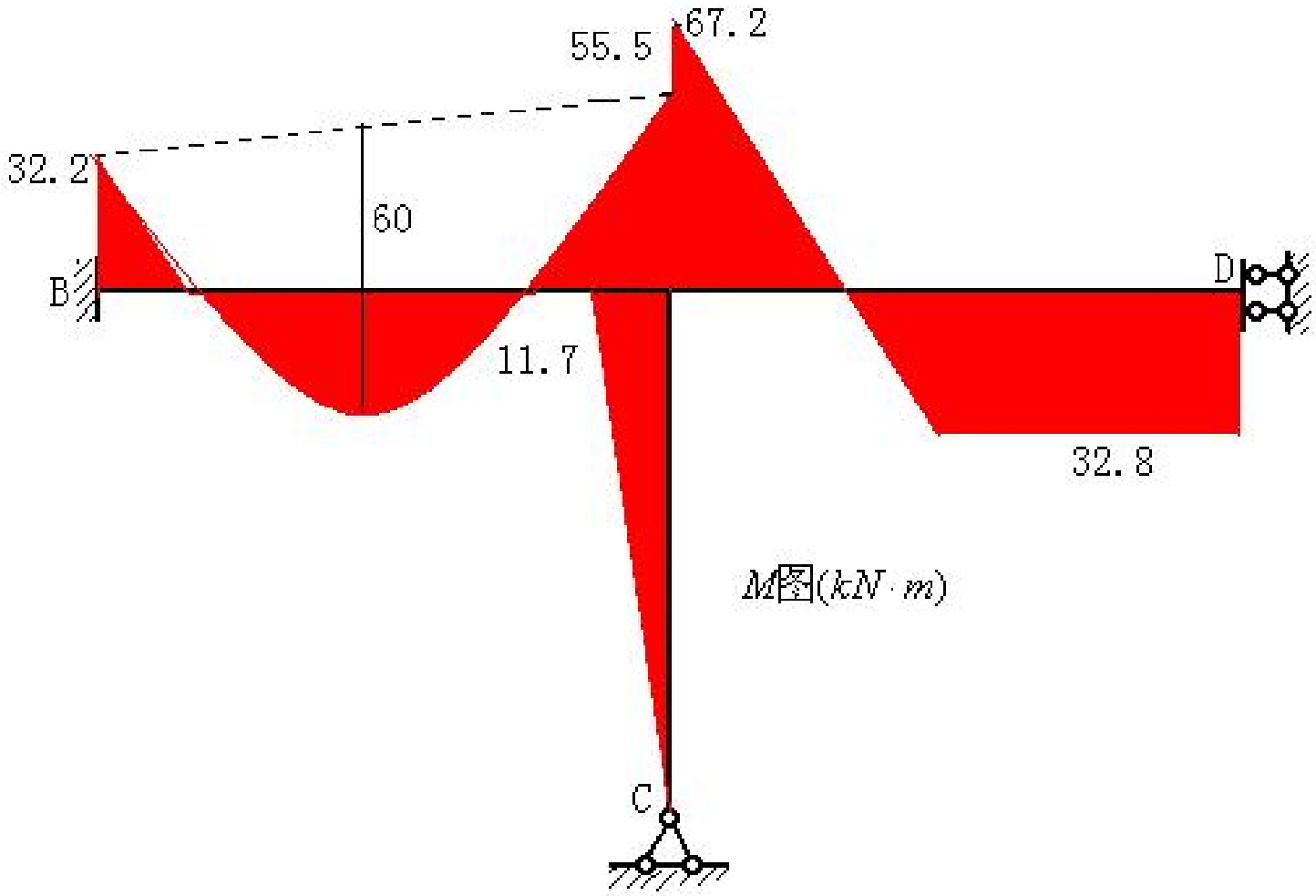
$$\mu_{AB}=\frac{S_{AB}}{\sum S_A}=\frac{S_{AB}}{S_{AB}+S_{AC}+S_{AD}}=\frac{4\times1}{4\times1+3\times1+2}=\frac{4}{9}=0.445,$$

$$\mu_{AC}=\frac{3}{9}=0.333,\qquad \mu_{AD}=\frac{2}{9}=0.222$$

计算过程


A
杆端
AB
AC
AD
分配系数
0.445
0.333
0.222
B
D
-40
固端弯矩
40
0
-75
-25
传递弯矩7.8
分配弯矩
15.5
11.7
7.8
传递弯矩-7.8
杆端弯矩-32.2
杆端弯矩
55.0
11.7
-67.2
杆端弯矩
-32.8
传递弯矩
0
C
0杆端弯矩

M图



§9-3　用力矩分配法计算连续梁和无侧移刚架

§9-3用力矩分配法计算连续梁和无侧移刚架

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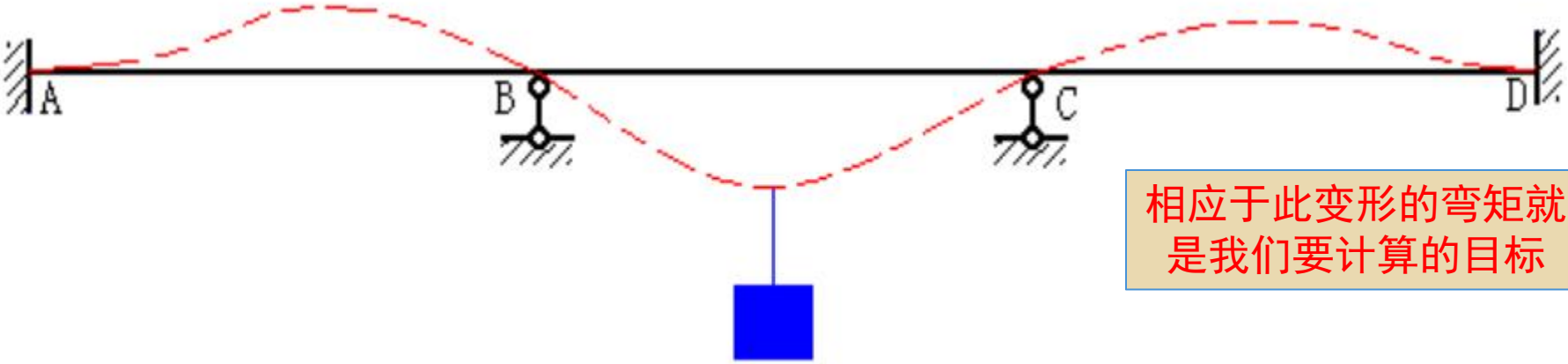
对于有多个结点连续梁和无侧移刚架，只需依次对各结点使用上节所述方法便可求出杆端弯矩。作法是：

（1）先将所有结点固定，计算固端弯矩和各结点的分配系数；

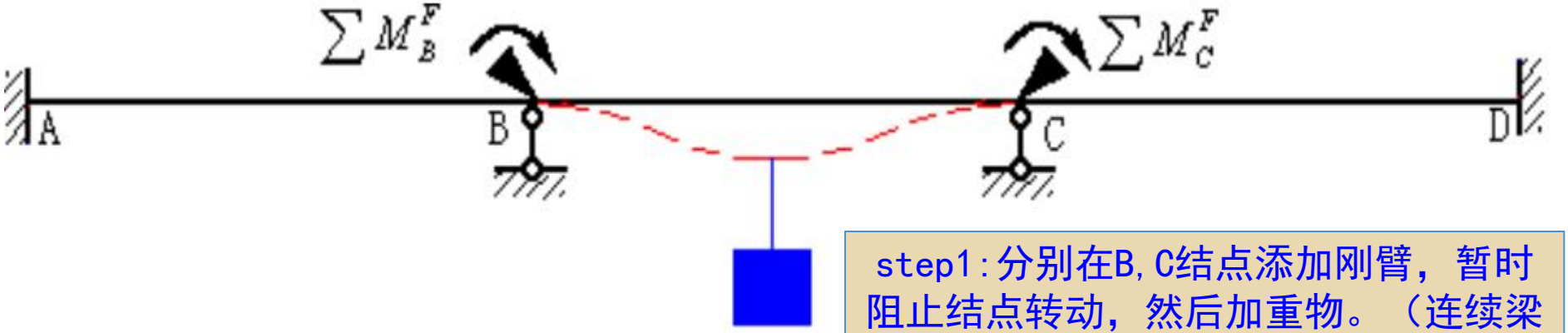
（2）然将各结点轮流地放松，即每次只放松一个结点，其它结点仍然暂时固定，这样把各结点的不平衡力矩轮流地进行分配、传递，直到传递弯矩小到可略去为止，这样的逐次渐近方法来计算杆端弯矩。

用力矩分配法计算连续梁

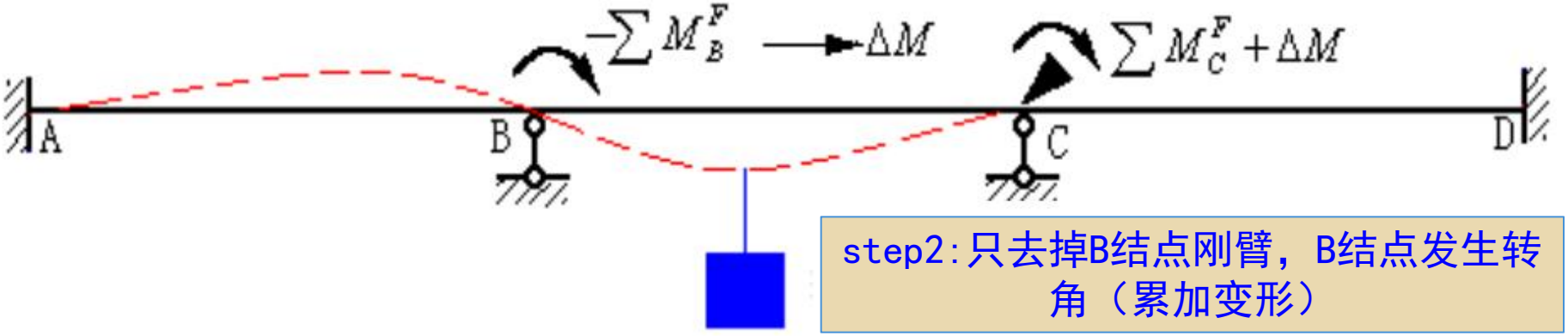
用一个三跨连续梁的模型来说明逐次渐近的过程。



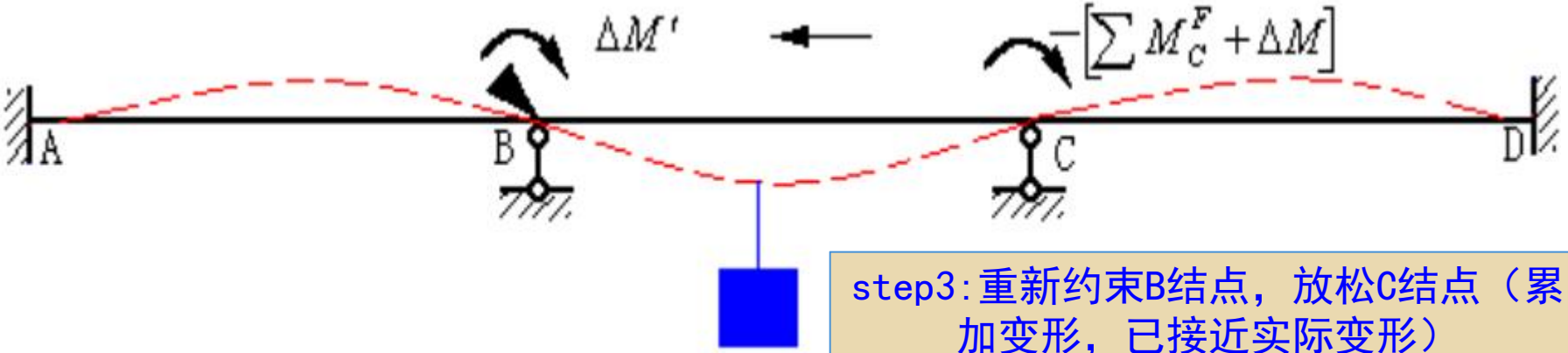
相应于此变形的弯矩就是我们要计算的目标



step1：分别在B, C结点添加刚臂，暂时阻止结点转动，然后加重物。（连续梁被分为三个单跨梁，仅BC一跨有变形）



step2:只去掉B结点刚臂，B结点发生转角（累加变形）



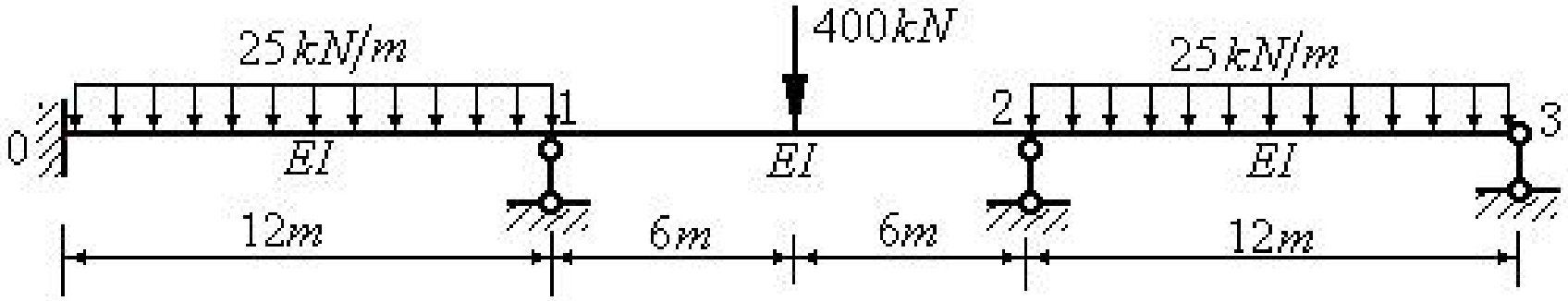
step3:重新约束B结点，放松C结点（累加变形，已接近实际变形）

依此类推，再重复上述第二步和第三步，即轮流去掉结点*B*和结点*C*的约束。连续梁的变形和内力很快就达到实际状态，但每次只放松一个结点，故每一步均为单结点的分配和传递运算。最后，将各项步骤所得杆端弯矩（弯矩增量）叠加，即得所求的杆端弯矩（总弯矩）。

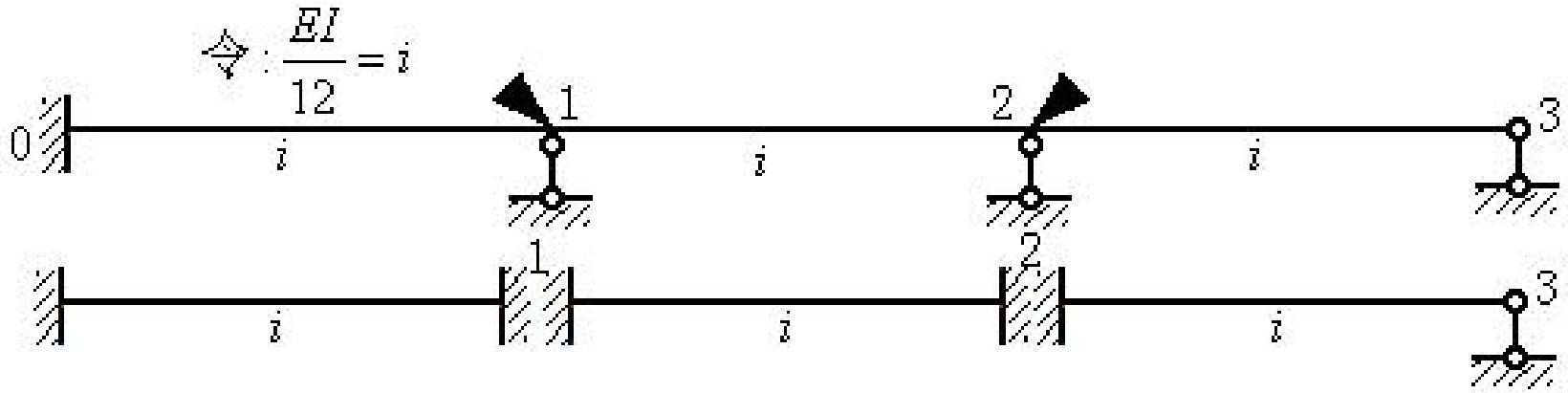
实际上，只需对各结点进行两到三个循环的运算，就能达到较好的精度。

例1 作图示连续梁的弯矩图

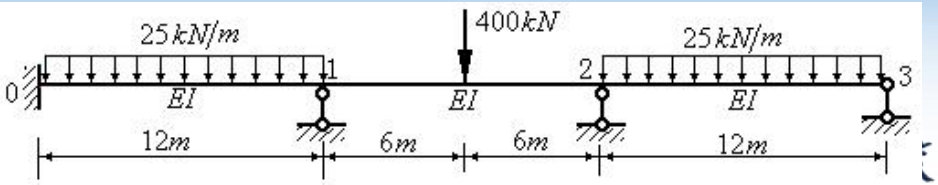
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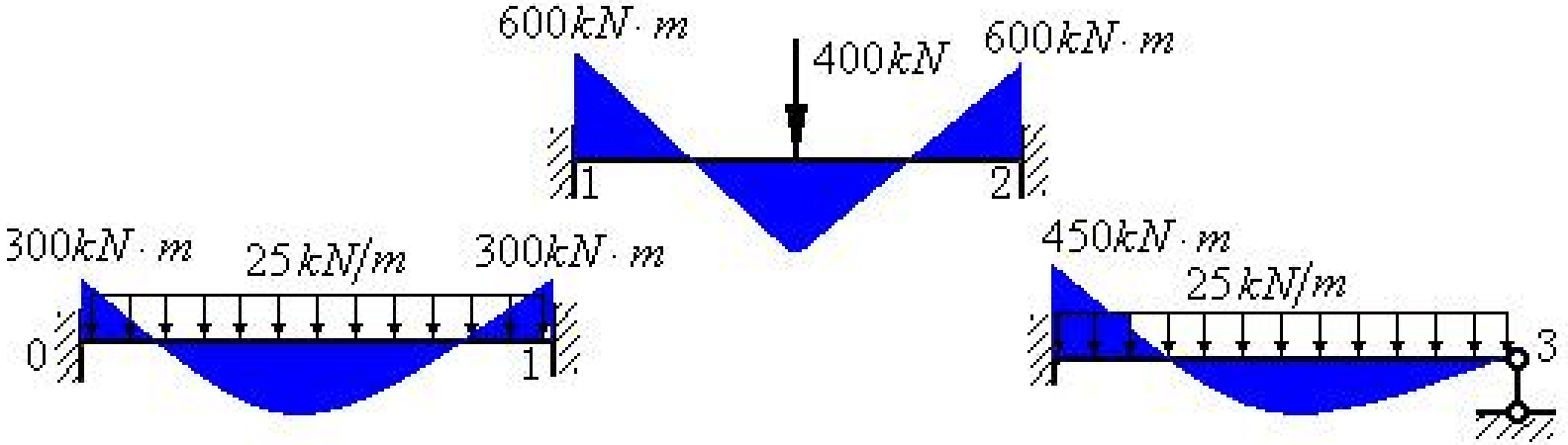


（1）将结点1、2固定。将该连续梁分成三个单跨梁。



计算固端弯矩



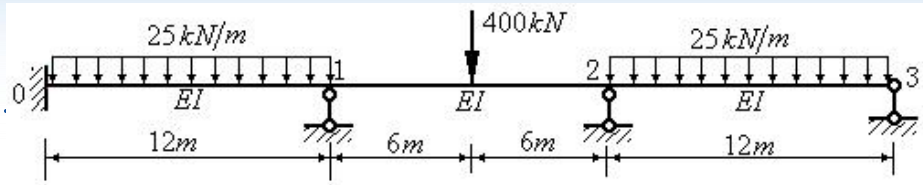


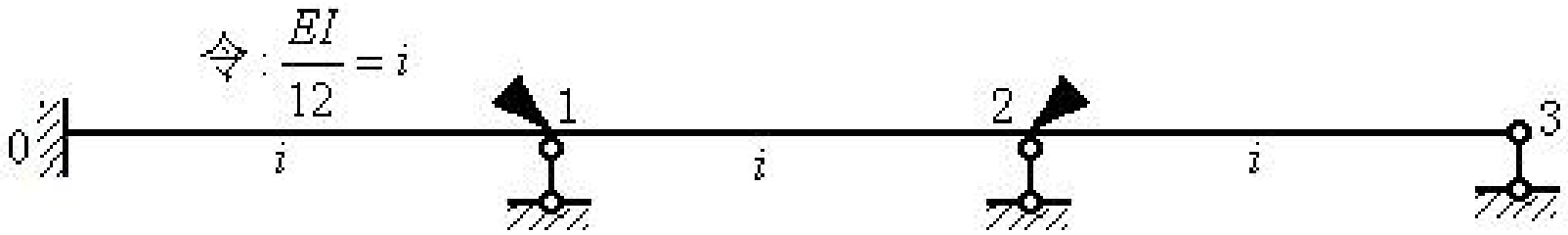
$$M_{01}^{F}=-\frac{ql^2}{12}=-\frac{25\,kN/m\times(12m)^2}{12}=-300kN\cdot m,\quad M_{10}^{F}=300kN\cdot m$$

$$M_{12}^{F}=-\frac{Pl}{8}=\frac{400kN\times12}{8}=-600kN\cdot m,\qquad M_{21}^{F}=600kN\cdot m$$

$$M_{23}^{F}=-\frac{ql^2}{8}=-\frac{25\,kN/m\times(12m)^2}{8}=-450kN\cdot m,\quad M_{32}^{F}=0$$

计算结点分配系数

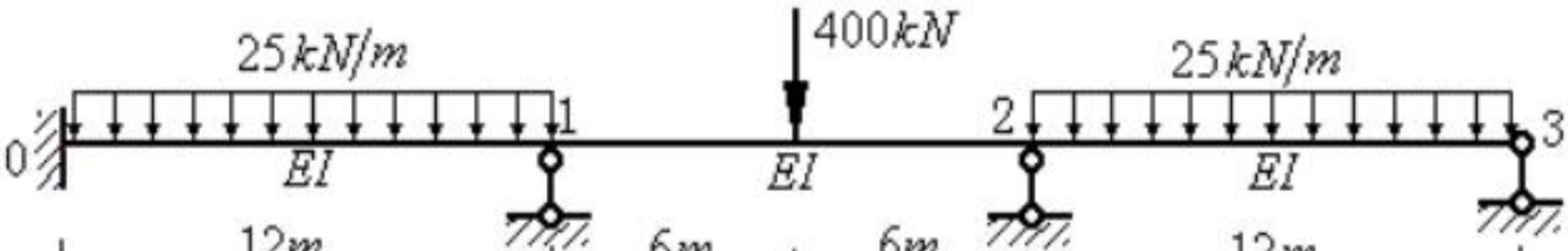




$$\mu_{10}=\frac{S_{10}}{S_{10}+S_{12}}=\frac{4i}{4i+4i}=\frac{1}{2},\qquad \mu_{12}=\frac{S_{12}}{S_{10}+S_{12}}=\frac{4i}{4i+4i}=\frac{1}{2}$$

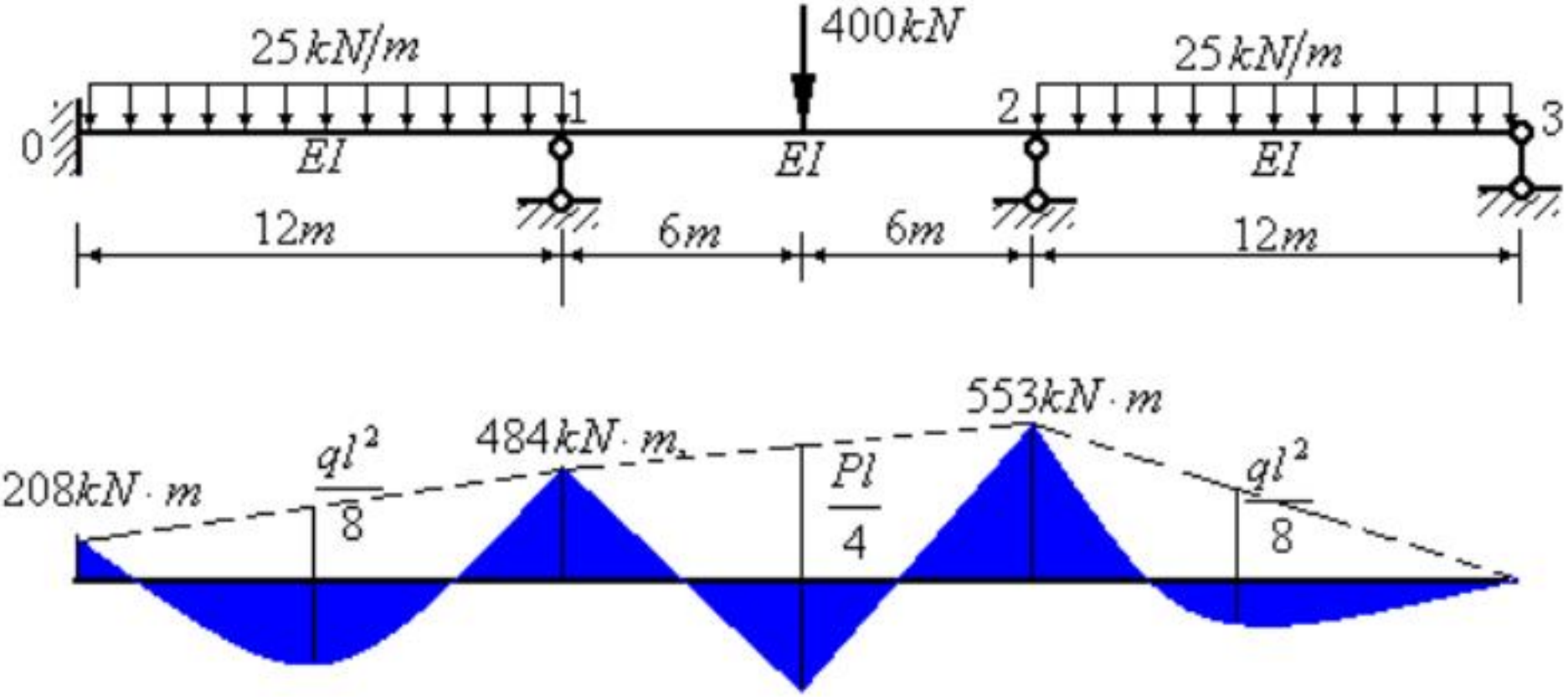
$$\mu_{21}=\frac{S_{21}}{S_{21}+S_{23}}=\frac{4i}{4i+3i}=\frac{4}{7},\qquad \mu_{23}=\frac{S_{23}}{S_{21}+S_{23}}=\frac{3i}{4i+3i}=\frac{3}{7}$$

分配、传递

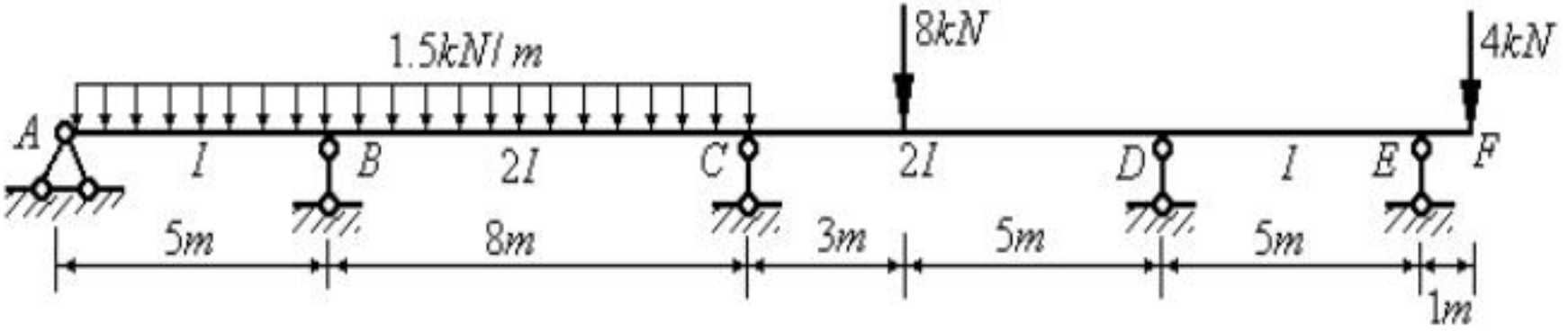


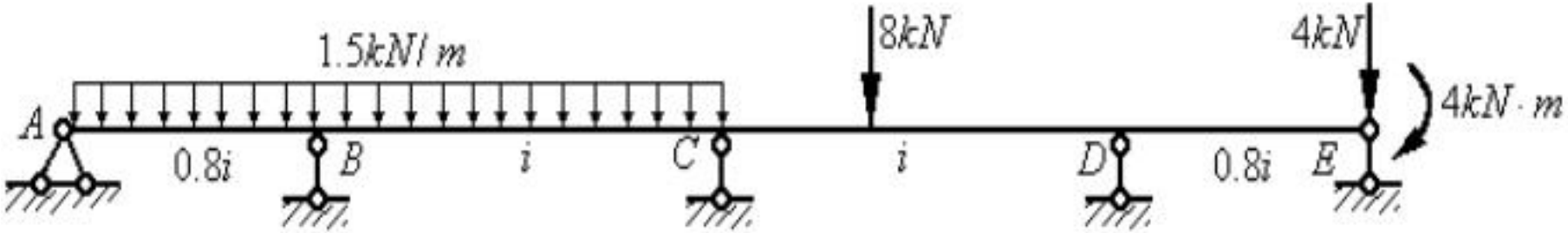
分配系数 μ			1/2	1/2	4/7	3/7	
固端弯矩 M^F	-300		300	-600	600	-450	
分配传递	75	←	150	150	→	75	
			-64	←	-129	-96	→ 0
	16	←	32	32	→	16	
			-5	←	-9	-7	→ 0
	1	←	2	3	→	1	
					-1	0	
杆端弯矩	-208		484	-484	553	-553	0

M图

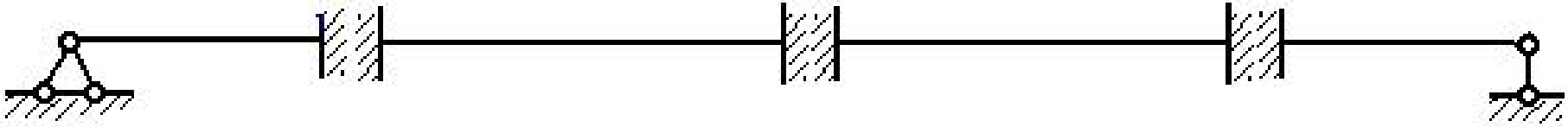

25kN/m
400kN
25kN/m
0
1
2
3
EI
EI
EI
12m
6m
6m
12m
208kN·m
ql²/8
484kN·m
Pl/4
553kN·m
ql²/8

例2 作图示连续梁的弯矩图

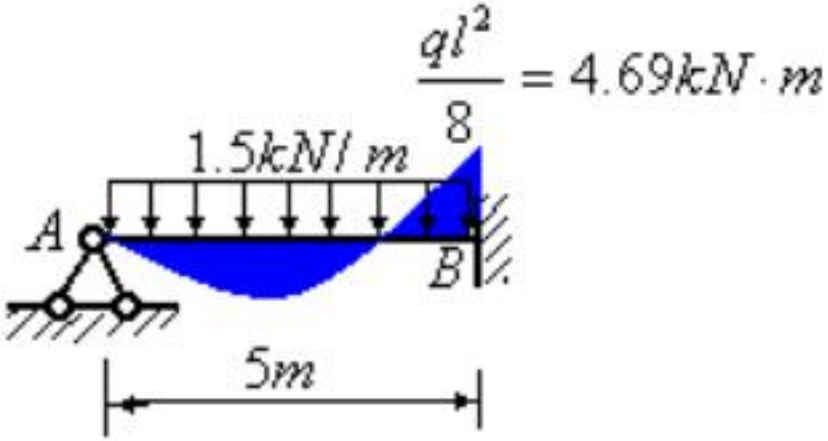


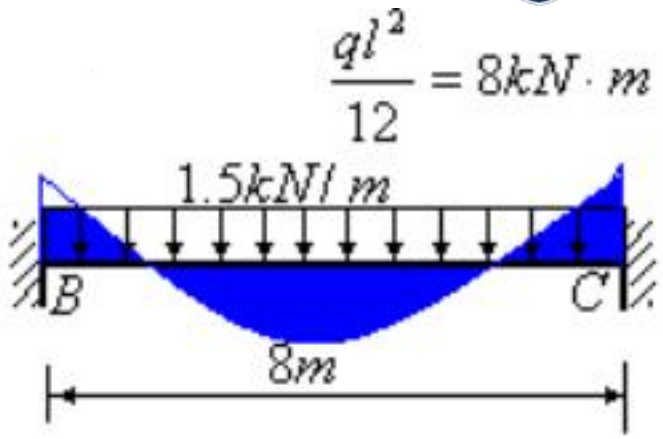


（1）将结点B、C、D固定。将该连续梁分成四个单跨梁



计算固端弯矩



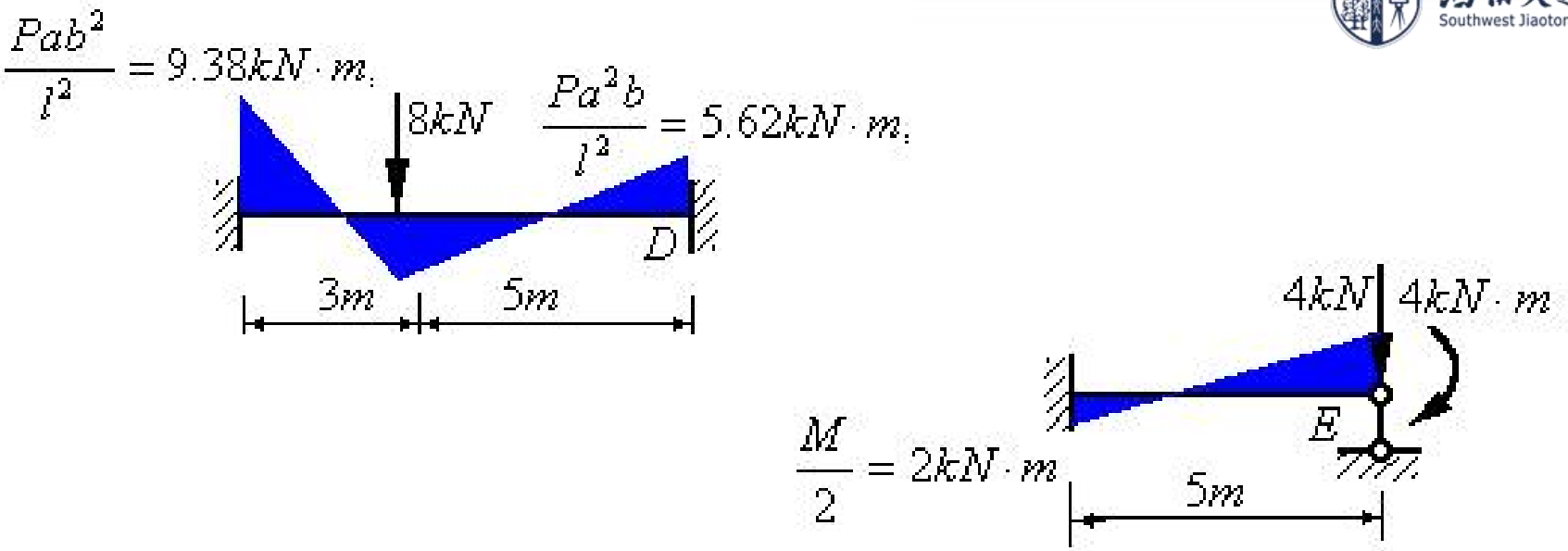


$$M_{BA}^{F}=\frac{ql^2}{8}=\frac{1.5kN/m\times(5m)^2}{8}=4.69kN\cdot m$$

$$M_{BC}^{F}=-\frac{ql^2}{12}=-\frac{1.5kN/m\times(8m)^2}{12}=-8kN\cdot m,$$

$$M_{CB}^{F}=8kN\cdot m$$

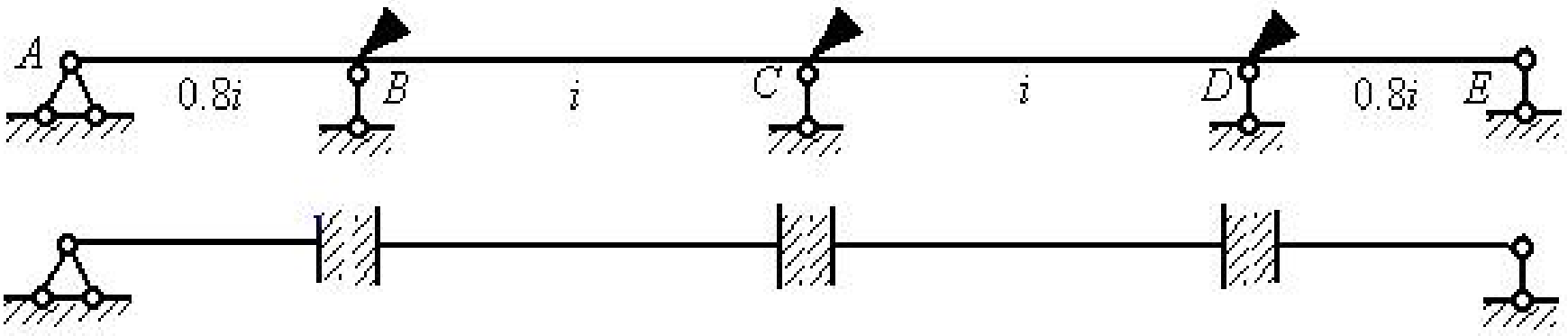
计算固端弯矩



$$M_{CD}^{F} = -\frac{Pab^2}{l^2} = -\frac{8kN \times 3m \times (5m)^2}{(8m)^2} = -9.38kN \cdot m,$$

$$M_{DC}^{F} = \frac{Pa^2b}{l^2} = 5.63kN \cdot m \qquad M_{DE}^{F} = \frac{M}{2} = 2kN \cdot m$$

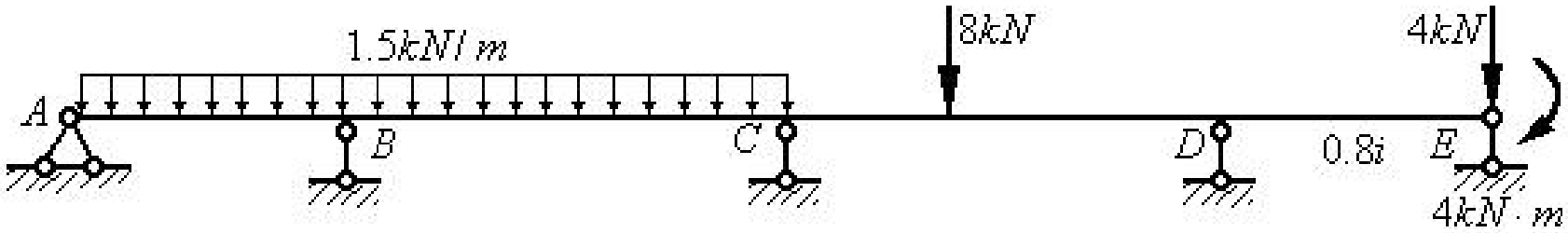
计算结点分配系数



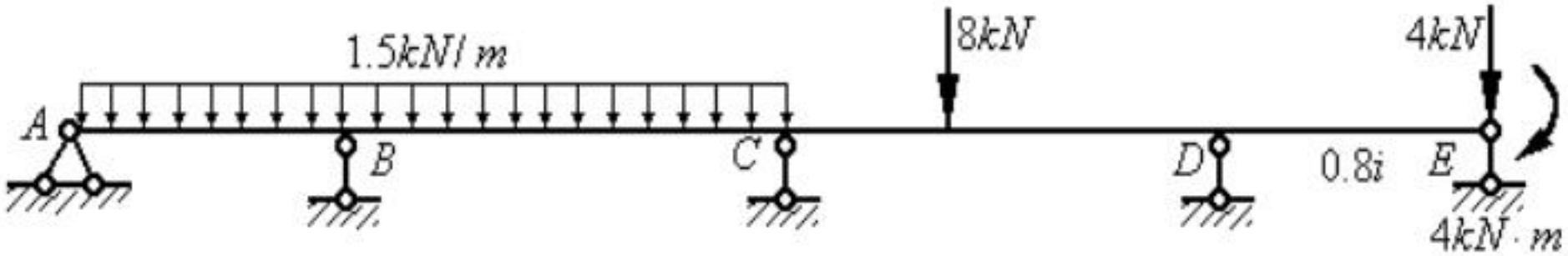
$$\mu_{BA}=\frac{S_{BA}}{S_{BA}+S_{BC}}=\frac{3\times0.8i}{3\times0.8i+4i}=0.375,\qquad \mu_{BC}=\frac{S_{BC}}{S_{BA}+S_{BC}}=\frac{4i}{3\times0.8i+4i}=0.625$$

$$\mu_{CB}=\frac{S_{CB}}{S_{CB}+S_{CD}}=\frac{4i}{4i+4i}=0.5\qquad \mu_{CD}=\frac{S_{CD}}{S_{CB}+S_{CD}}=\frac{4i}{4i+4i}=0.5$$

$$\mu_{DC}=\frac{S_{DC}}{S_{DC}+S_{DE}}=\frac{4i}{4i+3\times0.8i}=0.625,\qquad \mu_{DE}=\frac{S_{DE}}{S_{BD}+S_{DE}}=\frac{3\times0.8i}{4i+3\times0.8i}=0.375$$

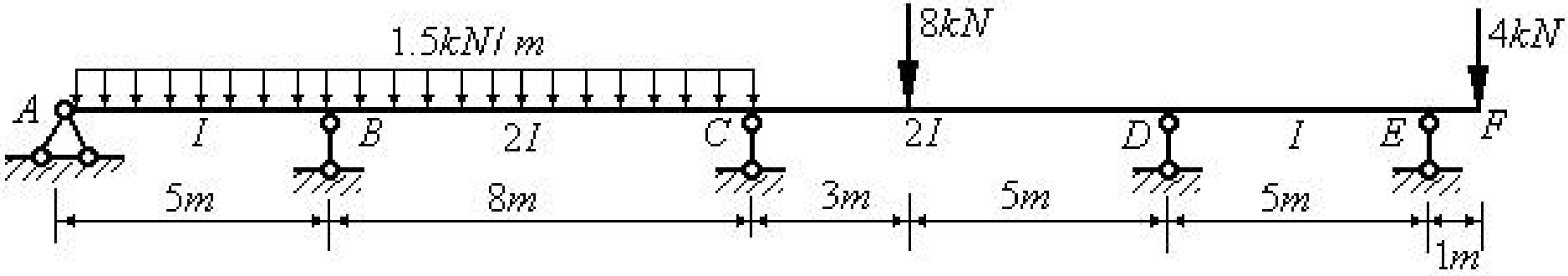


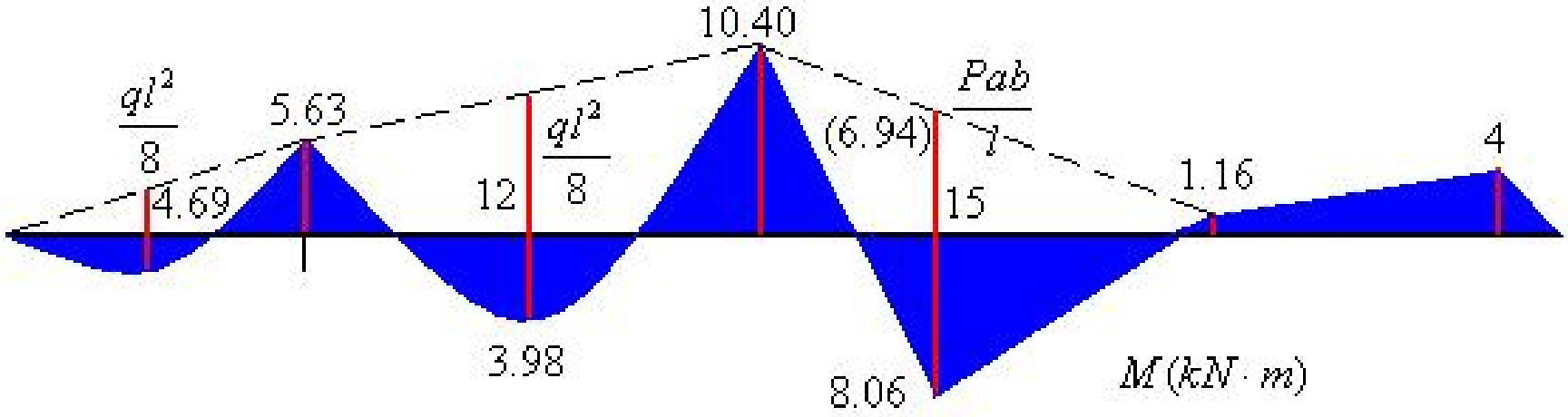
为了使计算时收敛较快，分配宜从不平衡力矩较大的节点开始，不相邻的节点可同时放松。



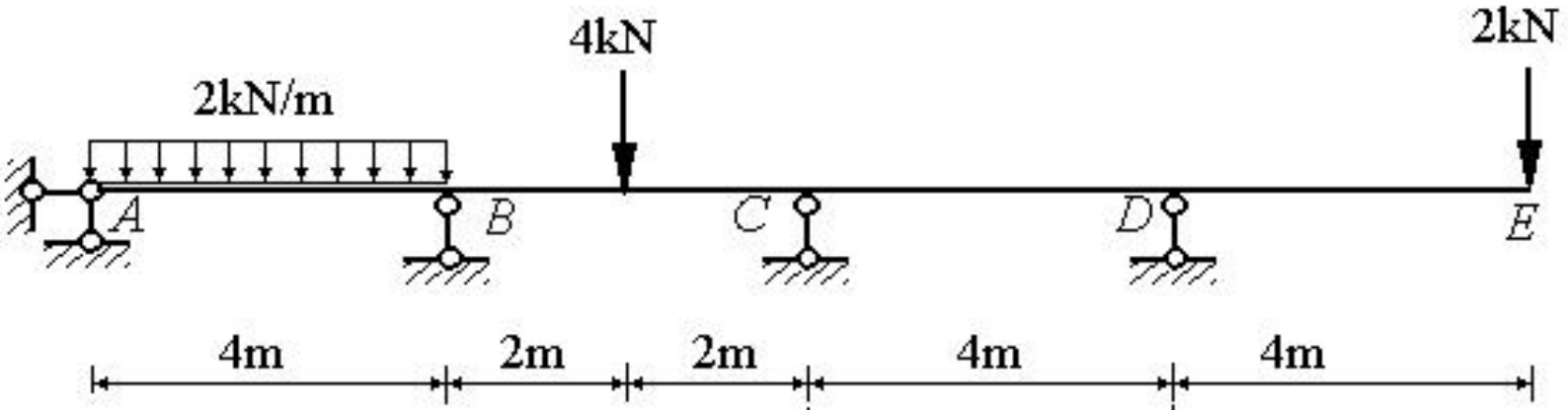
μ		0.375	0.625	0.5	0.5	0.625	0.375	
M^F	0	4.69	-8	8	-9.38	5.62	2	4
分配传递	0	1.24	2.07	1.03	-2.38	-4.76	-2.68	0
		0.68	1.37	1.36	0.68			
		-0.25	-0.43	-0.21	-0.21	-0.43	-0.25	
		0.11	0.21	0.21	0.11			
		-0.04	-0.07	-0.03	-0.03	-0.07	-0.04	
		0.02	0.03	0.03	0.02			
		-0.01	-0.01			-0.01	-0.01	
	0	5.63	-5.63	10.40	-10.40	1.16	-1.16	4

M图


1.5kN/m
8kN
4kN
A
I
B
2I
C
2I
D
I
E
F
5m
8m
3m
5m
5m
1m

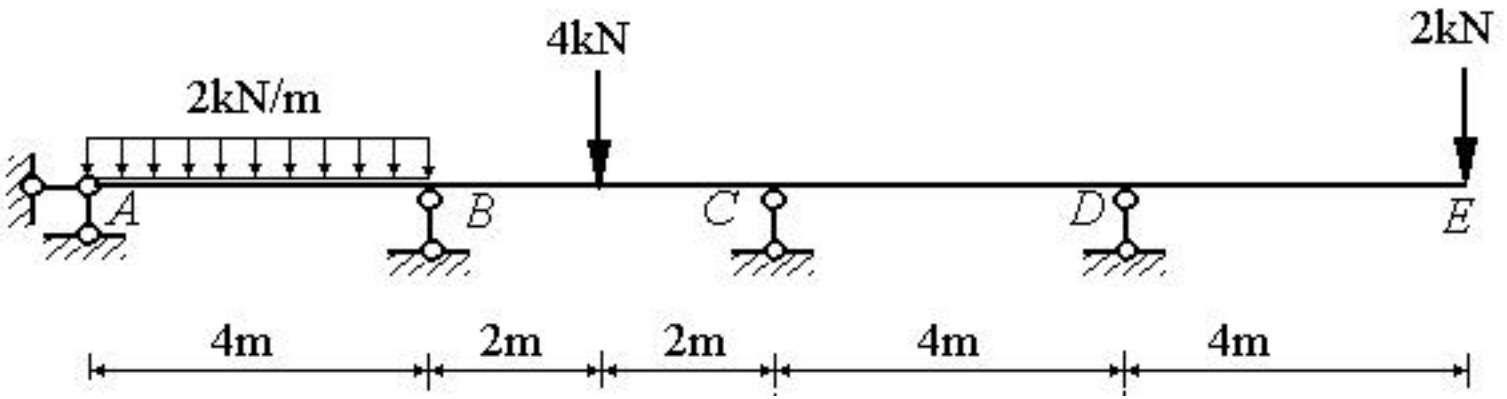

10.40
ql²/8
5.63
ql²/8
Pab/l
(6.94)
4
1.16
4.69
12
15
3.98
8.06
M(kN·m)

例3 作图示连续梁*M*图。*EI*=常数。



计算分配系数

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结点B：

$$\mu_{BA}=\frac{S_{AB}}{\sum_{B}S}=\frac{\frac{3EI}{4}}{\frac{3EI}{4}+\frac{4EI}{4}}=\frac{3}{7}$$

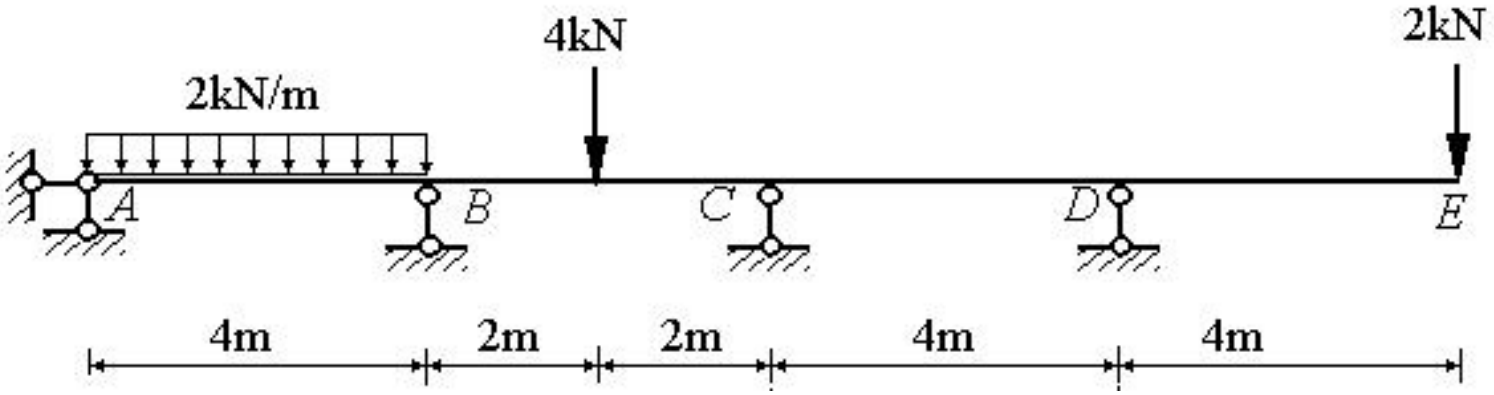
$$\mu_{BC}=\frac{S_{AB}}{\sum_{B}S}=\frac{\frac{4EI}{4}}{\frac{3EI}{4}+\frac{4EI}{4}}=\frac{4}{7}$$

结点C：

$$\mu_{CB}=\frac{4}{7}$$

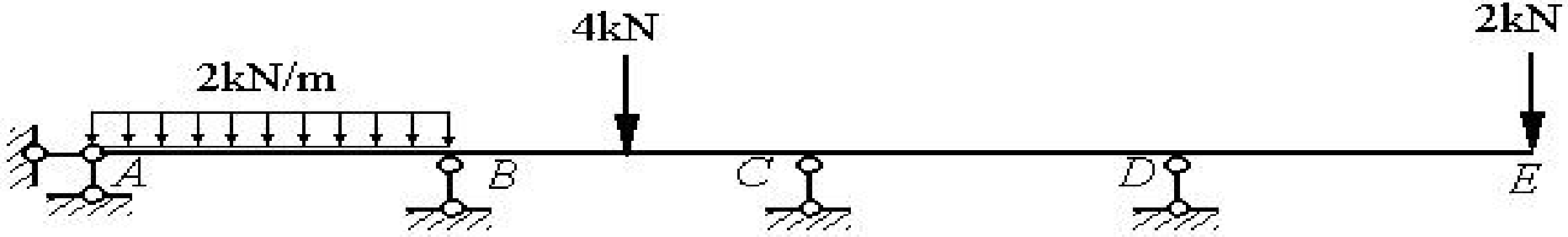
$$\mu_{CD}=\frac{3}{7}$$

计算固端弯矩：

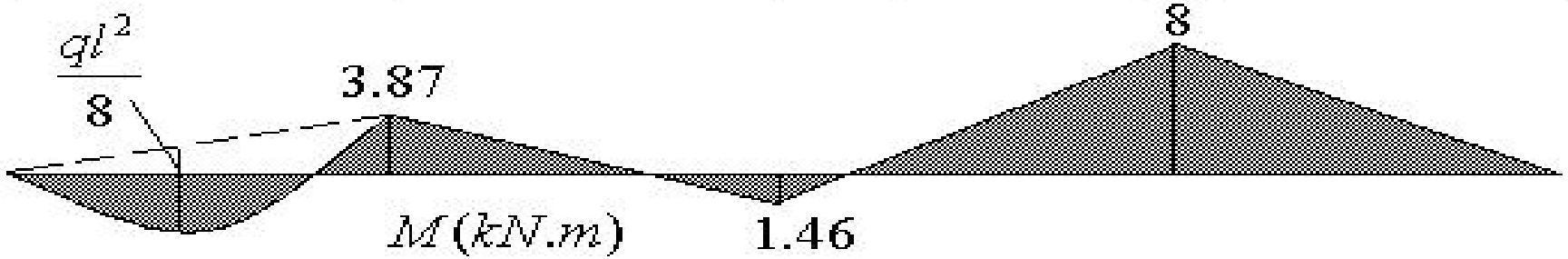


$$M^F_{BA} = \frac{ql^2}{8} = 4kN.m \qquad M^F_{CB} = \frac{Pl}{8} = 2kN.m = -M^F{}_{BC}$$

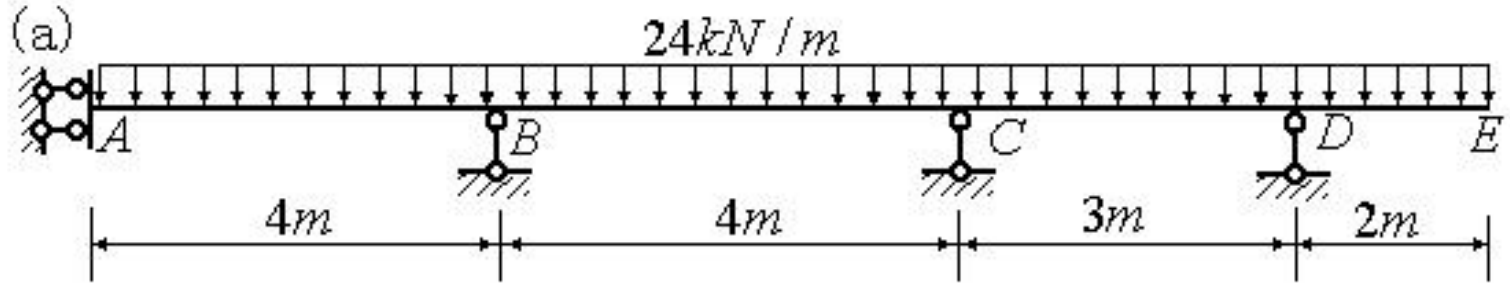
$$M^F_{CD} = 4kN.m \qquad M^F_{DC} = 8kN.m$$



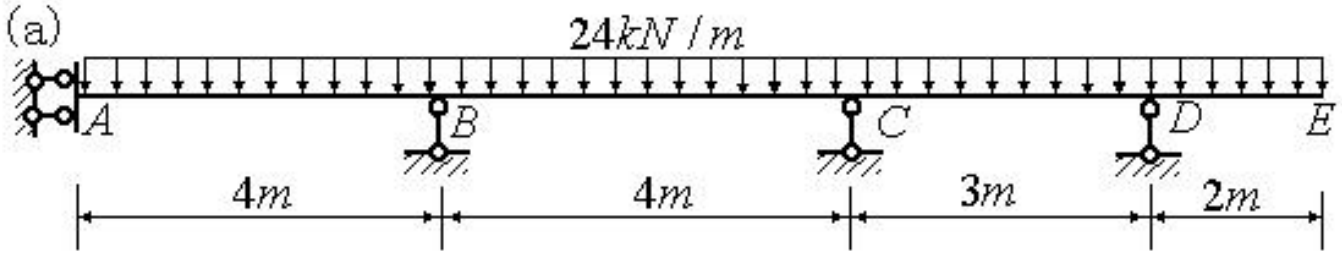
	4m	2m	2m	4m	4m
μ		3/7	4/7	4/7	3/7	
M^F	4	−2	2	4	8	8
		−1.71 ←	−3.43	−2.57 →	0	
	0 ←	−0.12	−0.17 →	−0.08		
		0.02 ←	0.05	0.03 →	0	
	0 ←	−0.01	−0.01			
Σ	3.87	−3.87	−1.46	1.46	8	8



例4 用力矩分配法计算连续梁，作图并求结点B的转角。EI=常数 （循环两次）.



计算分配系数



结点**B**：

$$u_{BA}=\frac{S_{BA}}{\sum_{B}S}=\frac{\frac{EI}{4}}{\frac{EI}{4}+\frac{4EI}{4}}=\frac{4EI}{20EI}=0.2$$

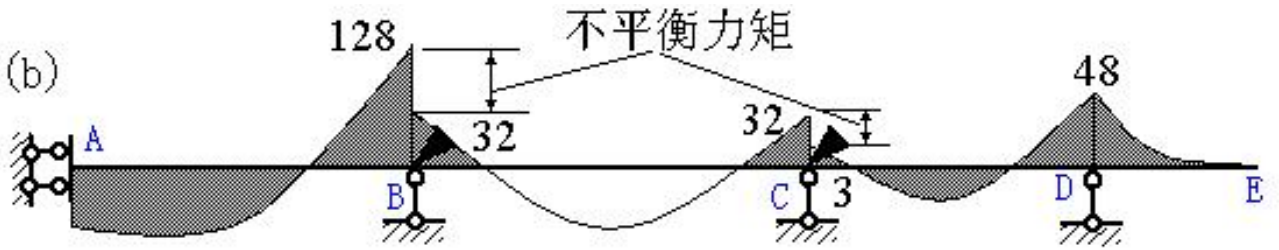
$$u_{BC}=\frac{S_{BC}}{\sum_{B}S}=\frac{\frac{4EI}{4}}{\frac{EI}{4}+\frac{4EI}{4}}=0.8$$

结点**C**：

$$u_{CB}=\frac{S_{CB}}{\sum_{C}S}=\frac{\frac{4EI}{4}}{\frac{4EI}{4}+\frac{3EI}{3}}=\frac{EI}{2EI}=0.5$$

$$u_{CD}=\frac{S_{CD}}{\sum_{C}S}=\frac{\frac{3EI}{3}}{\frac{4EI}{4}+\frac{3EI}{3}}=\frac{EI}{2EI}=0.5$$

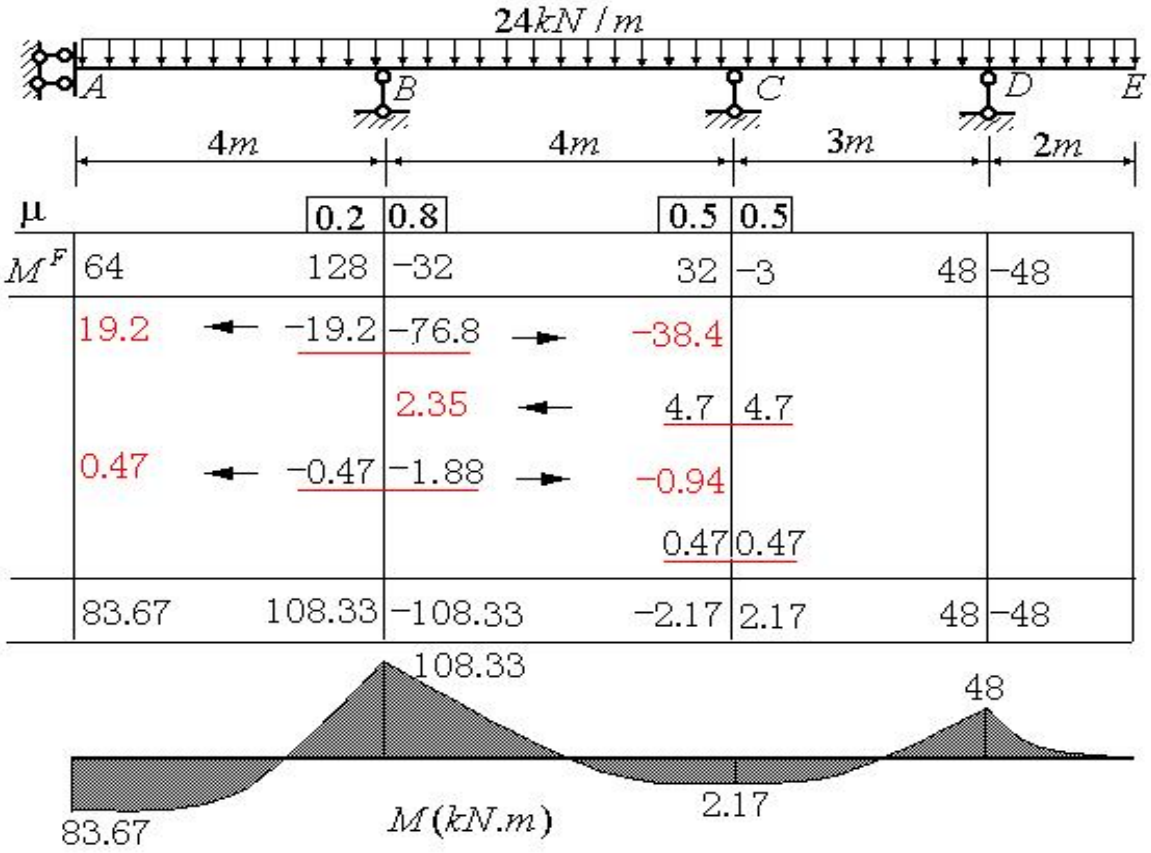
计算固端弯矩：



$$M_{AB}^{F}=\frac{ql^2}{6}=\frac{24\times4\times4}{6}=64kN.m \qquad M_{BA}^{F}=\frac{ql^2}{3}=\frac{24\times4\times4}{3}=128kN.m$$

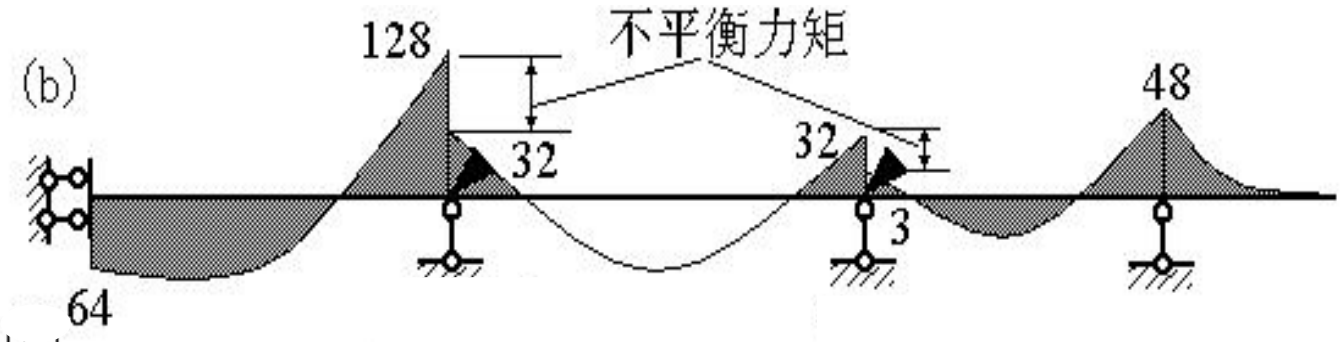
$$M_{BC}^{F}=-\frac{ql^2}{12}=-\frac{24\times4\times4}{12}=-32kN.m=-M_{CB}^{F}$$

$$M_{CD}^{F}=-\frac{ql^2}{8}+\frac{1}{2}\times48=-\frac{24\times3\times3}{8}+24=-27+24=-3kN.m$$

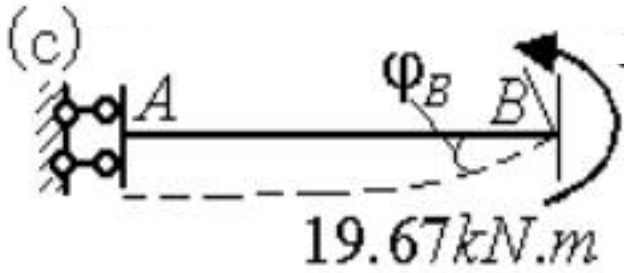

24kN / m
A
B
C
D
E
4m
4m
3m
2m
μ
0.2
0.8
0.5
0.5
M^F
64
128
−32
32
−3
48
−48
19.2
−19.2
−76.8
−38.4
2.35
4.7
4.7
0.47
−0.47
−1.88
−0.94
0.47
0.47
83.67
108.33
−108.33
−2.17
2.17
48
−48
108.33
48
83.67
2.17
M(kN.m)

结点*B*的转角

结点角位移是由不平衡弯矩所产生的，对于每一杆件的杆端转角来说，则是由分配弯矩引起的。

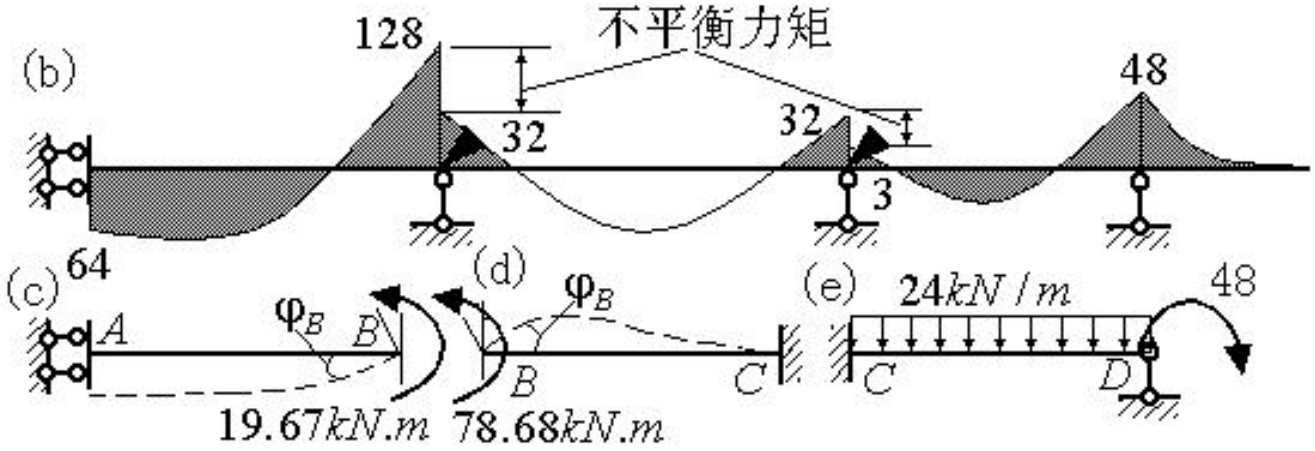


如由杆件AB求φ_B，由图c（此时视B端为固定，A端为滑动的等截面梁），则



$$\frac{EI}{4}\times\varphi_B=-(19.2+0.47)=-19.67kN.m$$

$$\varphi_B=-\frac{78.68kN.m^2}{EI}$$

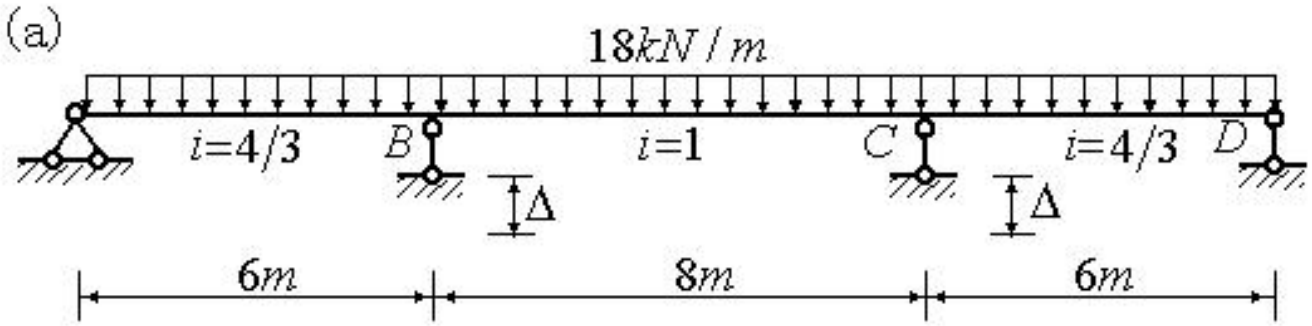


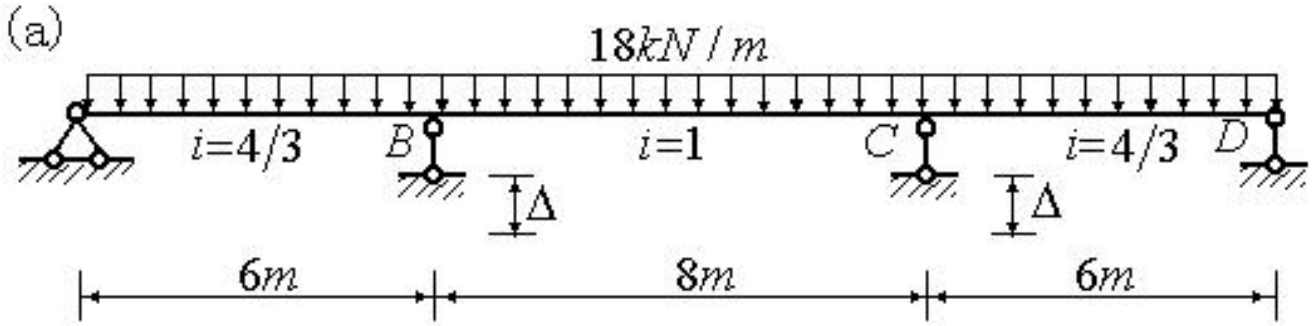
若由杆BC件求φ_B，由图d（视为两端固定梁），则

$$\frac{4EI}{4}\times\varphi_B=-(76.8+1.88)=-78.68kN.m$$

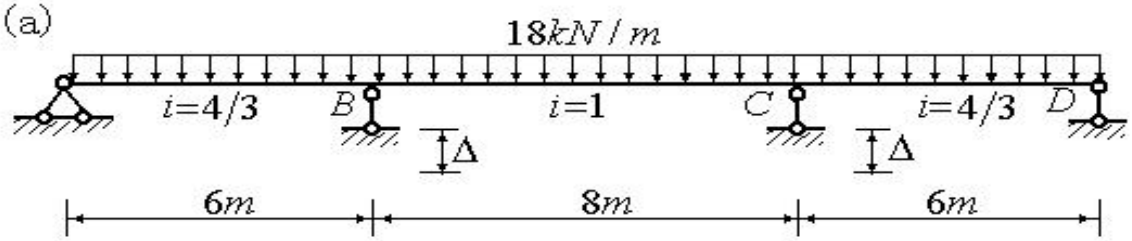
$$\varphi_B=-\frac{78.68kN.m^2}{EI}$$

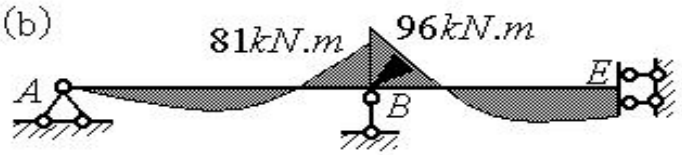
例5　图a所示等截面连续梁$EI=36000kN.m^2$，若欲使梁中最大正、负弯矩的绝对值相等，应将B、C两支座同时升降若干？



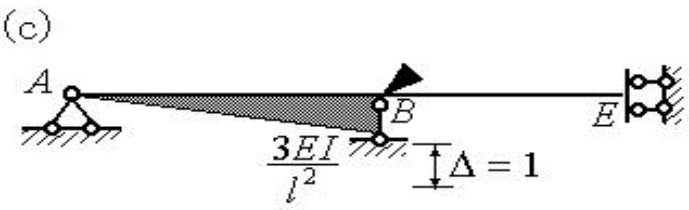


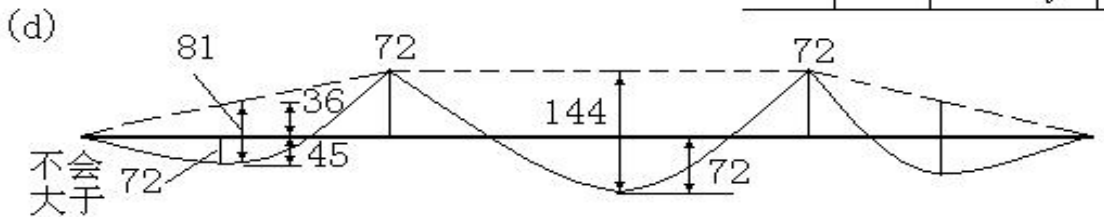
解：取半结构计算。应用叠加法，对荷载和支座沉降分别考虑（图b、c）。令 $\Delta=1$，两种影响按力矩分配法进行计算，其过程如分配表所示。





	μ		2/3	1/3
荷载弯矩	M^F		81	−96
	分配		10	5
	M_P		91	−91
$\Delta=1$ 时之弯矩	$\overline{M}^F_\Delta$		$-\frac{3EI}{l^2}$	
	分配		$\frac{2EI}{l^2}$	$\frac{EI}{l^2}$
	$\overline{M}_\Delta$		$-\frac{EI}{l^2}$	$\frac{EI}{l^2}$





若令中跨中点E处之弯矩与B、C处的相等，则有

$$M_E = M_B \quad M_E + M_B = 144 \quad M_E = \frac{144}{2} = 72$$

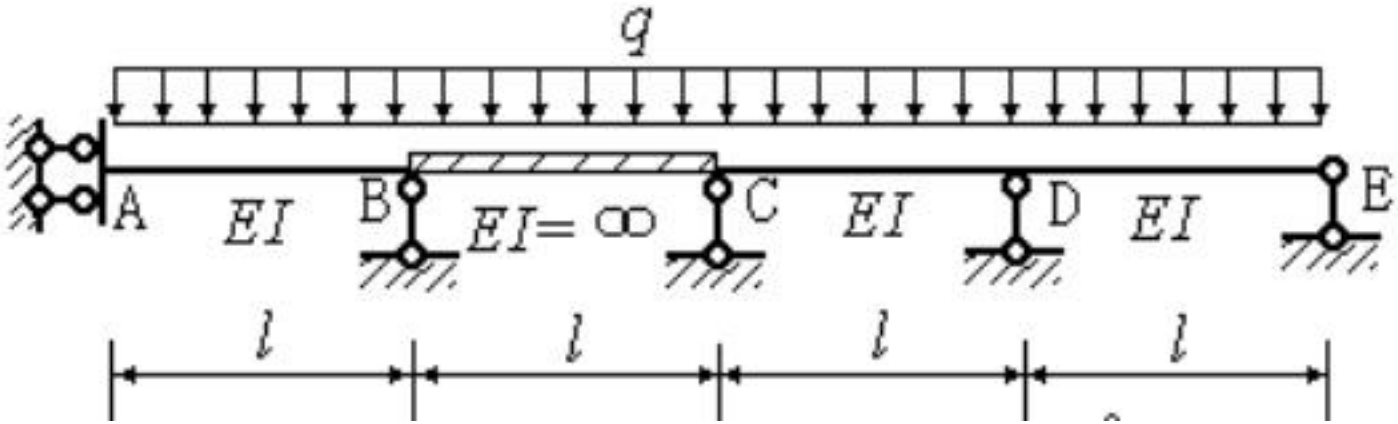
全梁最大正弯矩即在E处。

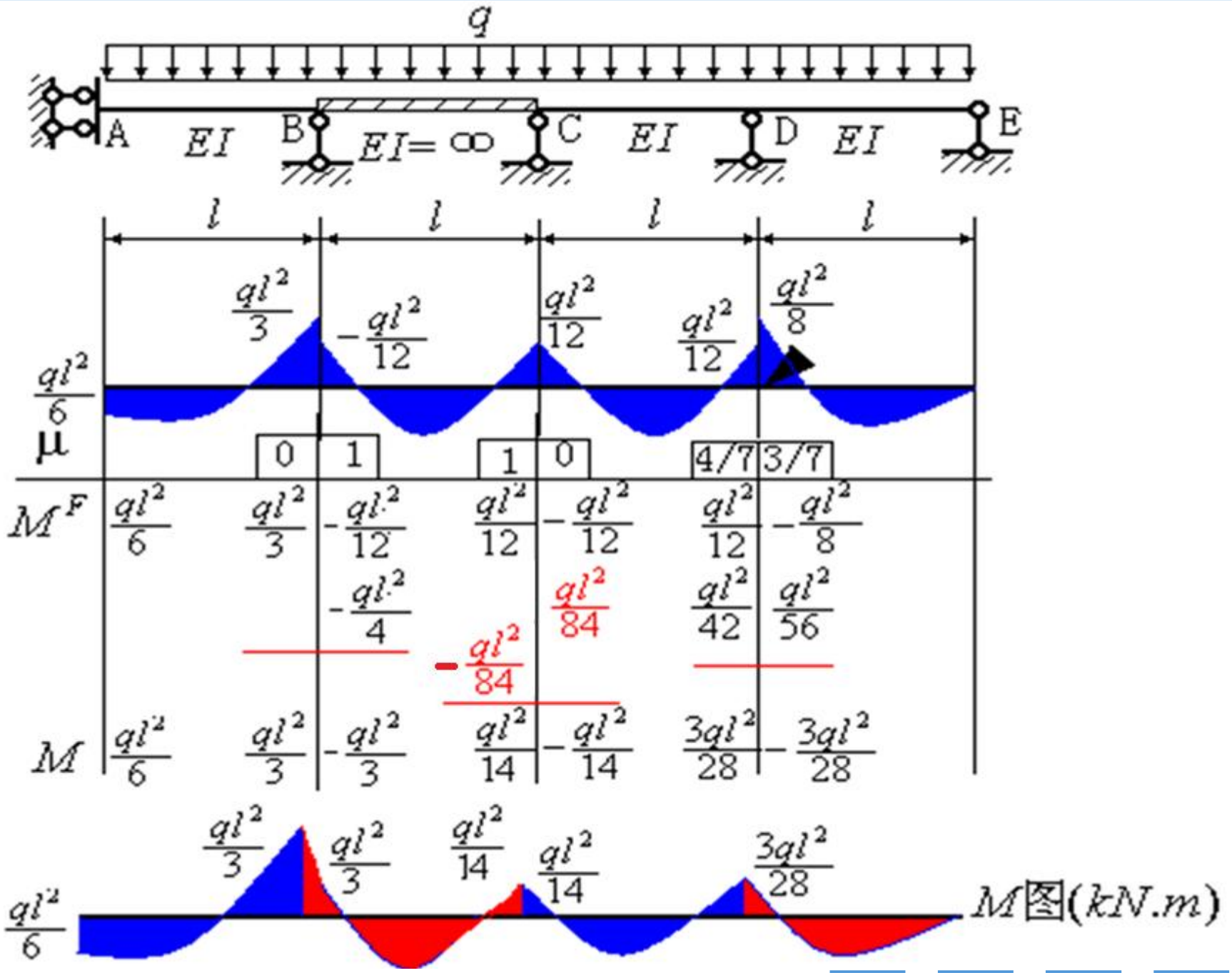
叠加：

$$M_{BA} = 91 - \frac{EI}{l^2} \times \Delta = 72kN.m$$

$$\Delta = \frac{l^2}{EI}(91 - 72) = \frac{36}{36000} \times 19 = 0.019m \downarrow$$

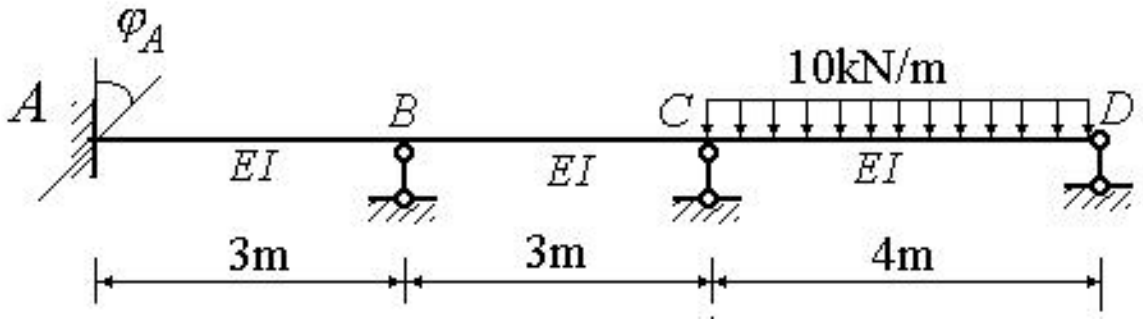
例6 用力矩分配法计算图示结构，并作 M 图。



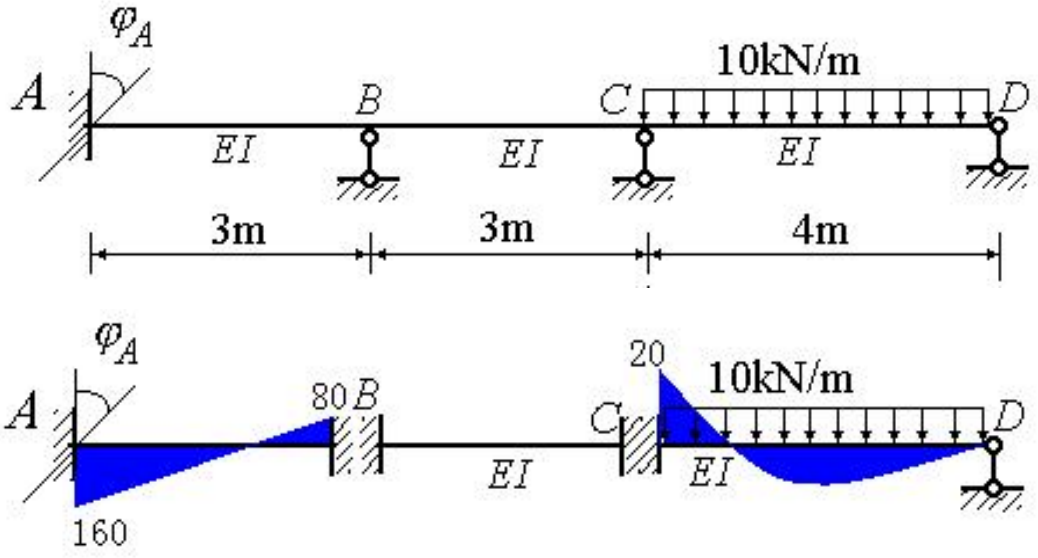

q
A
EI
B
EI= ∞
C
EI
D
EI
E
l
l
l
l
μ
0
1
1
0
4/7
3/7
M F
M
M图(kN.m)

例7 用力矩分配法作图示连续梁的图。(计算二轮)

已知 $\varphi_A = 0.06$， 各杆 $EI = 2.0 \times 10^3\ \mathrm{kN \cdot m^2}$

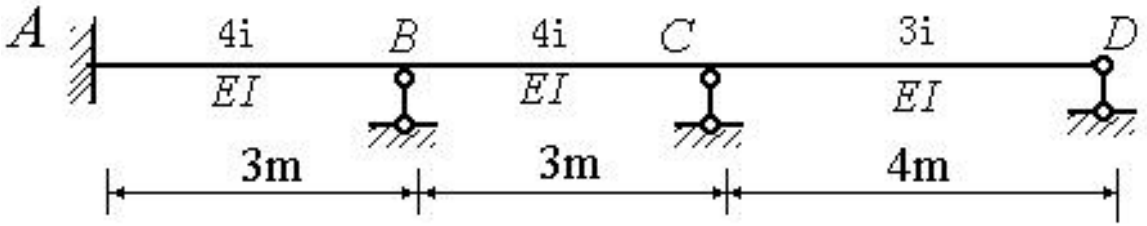


计算固端弯矩：



$$M_{AB}^{F}=4\times\frac{EI}{l}\times\varphi_A=\frac{4\times2.0\times10^{3}\,kN\cdot m^{2}\times0.06}{3}=160kN\cdot m$$

$$M_{BA}^{F}=80kN\cdot m,\qquad M_{CD}^{F}=-\frac{ql^{2}}{8}=-\frac{10kN/m\times(4m)^{2}}{8}=-20kN\cdot m$$



计算分配系数

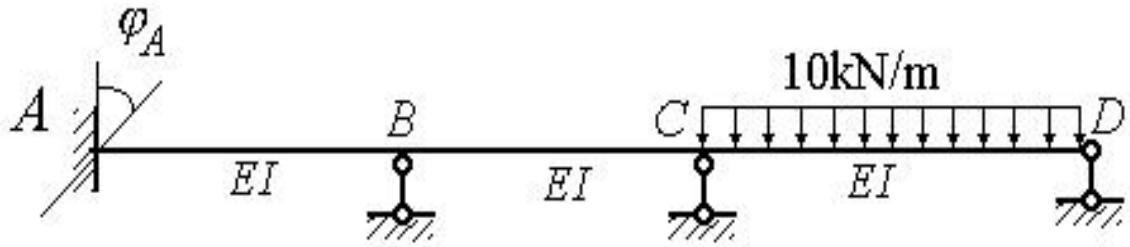
$$\frac{EI}{12}=i$$

$$\mu_{BA}=\frac{S_{BA}}{S_{BA}+S_{BC}}=\frac{4\times 4i}{4\times 4i+4\times 4i}=0.5$$

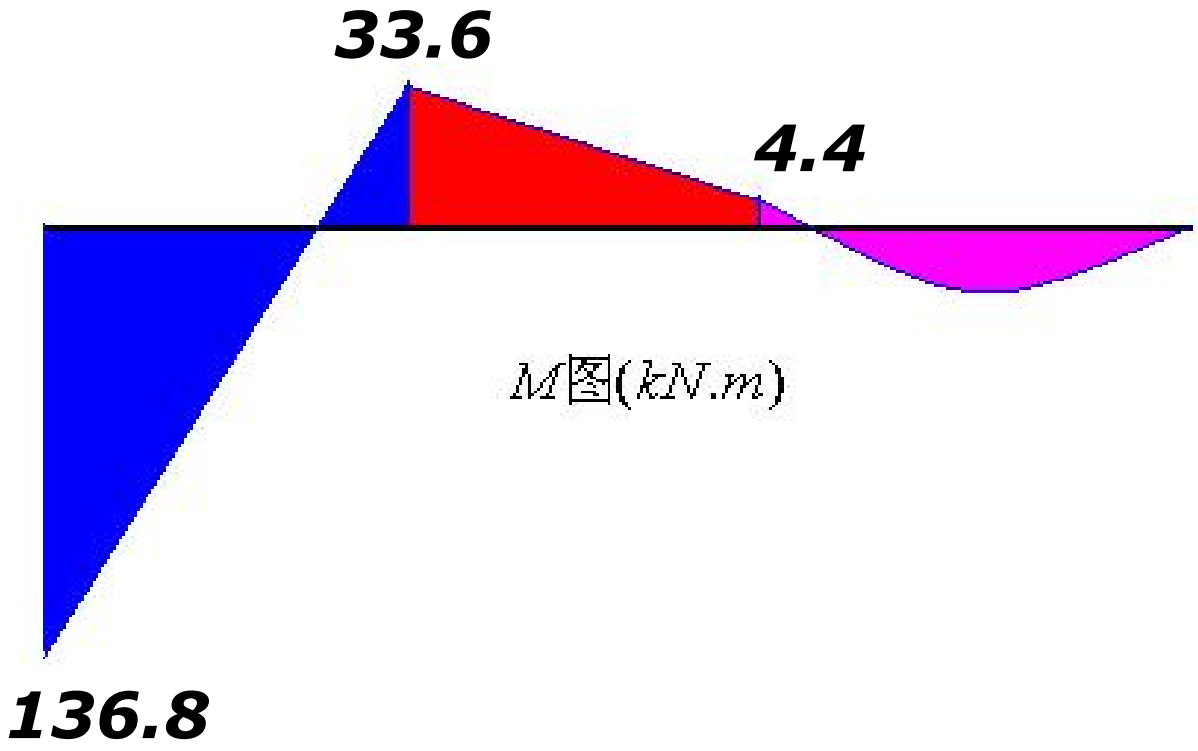
$$\mu_{BC}=\frac{S_{BC}}{S_{BA}+S_{BC}}=\frac{4\times 4i}{4\times 4i+4\times 4i}=0.5$$

$$\mu_{CB}=\frac{S_{CB}}{S_{CB}+S_{CD}}=\frac{4\times 4i}{4\times 4i+3\times 3i}=0.64$$

$$\mu_{CD}=\frac{S_{CD}}{S_{CB}+S_{CD}}=\frac{3\times 3i}{4\times 4i+3\times 3i}=0.36$$



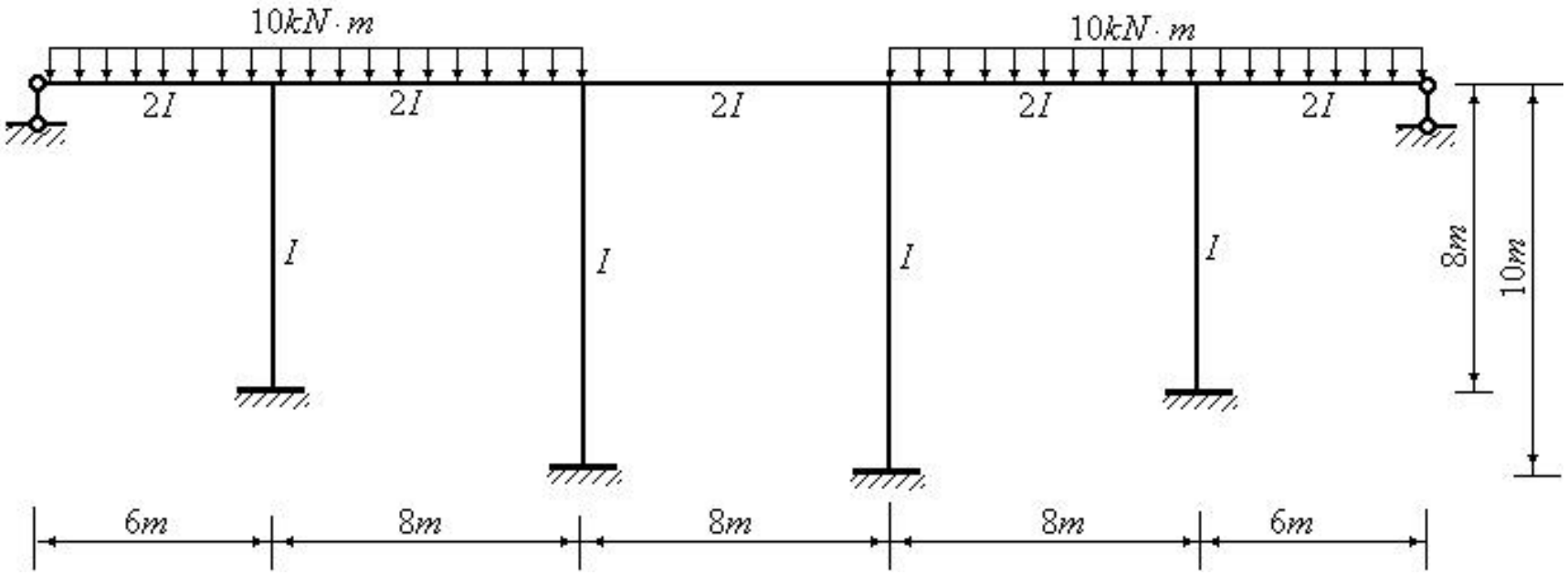
μ		0.5	0.5	0.64	0.36
M^F	160	80			-20
	-20	-40	-40	-20	
			12.8	25.6	14.4
	-3.2	-6.4	-6.4	-3.2	
				2	1.2
M	136.8	33.6	-33.6	4.4	-4.4

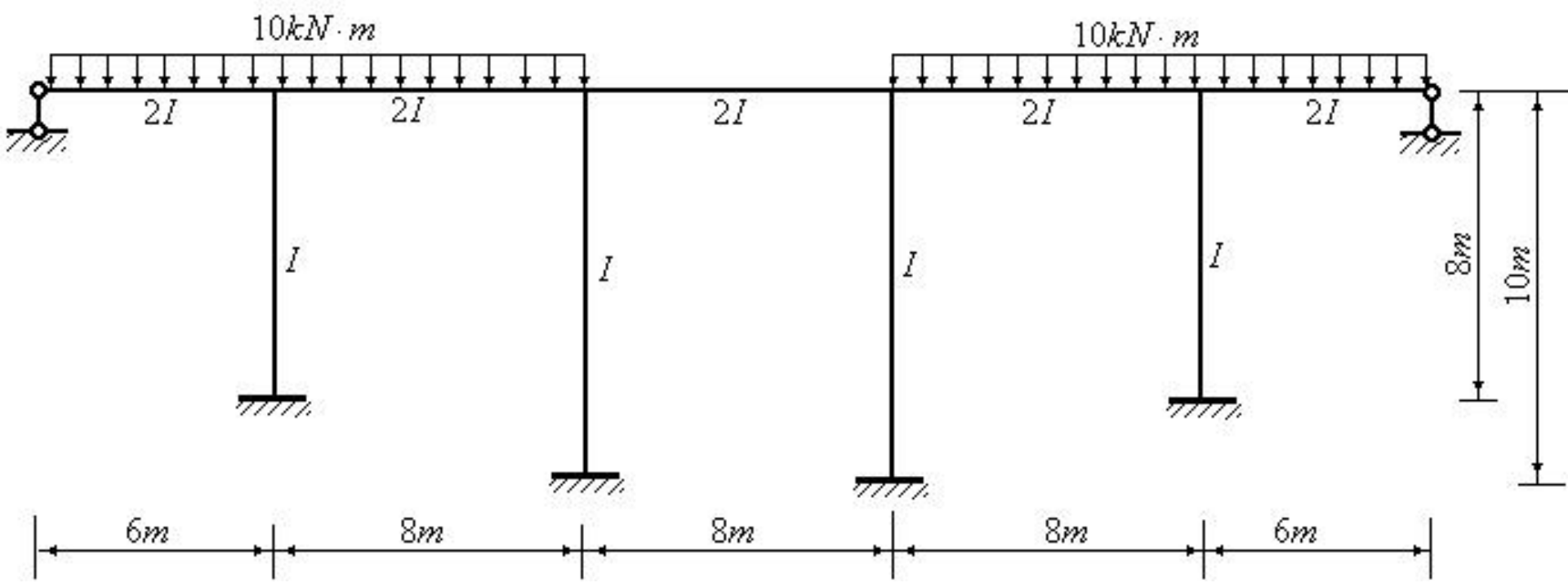

33.6
4.4
M图(kN.m)
136.8

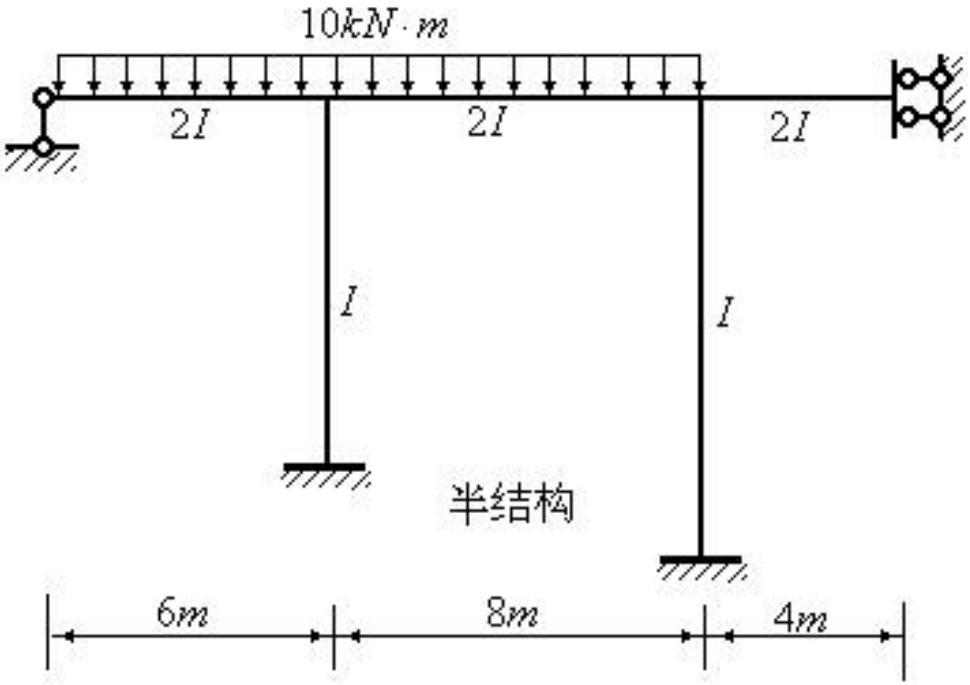
用力矩分配法计算无侧移刚架

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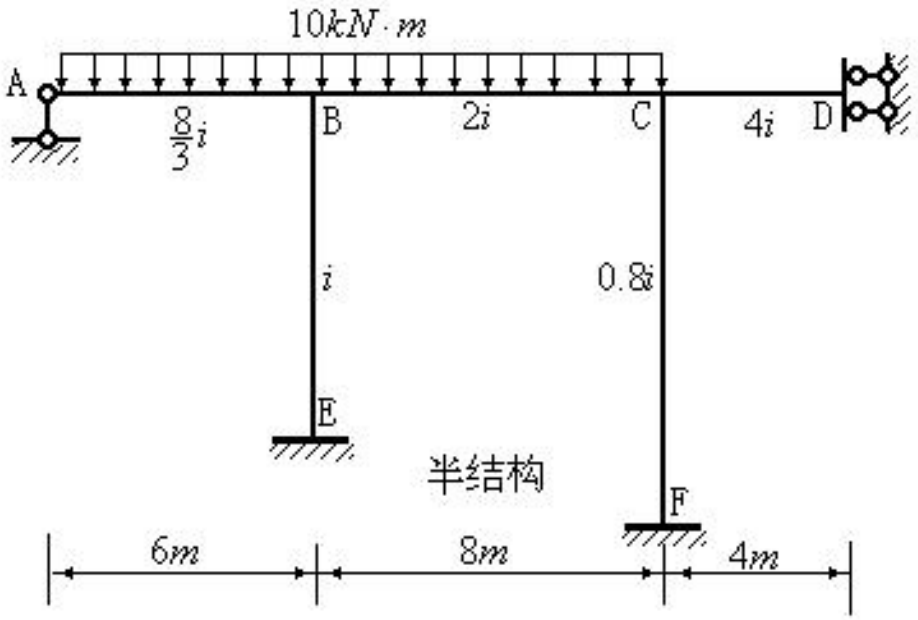
例8 用力矩分配法计算图示对称刚架

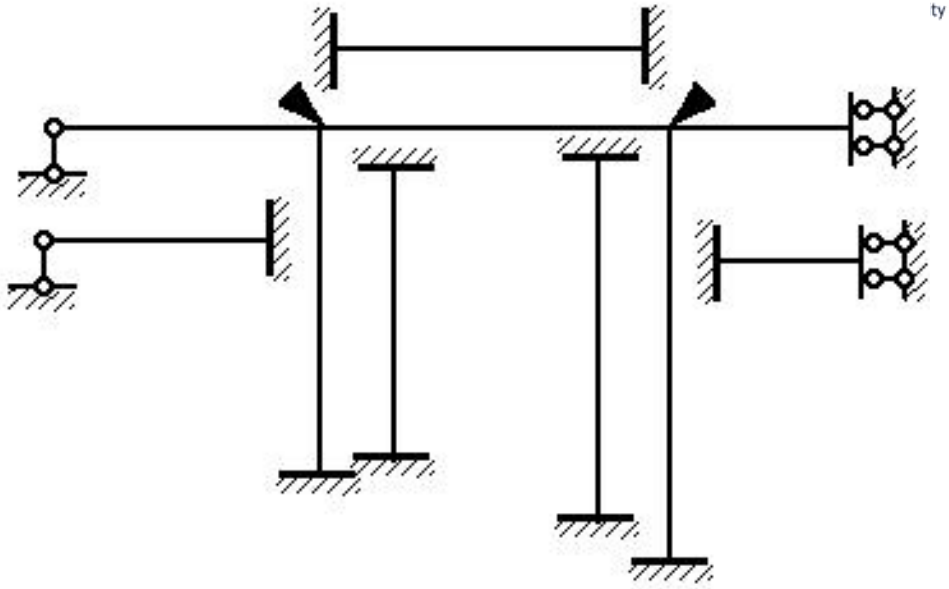



10kN·m
10kN·m
2I
2I
2I
2I
2I
I
I
I
I
8m
10m
6m
8m
8m
8m
6m


10kN·m
2I
2I
2I
I
I
半结构
6m
8m
4m

计算固端弯矩



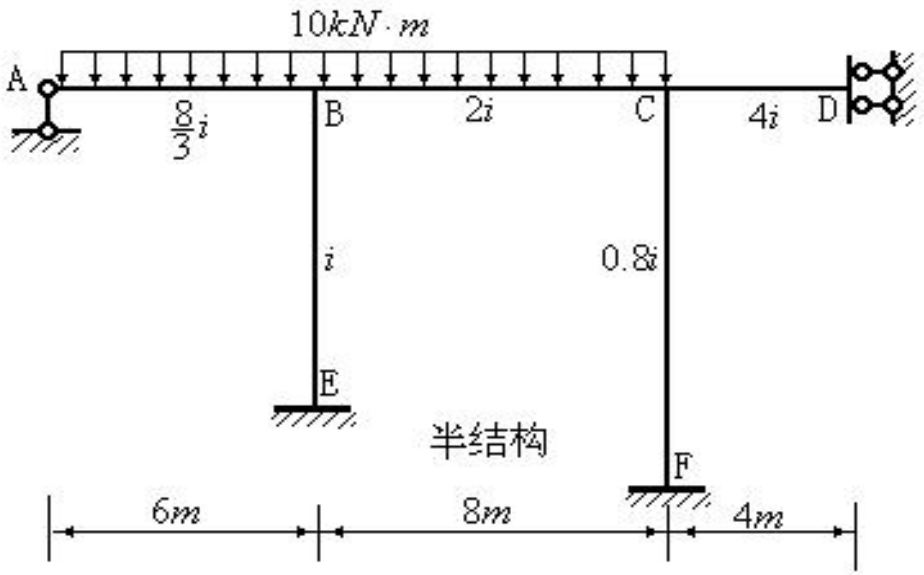


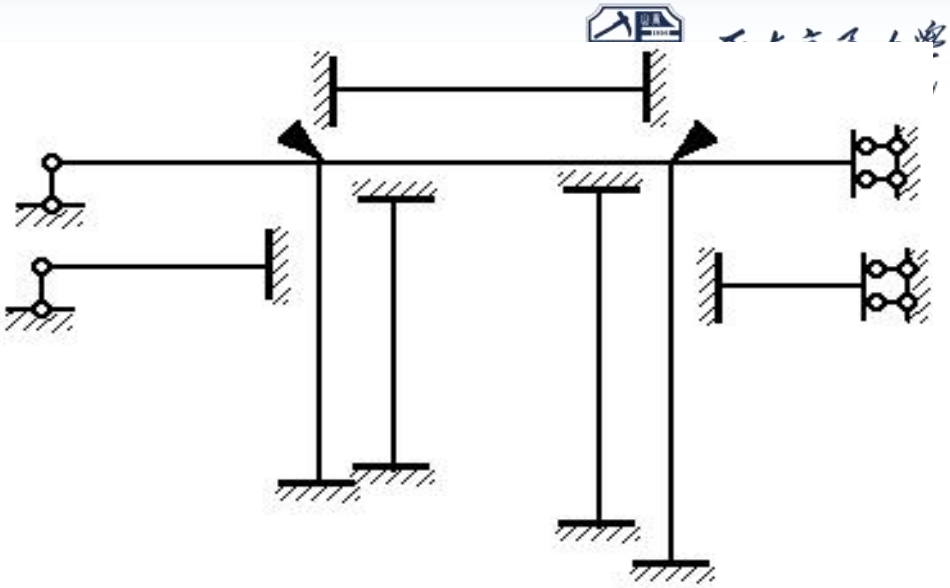
$$M_{BA}^{F}=\frac{ql^2}{8}=\frac{10kN/m\times(6m)^2}{8}=45kN\cdot m$$

$$M_{BC}^{F}=-\frac{ql^2}{12}=-\frac{10kN/m\times(8m)^2}{12}=-53.3kN\cdot m$$

$$M_{CB}^{F}=\frac{ql^2}{12}=\frac{10kN/m\times(8m)^2}{12}=53.3kN\cdot m$$

分配系数计算



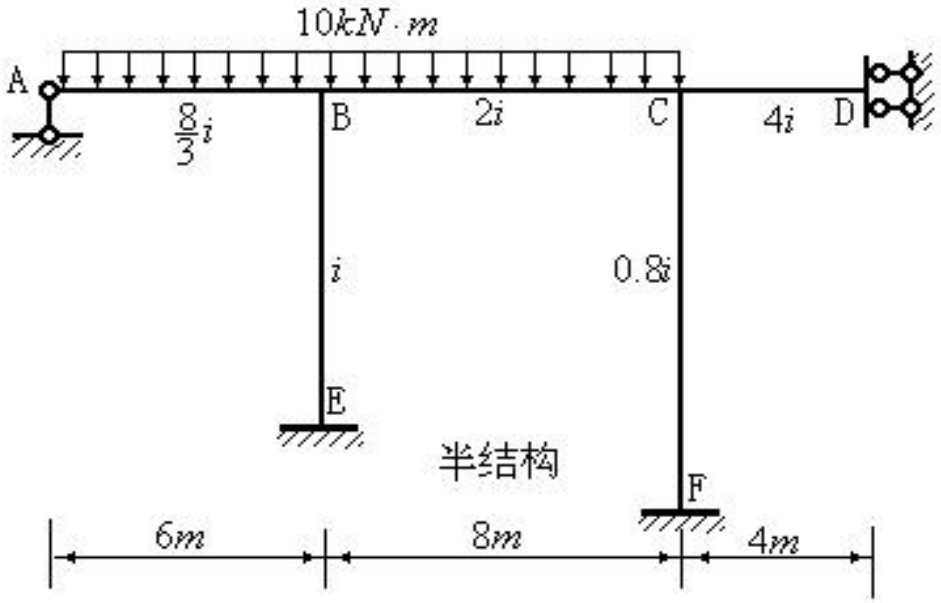


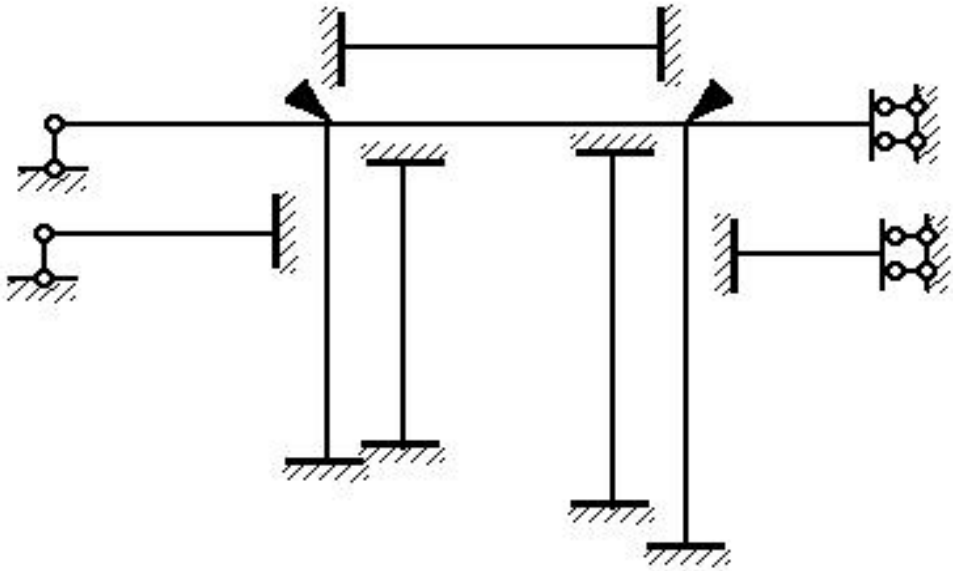
B结点

$$\mu_{BA}=\frac{S_{BA}}{S_{BA}+S_{DE}+S_{BC}}=\frac{3\times\frac{8}{3}}{3\times\frac{8}{3}+4i+4\times 2i}=0.4$$

$$\mu_{BE}=\frac{S_{BE}}{S_{BA}+S_{DE}+S_{BC}}=\frac{4i}{3\times\frac{8}{3}+4i+4\times 2i}=0.2$$

$$\mu_{BC}=\frac{S_{BC}}{S_{BA}+S_{DE}+S_{BC}}=\frac{4\times 2i}{3\times\frac{8}{3}+4i+4\times 2i}=0.4$$



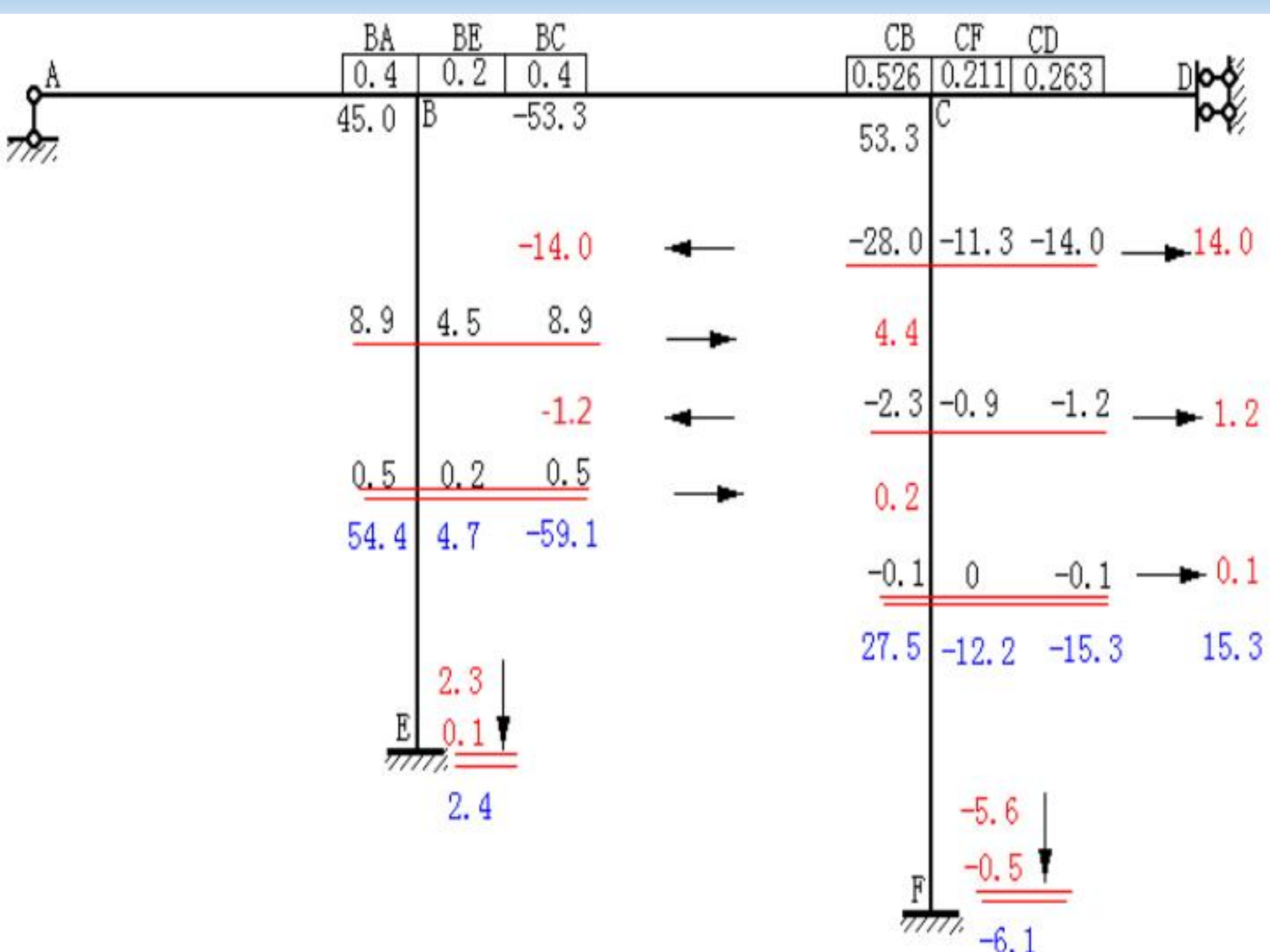


C结点

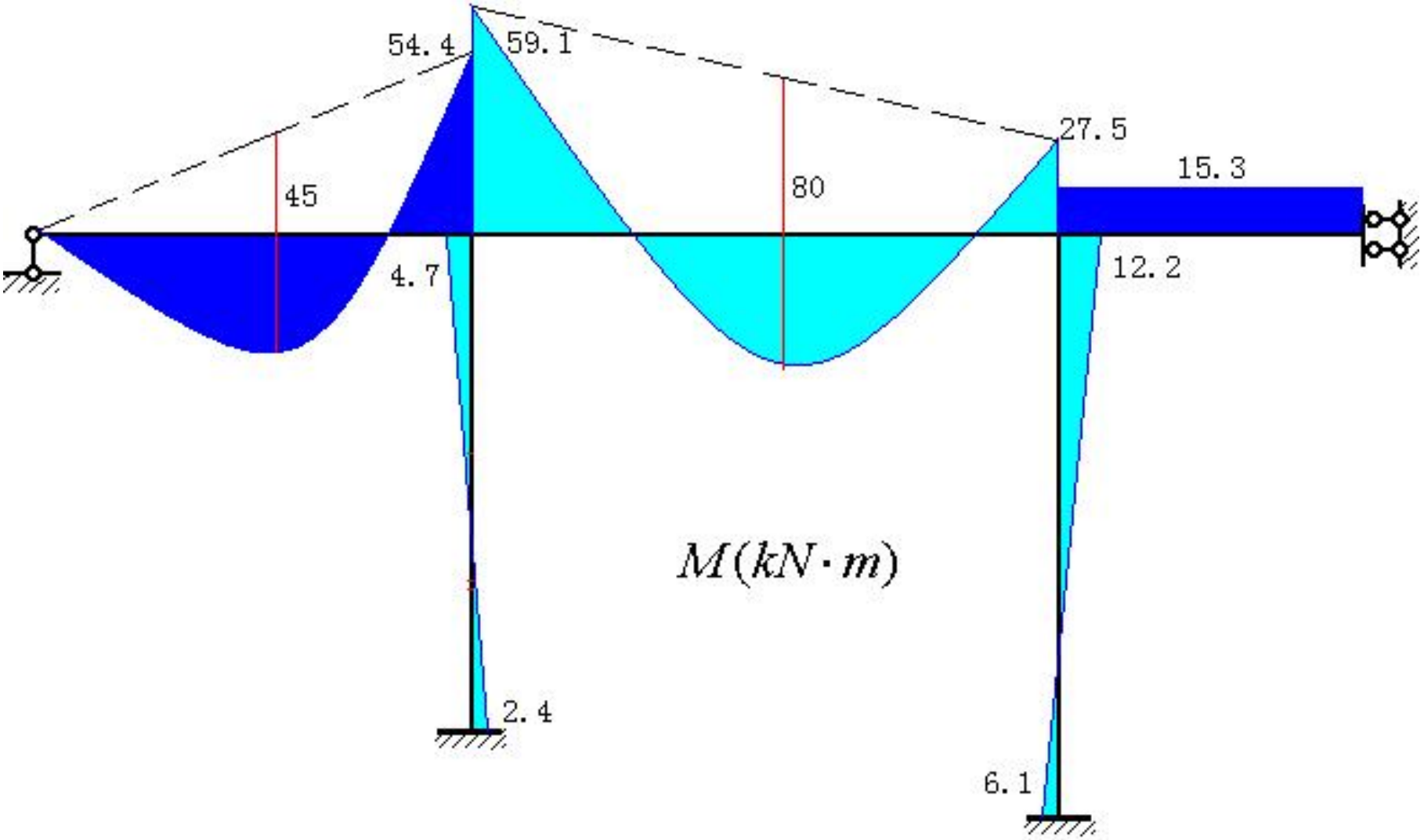
$$\mu_{CB}=\frac{S_{CB}}{S_{CB}+S_{CF}+S_{CD}}=\frac{4\times 2i}{4\times 2i+4\times 0.8i+1\times 4i}=0.526$$

$$\mu_{CF}=\frac{S_{CF}}{S_{CB}+S_{CF}+S_{CD}}=\frac{4\times 0.8i}{4\times 2i+4\times 0.8i+1\times 4i}=0.211$$

$$\mu_{CD}=\frac{S_{CD}}{S_{CB}+S_{CF}+S_{CD}}=\frac{1\times 4i}{4\times 2i+4\times 0.8i+1\times 4i}=0.263$$


BA
BE
BC
CB
CF
CD
A
B
C
D
E
F
0.4
0.2
0.4
0.526
0.211
0.263
45.0
-53.3
53.3
-14.0
-28.0
-11.3
-14.0
14.0
8.9
4.5
8.9
4.4
-1.2
-2.3
-0.9
-1.2
1.2
0.5
0.2
0.5
0.2
54.4
4.7
-59.1
-0.1
0
-0.1
0.1
27.5
-12.2
-15.3
15.3
2.3
0.1
2.4
-5.6
-0.5
-6.1

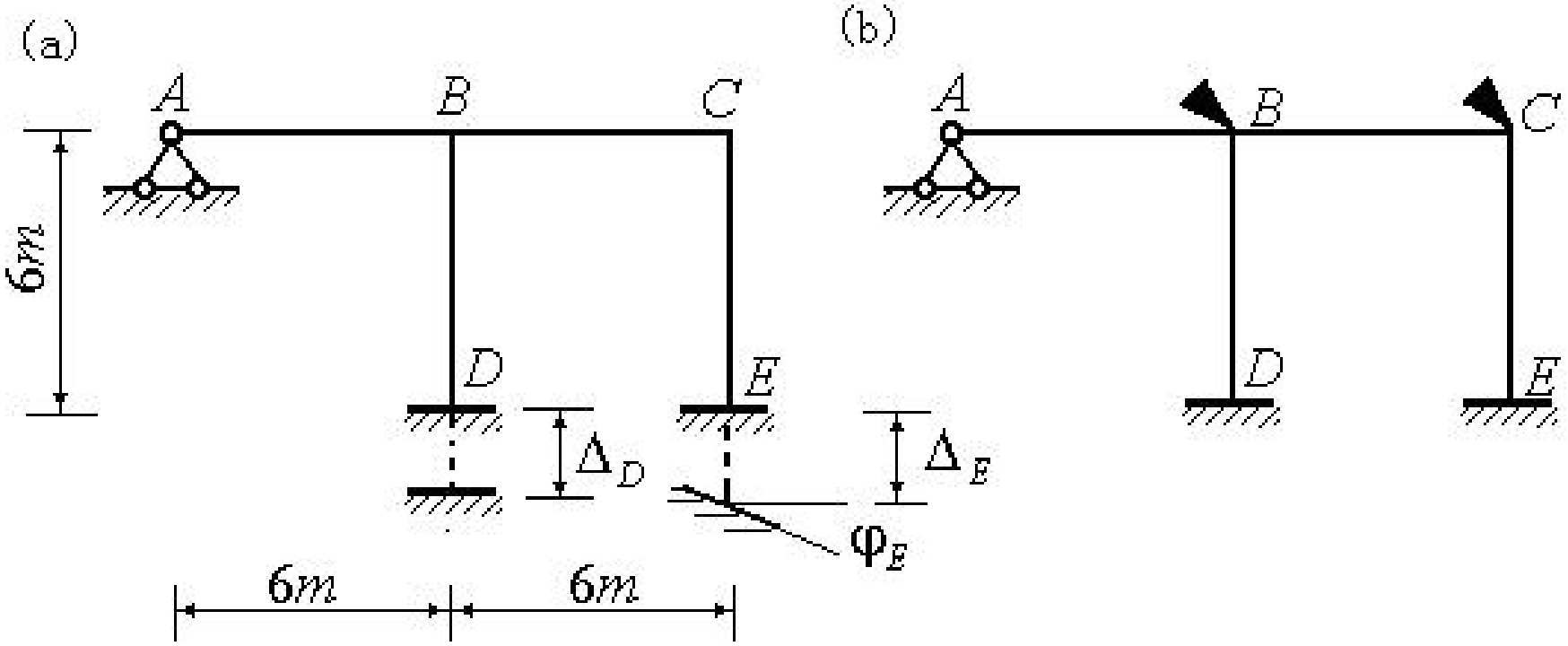
M图



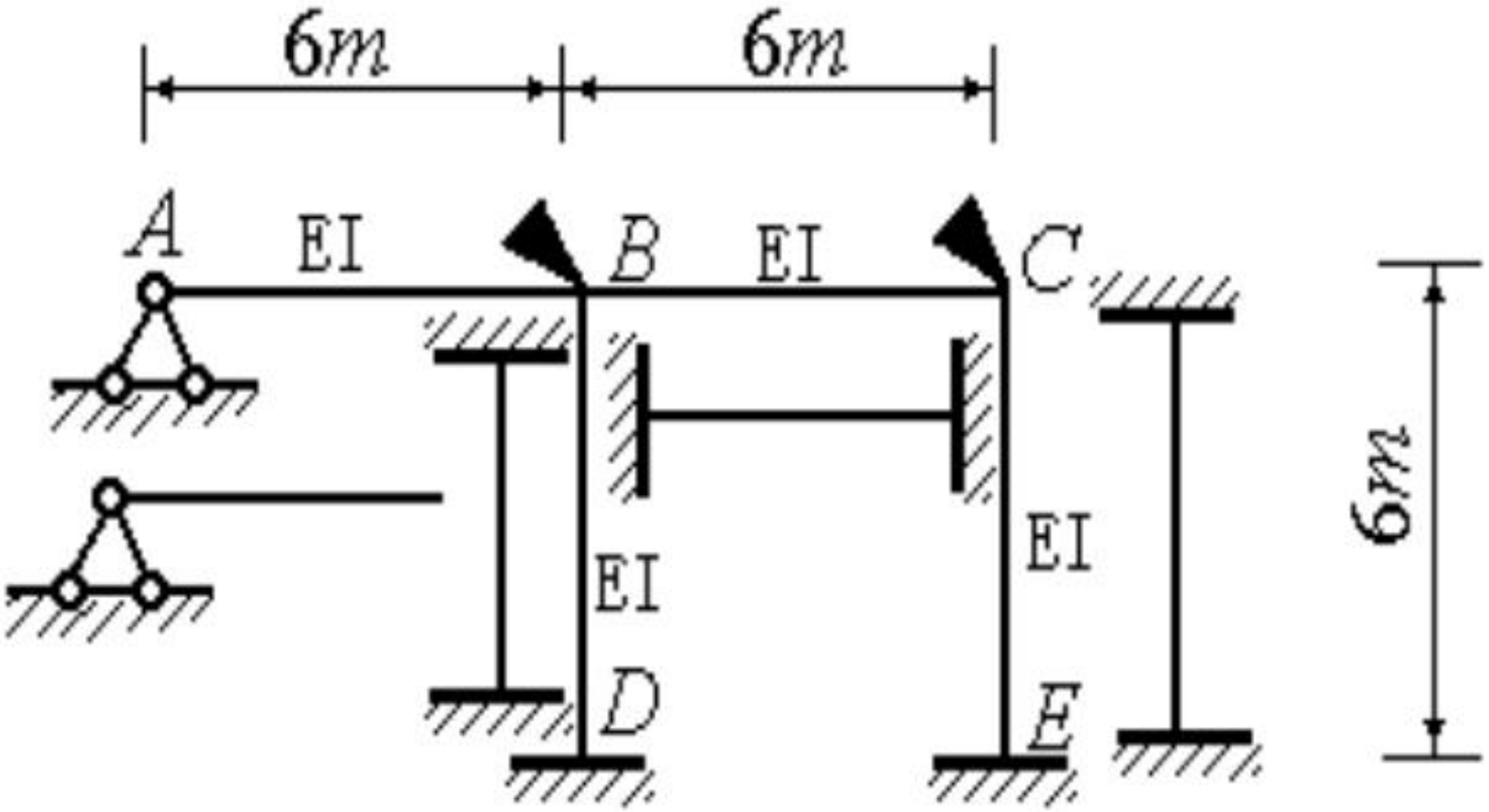
例9

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图*a*所示刚架支座*D*下沉了$\Delta_D = 0.08m$，支座*E*下沉了$\Delta_E = 0.05m$并发生了顺时针方向的转角$\varphi_E = 0.01rad$，试计算由此引起的各杆端弯矩。已知各杆$EI = 6\times10^4 kN.m^2$



计算分配系数



计算分配系数

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结点B：

$$\mu_{BA}=\frac{S_{BA}}{\sum_{B}S}=\frac{\frac{3EI}{6}}{\frac{3EI}{6}+\frac{4EI}{6}+\frac{4EI}{6}}=\frac{\frac{3EI}{6}}{\frac{11EI}{6}}=\frac{3}{11}$$

$$\mu_{BC}=\frac{S_{BC}}{\sum_{B}S}==\frac{\frac{4EI}{6}}{\frac{11EI}{6}}=\frac{4}{11}$$

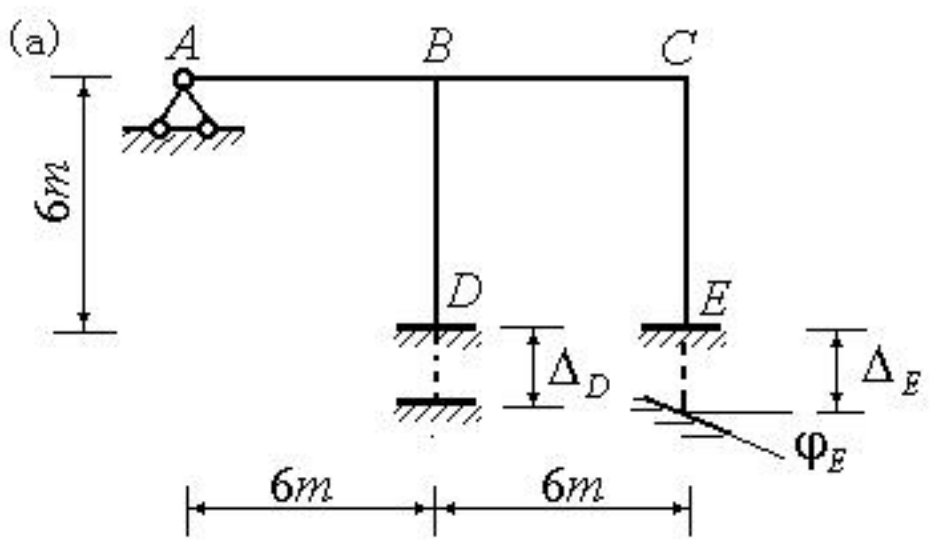
$$\mu_{BD}=\frac{S_{BD}}{\sum_{B}S}==\frac{\frac{4EI}{6}}{\frac{11EI}{6}}=\frac{4}{11}$$

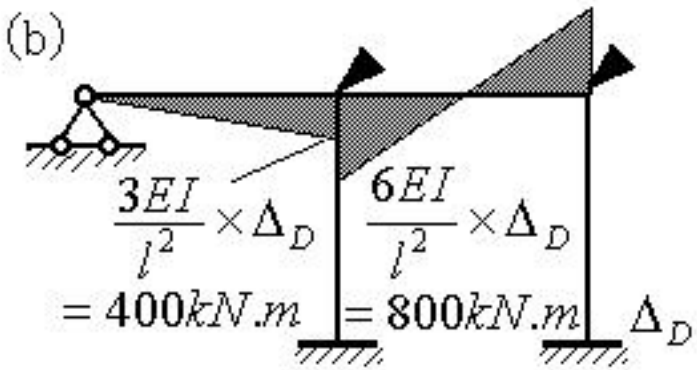
结点C：

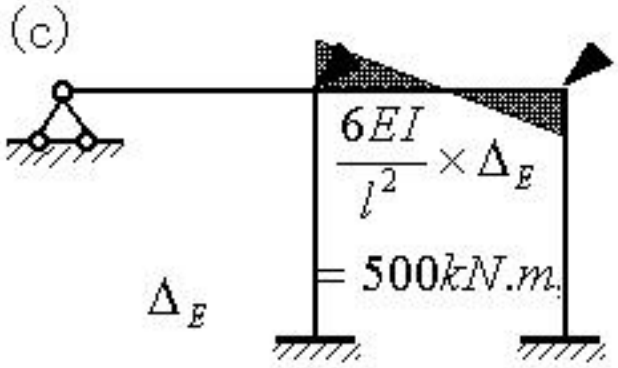
$$\mu_{CB}=\frac{S_{CB}}{\sum_{C}S}=\frac{\frac{4EI}{6}}{\frac{4EI}{6}+\frac{4EI}{6}}=\frac{\frac{4EI}{6}}{\frac{8EI}{6}}=\frac{1}{2}$$

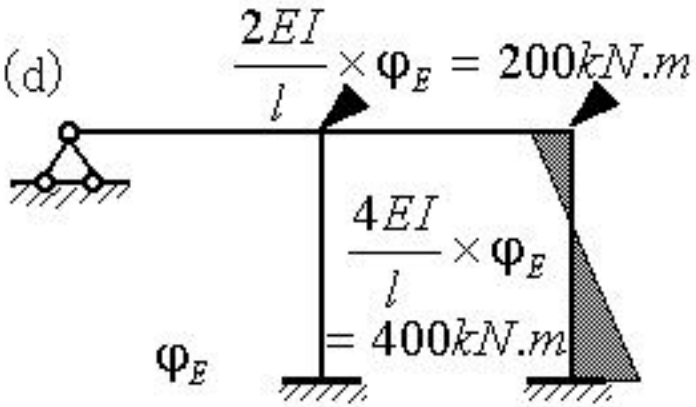
$$\mu_{CE}=\frac{S_{CE}}{\sum_{C}S}==\frac{\frac{4EI}{6}}{\frac{8EI}{6}}=\frac{1}{2}$$

计算固端弯矩

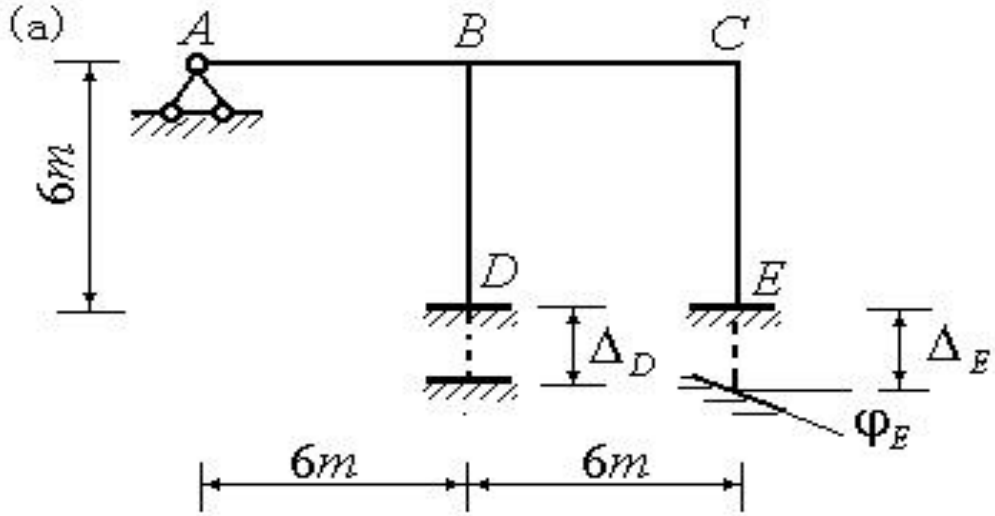


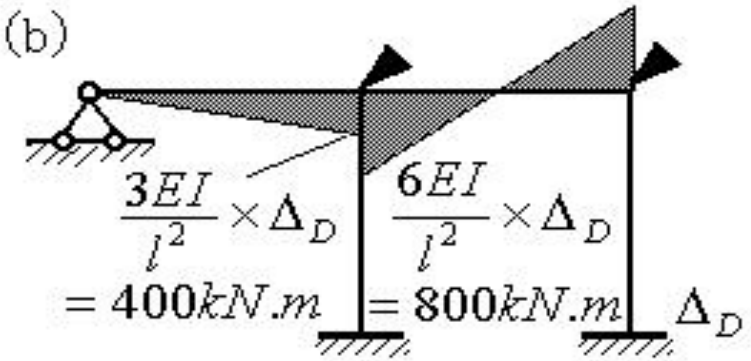


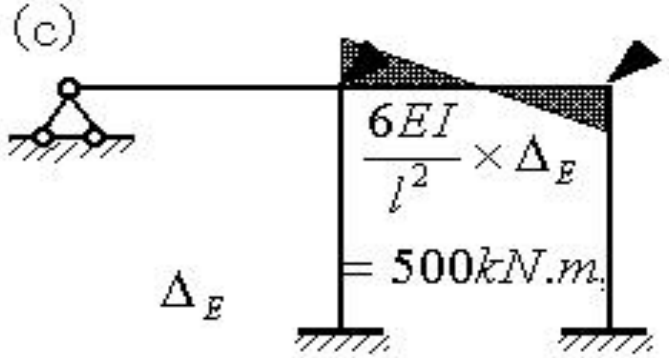


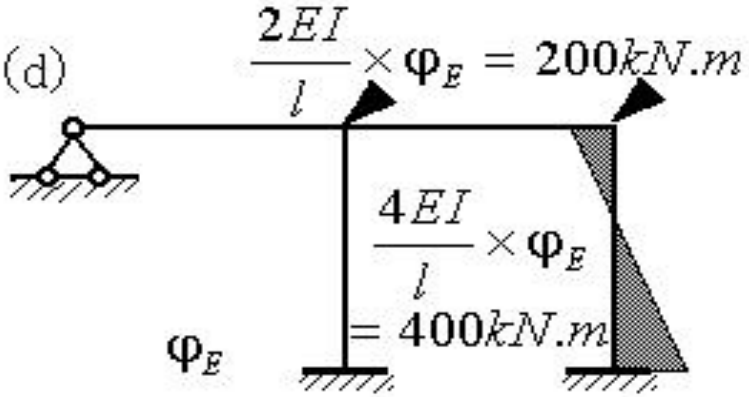


$$M_{BA}^{F}=-\frac{3EI}{l^2}\times\Delta_D=-\frac{3\times6\times10^4\,kN.m^2\times0.08m}{6\times6m^2}=-400kN.m$$



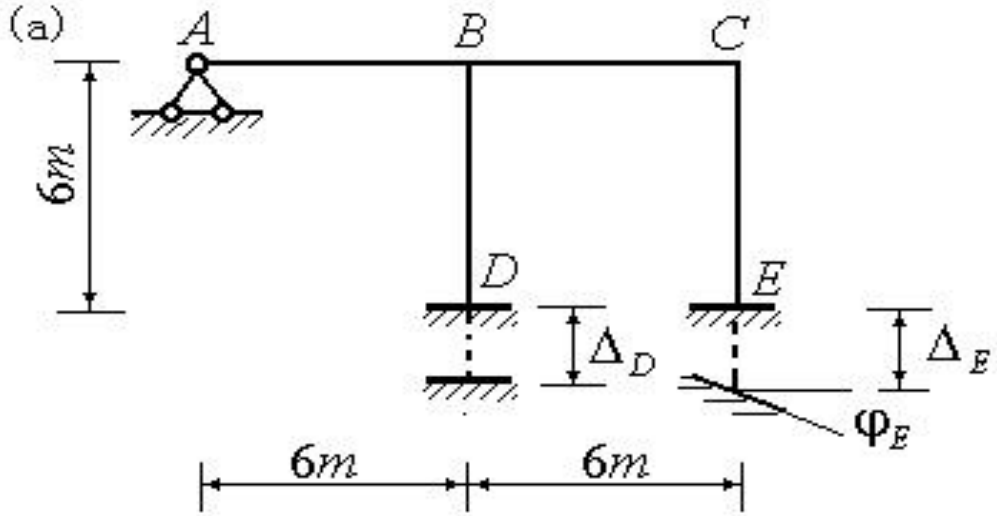


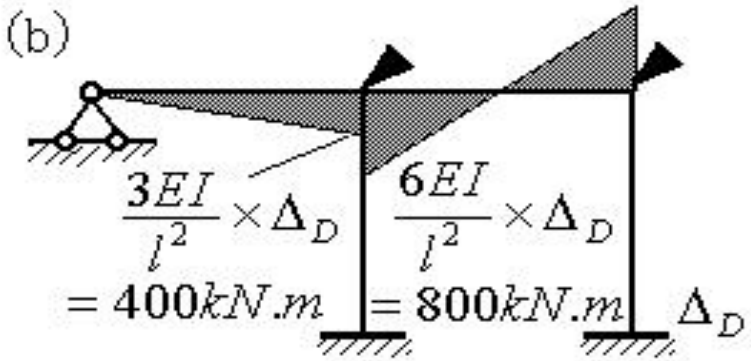


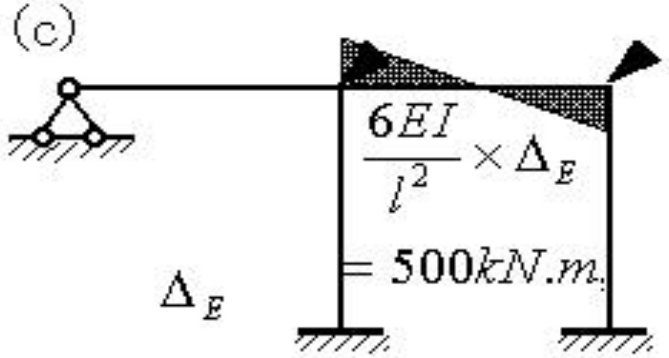


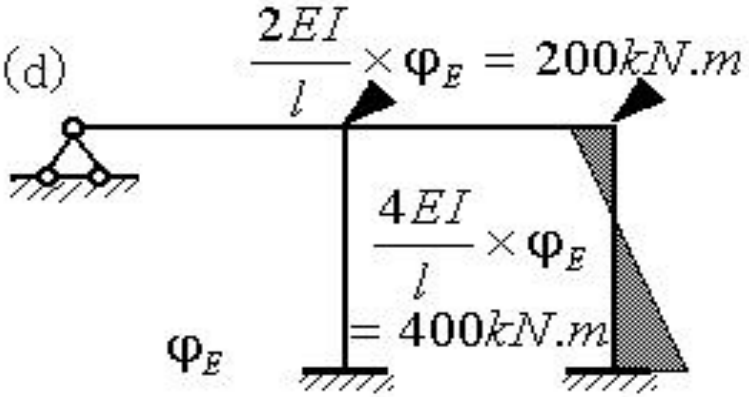
$$M_{BC}^F=\frac{6EI}{l^2}\times\Delta_D-\frac{6EI}{l^2}\times\Delta_E=\frac{6\times6\times10^4kN.m^2\times0.08m}{6\times6m^2}$$

$$-\frac{6\times6\times10^4kN.m^2\times0.05m}{6\times6m^2}=800kN.m-500kN.m=300kN.m=M_{CB}^F$$



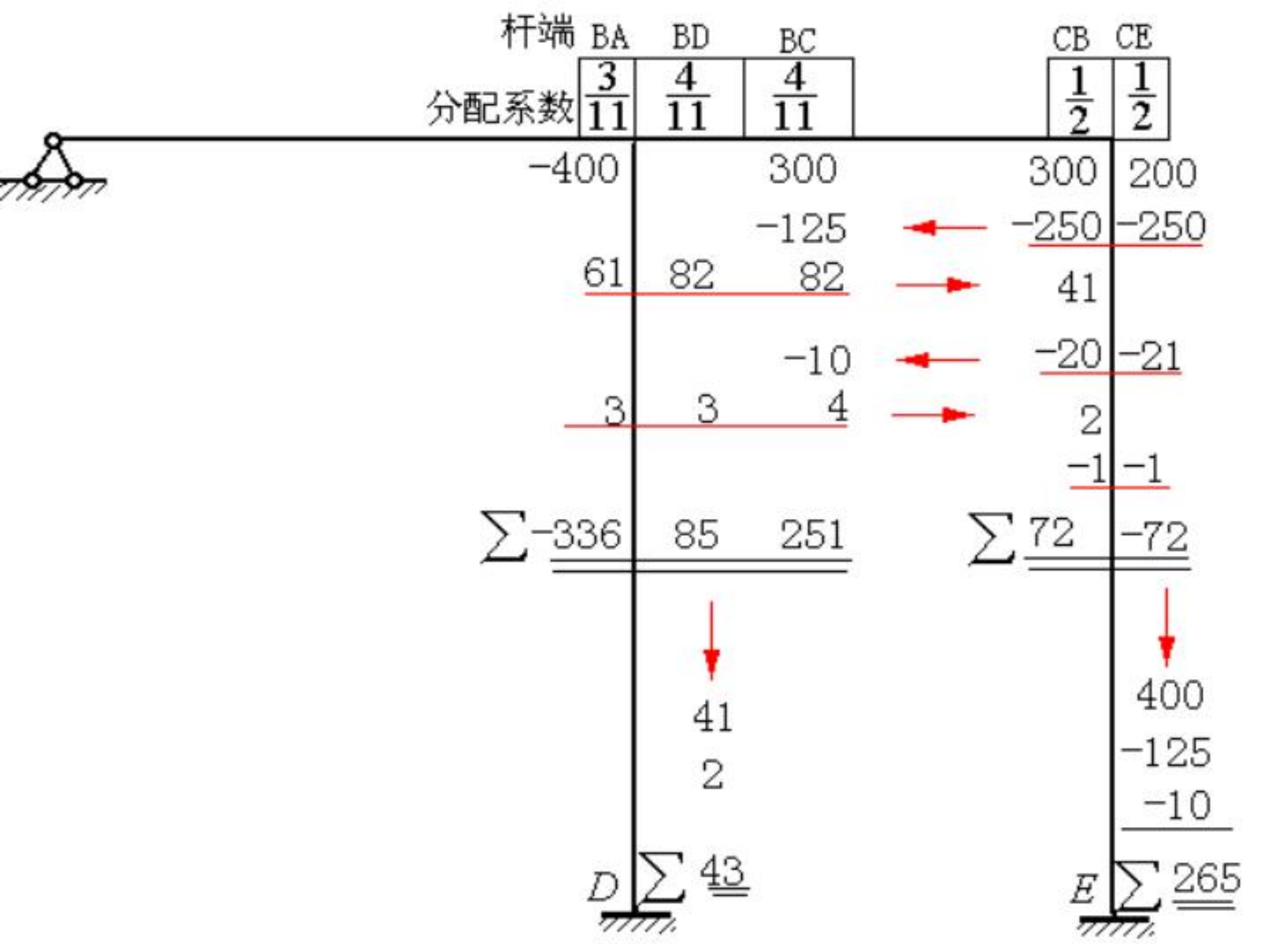




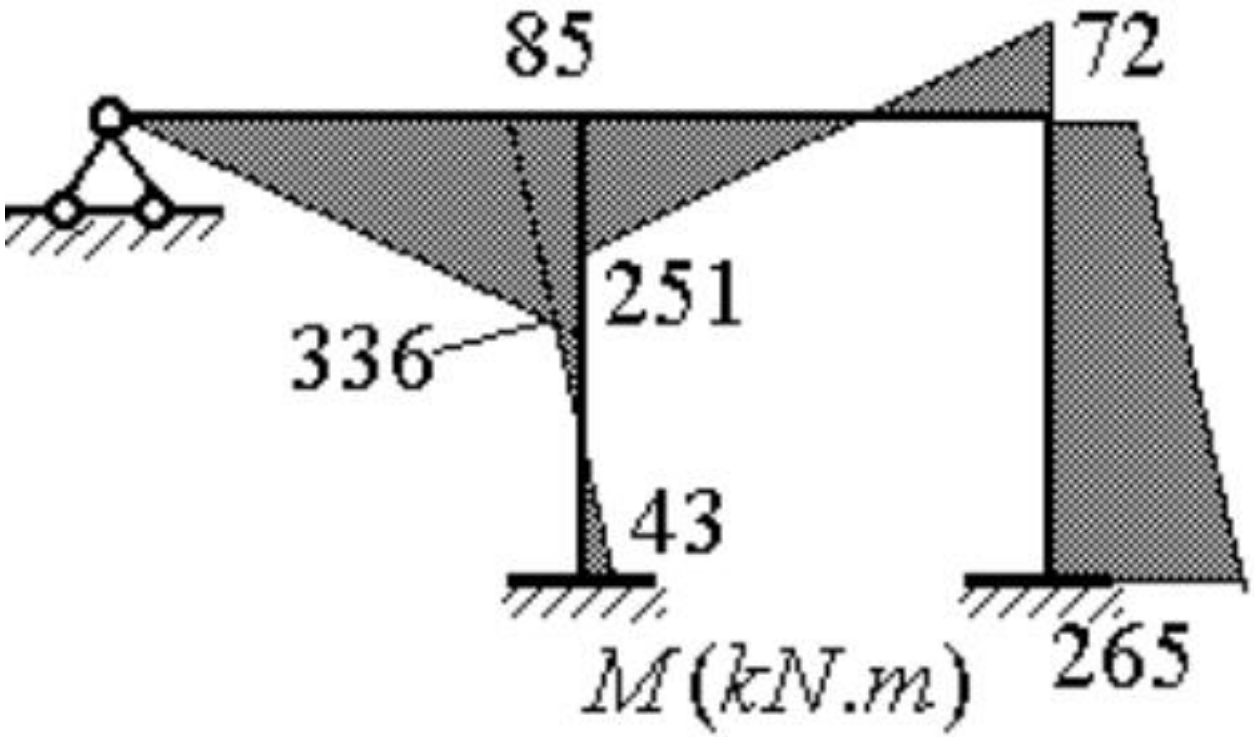


$$M_{CE}^{F}=\frac{2EI}{l}\times\varphi_E=\frac{2\times6\times10^4\,kN.m^2\times0.01}{6}=200kN.m$$

$$M_{EC}^{F}=\frac{4EI}{l}\times\varphi_E=\frac{4\times6\times10^4\,kN.m^2\times0.01}{6}=400kN.m$$


杆端 BA BD BC CB CE
分配系数 3/11 4/11 4/11 1/2 1/2
-400 300 300 200
-125 -250 -250
61 82 82 41
-10 -20 -21
3 3 4 2
-1 -1
∑ -336 85 251 ∑ 72 -72
41
2
400
-125
-10
D ∑ 43
E ∑ 265

M图



讨论：

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从上例可知，超静定结构对支座移动是很敏感的，不大的支座移动可以引起相当大的内力。抗弯刚度EI愈大，则引起的内力愈大。因此，在超静定结构设计中，应当注意地基情况。

支座移动、温度变化引起的内力与杆件的EI成正比。在给定的支座移动、给定的温度条件下，截面尺寸愈大，内力愈大。

所以为了改善结构在支座移动、温度变化作用下的受力状态，加大截面尺寸并不是一个有效的途经。

§9-4　无剪力分配法

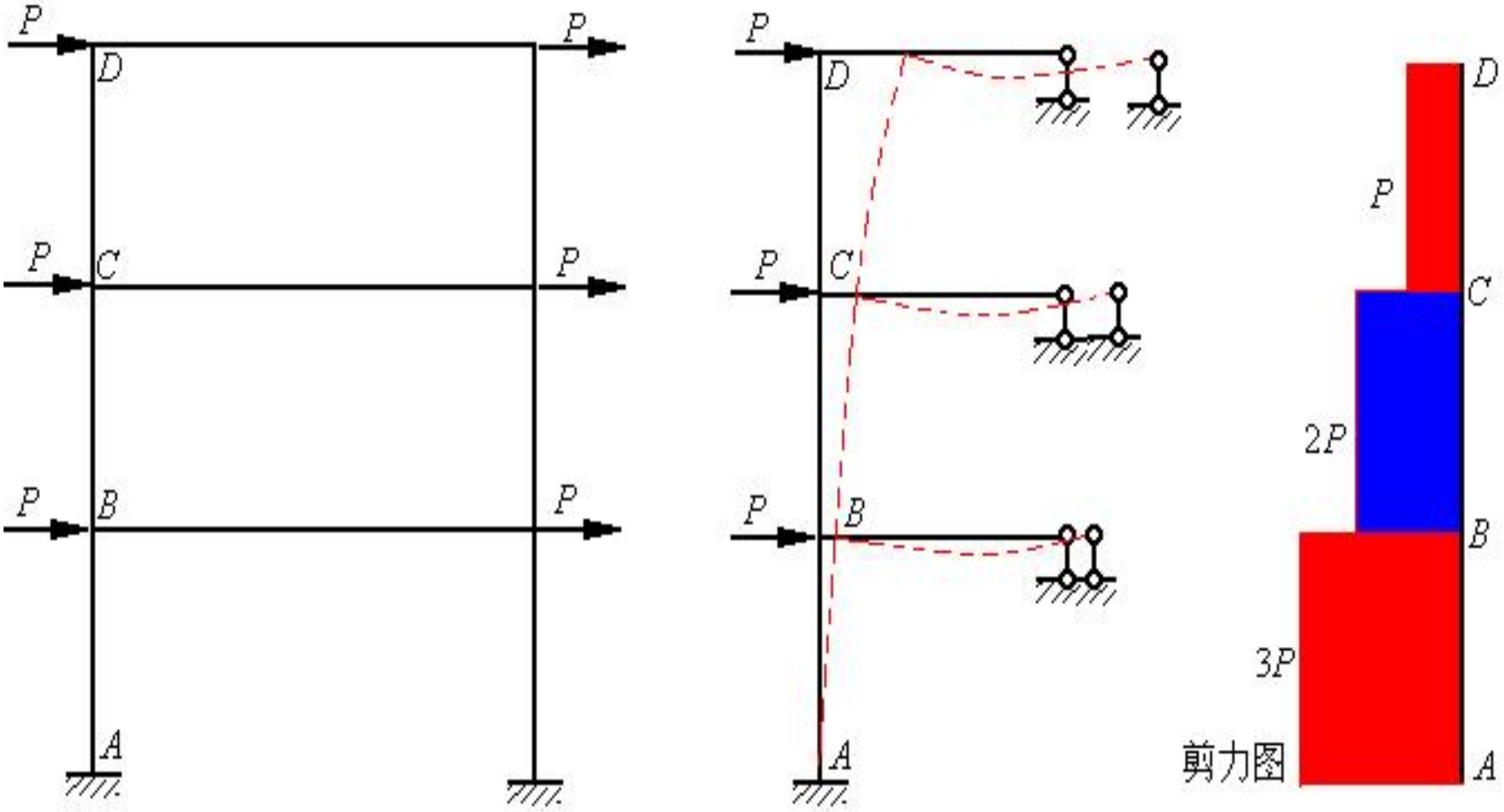
§ 9-4 无剪力分配法

在位移法中，刚架分为无侧移刚架与有侧移刚架两类。它们的区别是：在位移法的基本未知量中，前者只包含结点角位移，后者则还包含结点线位移。

力矩分配法适用于无侧移刚架，但对某些特殊的有侧移刚架，可以用与力矩分配法类似的无剪力分配法进行计算。

无剪力分配法的应用条件：用于剪力静定柱

例



单跨对称刚架在反对荷载作用下的计算为这类特定条件下的有侧移（即线位移）刚架。

无剪力分配法中的两类杆件：

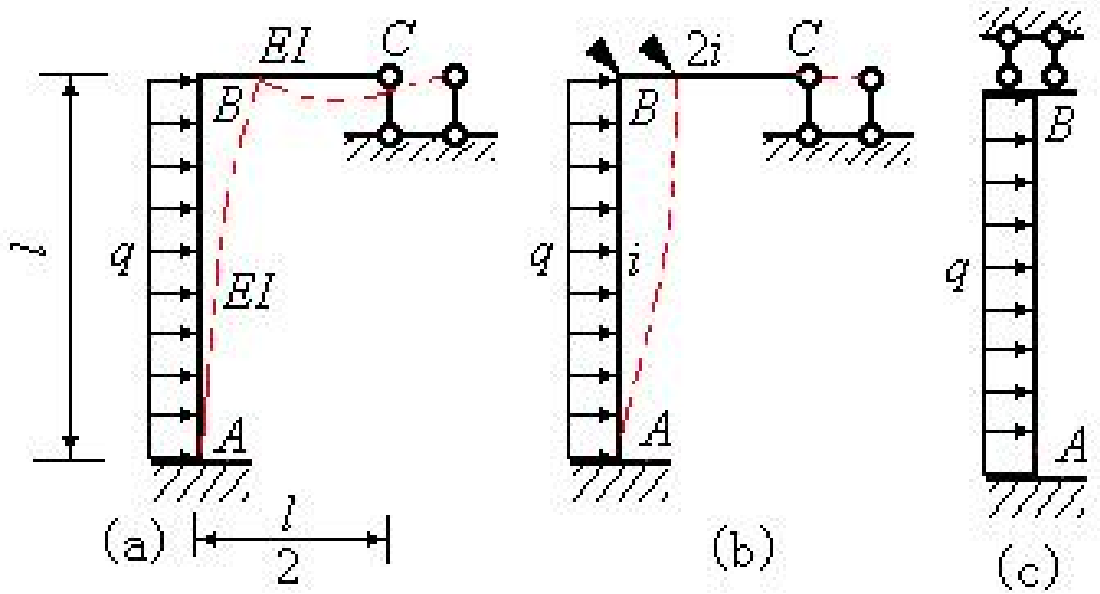
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在上图中，各梁的两端结点没有相对线位移（即没有垂直杆轴的相对位移），这种杆称为两端无相对线位移杆件（或称为**无侧移杆件**）；

各柱的两端结点虽然有侧移，但剪力是静定的（可根据平衡条件求出，如上图所示），这种杆件称为**剪力静定杆**。

无剪力分配法的应用条件是：刚架中除两端无相对线位移的杆件外，其余杆件都是剪力静定杆件。分配传递过程中剪力不变，仅弯矩可能变化。

无剪力分配法中固端弯矩的计算



（1）固定结点

只加刚臂阻止结点*B*的转动，而不加链杆阻止其线位移。

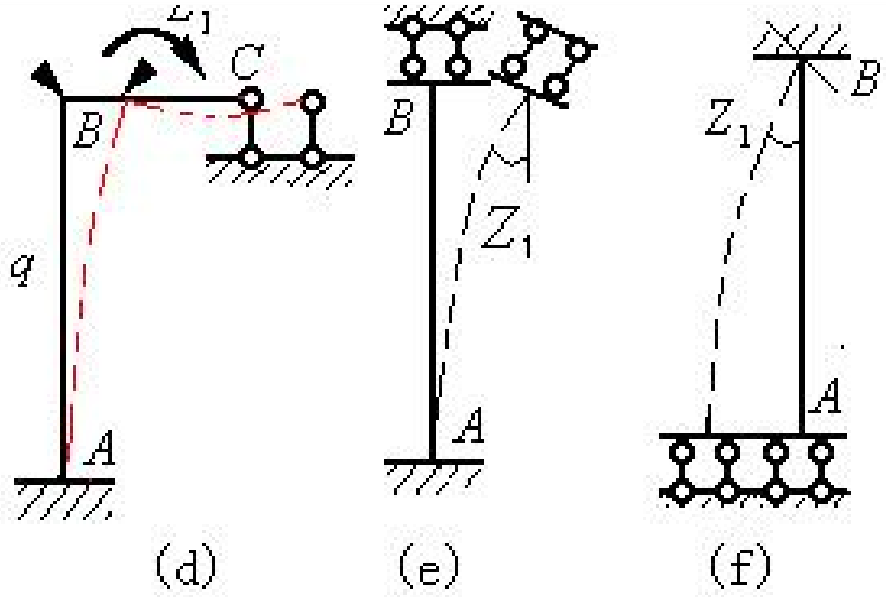
此时*AB*杆仍可自由地水平滑动，相当图*c*。而横梁仍视为*B*端固定*C*端为链杆支座的梁。

*AB*杆的固端弯矩为：

$$M_{AB}^{F} = -\frac{ql^2}{3}, \qquad M_{BA}^{F} = -\frac{ql^2}{6}$$

无剪力分配法中分配系数的计算

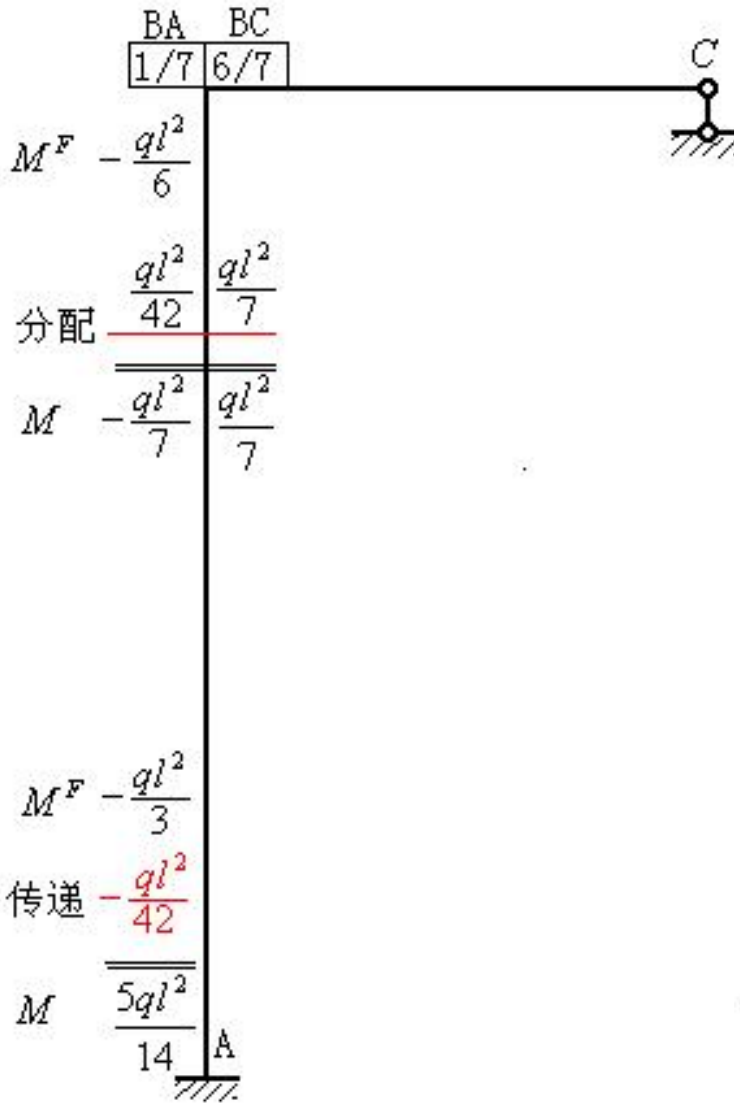
（2）放松结点

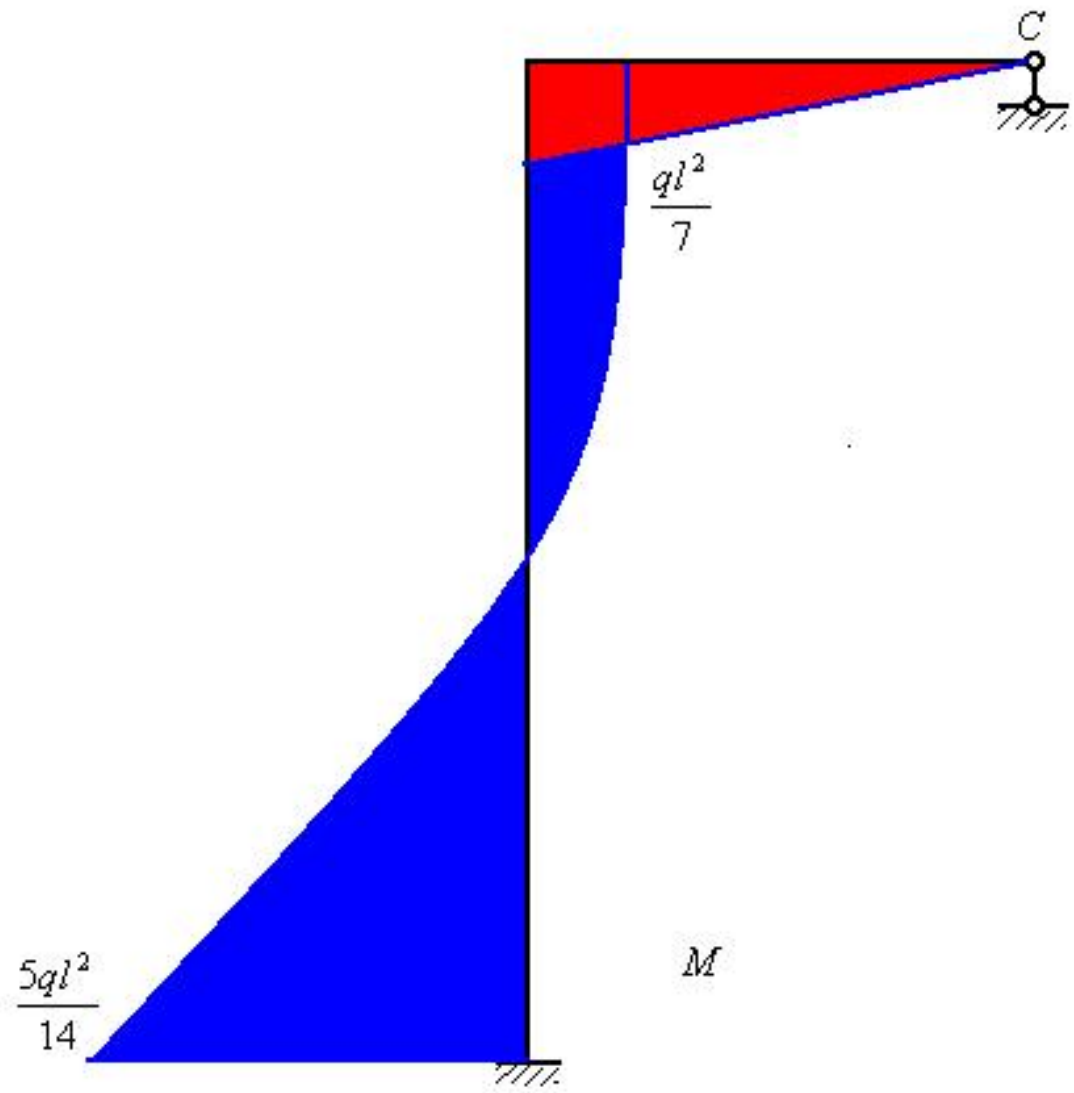


为了消除刚臂的不平衡力矩，需放松结点，然后进行力矩的分配和传递。为此需求结点的分配系数。

结点B不仅转动Z_1角，同时发生水平位移，如图d所示。此时杆AB的变形相当图e或图f*(力法推导知，ef弯矩相同)*。于是结点B的分配系数为

$$\mu_{BA}=\frac{S_{BA}}{S_{BA}+S_{BC}}=\frac{i}{i+3\times 2i}=\frac{1}{7},\qquad \mu_{BC}=\frac{S_{BC}}{S_{BA}+S_{BC}}=\frac{3\times 2i}{i+3\times 2i}=\frac{6}{7}$$


BA
BC
1/7
6/7
C
M^F $-\frac{ql^2}{6}$
$\frac{ql^2}{42}$ $\frac{ql^2}{7}$
分配
M $-\frac{ql^2}{7}$ $\frac{ql^2}{7}$
M^F $-\frac{ql^2}{3}$
传递 $-\frac{ql^2}{42}$
M $\frac{5ql^2}{14}$
A


C
$\frac{ql^2}{7}$
$\frac{5ql^2}{14}$
M

由上题的计算可知：

在固定结点时柱AB的剪力是静定的；

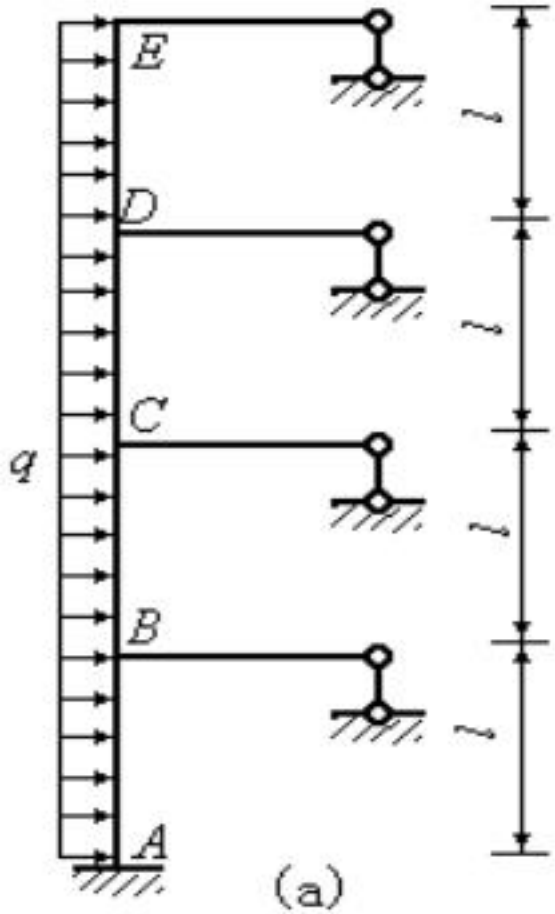
在放松结点时，柱B端得到的分配弯矩将乘以-1的传递系数传到A端，因此弯矩沿AB杆全长为常数而剪力为零。

在力矩的分配与传递中，柱中原剪力保持不变而不增加新的剪力，故这种方法称为无剪力分配法。

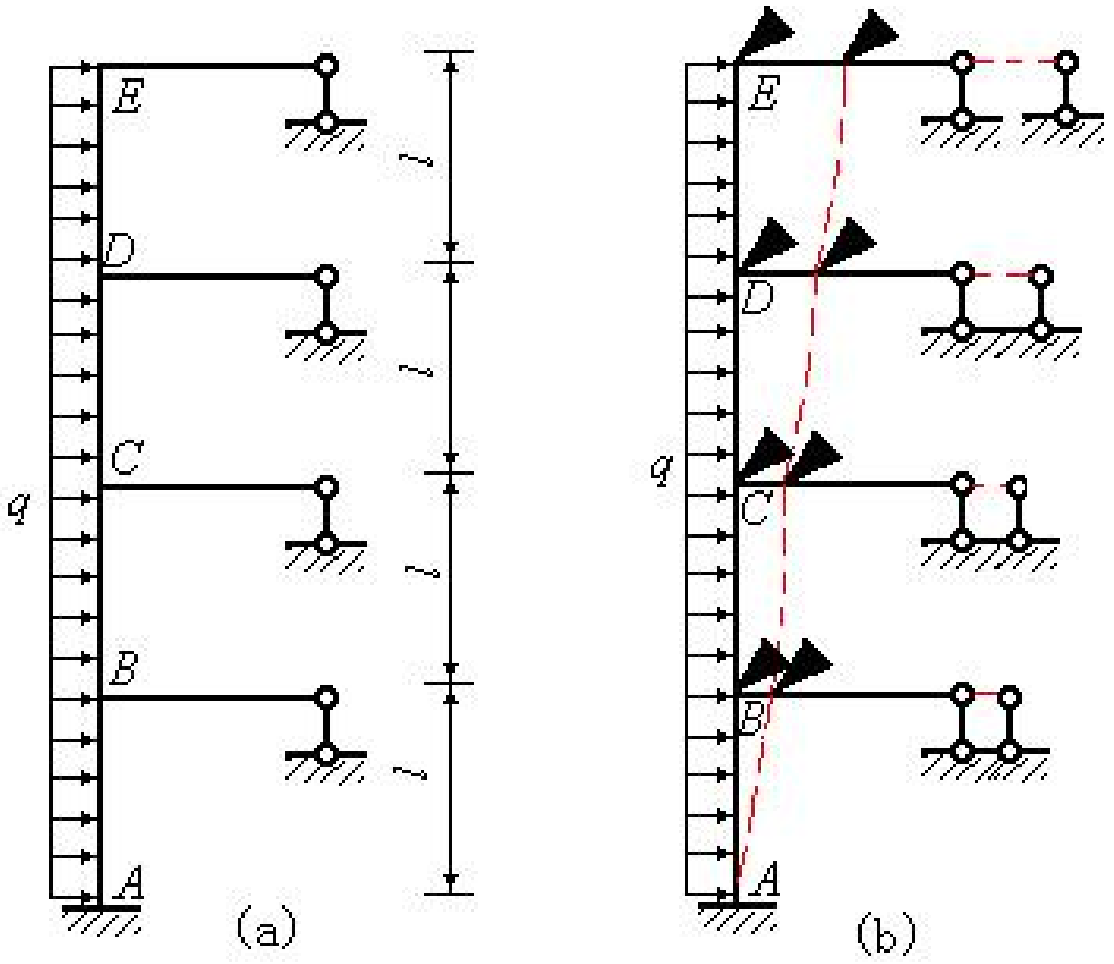
多层刚架的计算

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将上述方法推广到多层。如图示结构横梁为无侧移杆，竖柱为剪力静定杆。



（1）固定结点



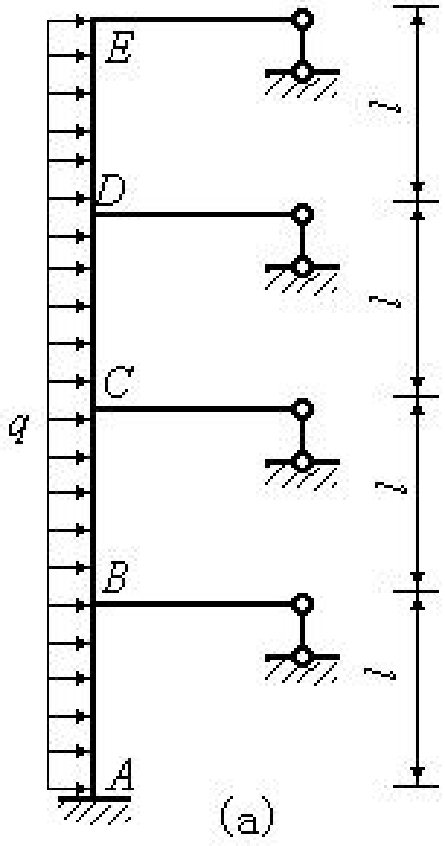
各柱两端无转角，但有侧移。

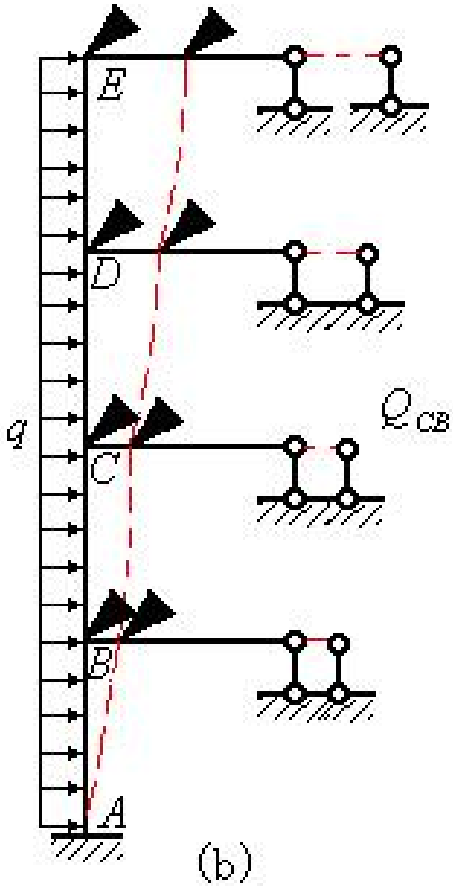
固端弯矩的计算

从图b中任取出任一层柱的约束形式如图c所示。

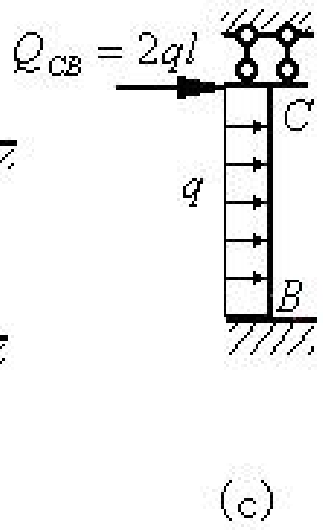
该层柱的上端所受剪力的数值为：

柱顶以上各所有水平荷载的代数和。



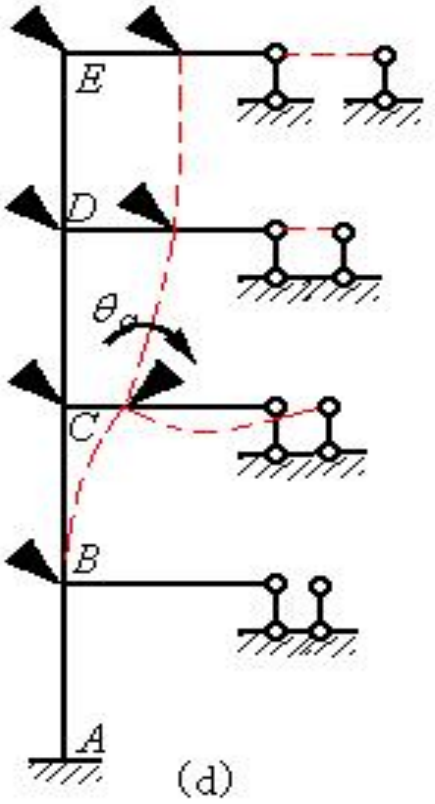


上端滑动下端固定



因此图*b*就可根据表8-1算出各柱的固端弯矩。

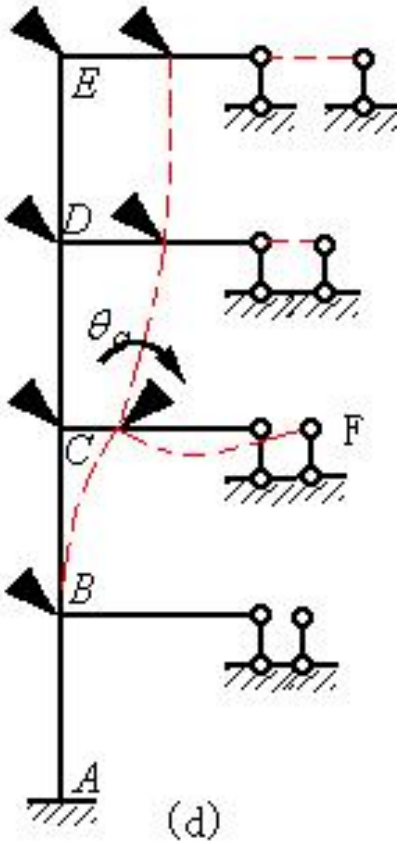
（2）轮流地放松结点



轮流地放松结点，进行力矩分配、传递。图*d* 即为放松某一结点*C*时的情形。

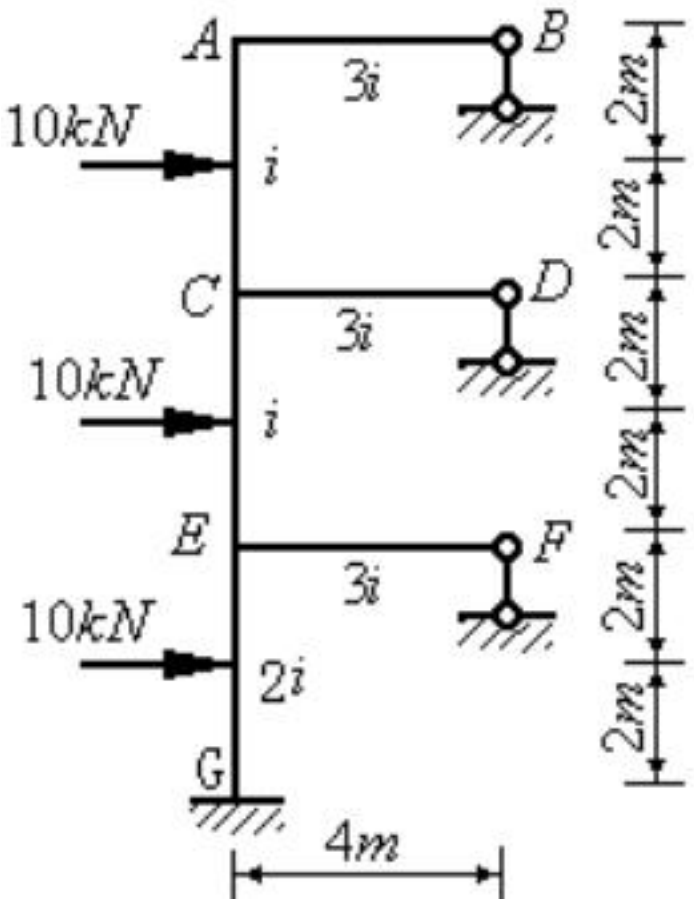
这相当于在该结点上的不平衡力矩反号做为力偶荷载施加于该结点上。此时结点C不仅转动一转角，同时BC、CD两柱还将产生相对侧移，但由平衡条件可知两柱剪力均为零，处于純弯曲状态，因而各柱的转动（劲度）系数为*i* 而传递为-1。

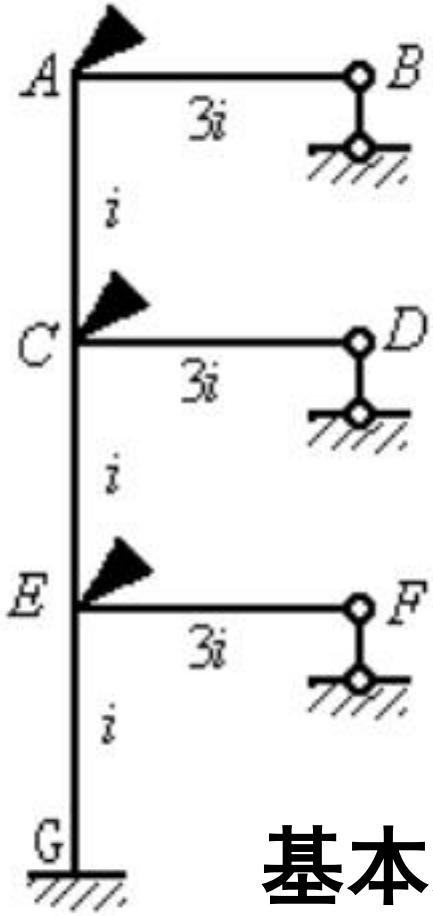
放松结点C



放松结点C时，力矩的分配、传递将只在CB、CF、CD三杆范围内进行。放松其它结点情况类似，至于力矩分配、传递的具体计算步骤于力矩分配相同。

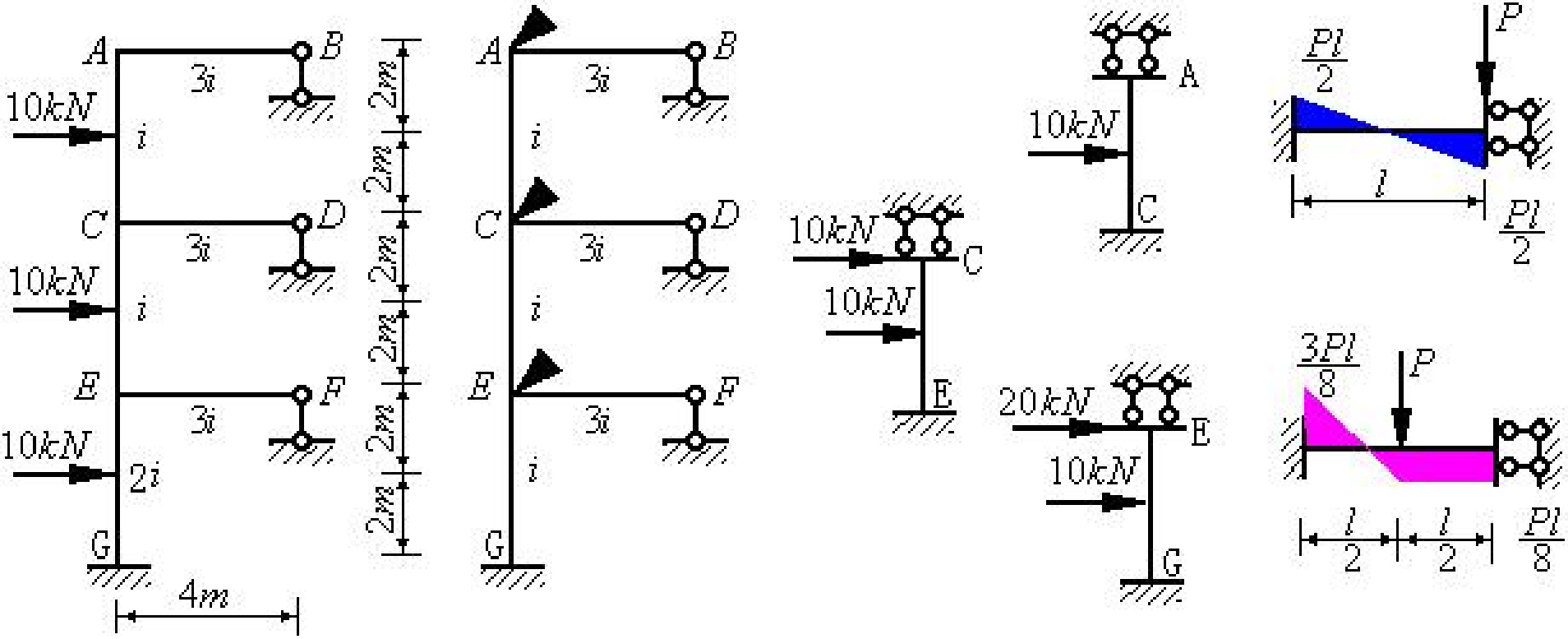
例1 用无剪力分配法计算图示刚架。





基本结构

计算固端弯矩

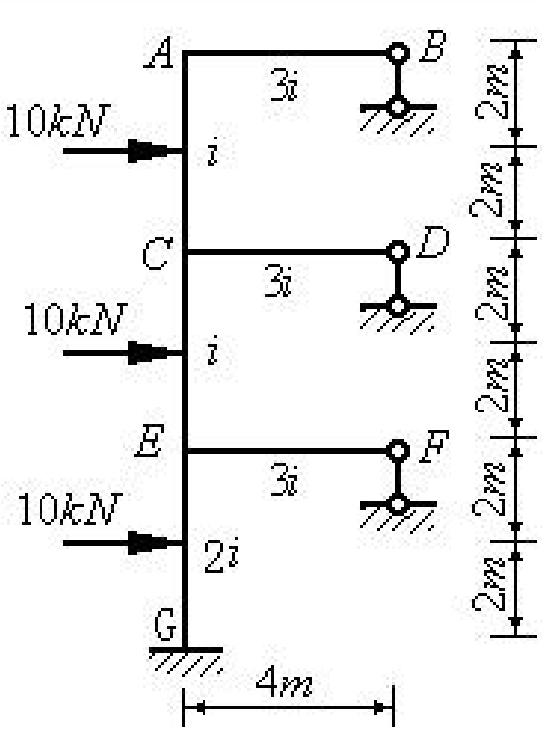


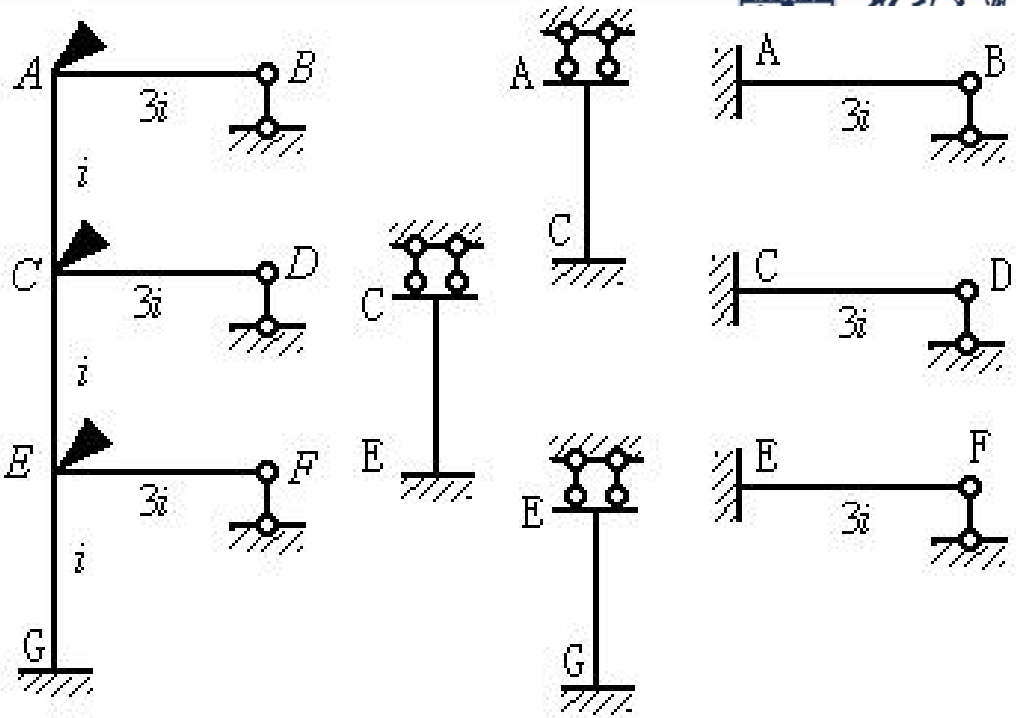
$$M_{AC}^{F} = -\frac{Pl}{8} = -\frac{10kN \times 4m}{8} = -5kN \quad m, \qquad M_{CA}^{F} = -\frac{3Pl}{8} = -15kN$$

$$M_{CE}^{F} = -\frac{Pl}{8} - \frac{Pl}{2} = -25kN \quad m, \qquad M_{EC}^{F} = -\frac{3Pl}{8} - \frac{Pl}{2} = -35kN \quad m$$

$$M_{EG}^{F} = -\frac{Pl}{8} - \frac{2Pl}{2} = -45kN \quad m, \qquad M_{GE}^{F} = -\frac{Pl}{8} - \frac{Pl}{2} = -55kN \quad m$$

计算分配系数

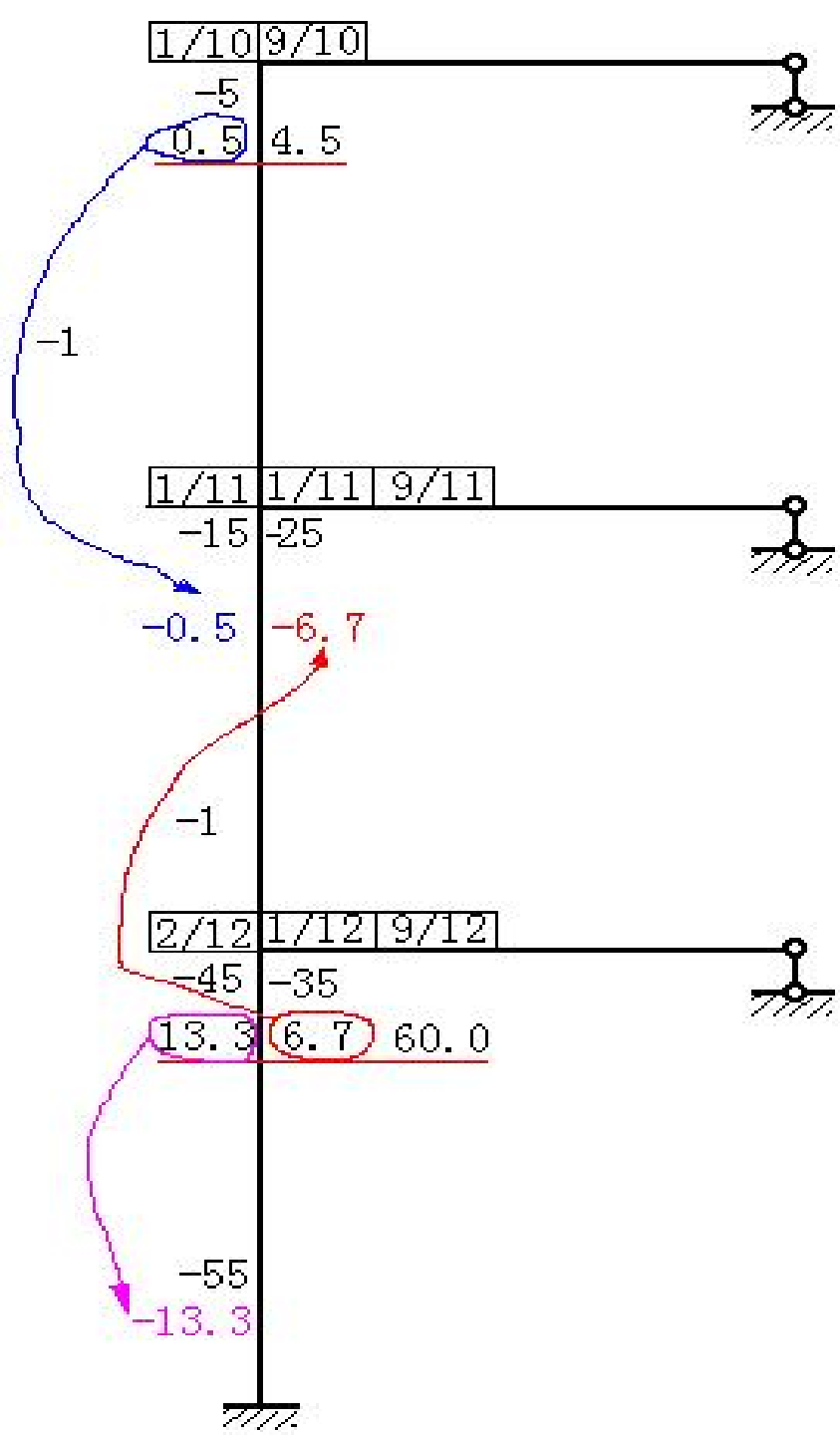


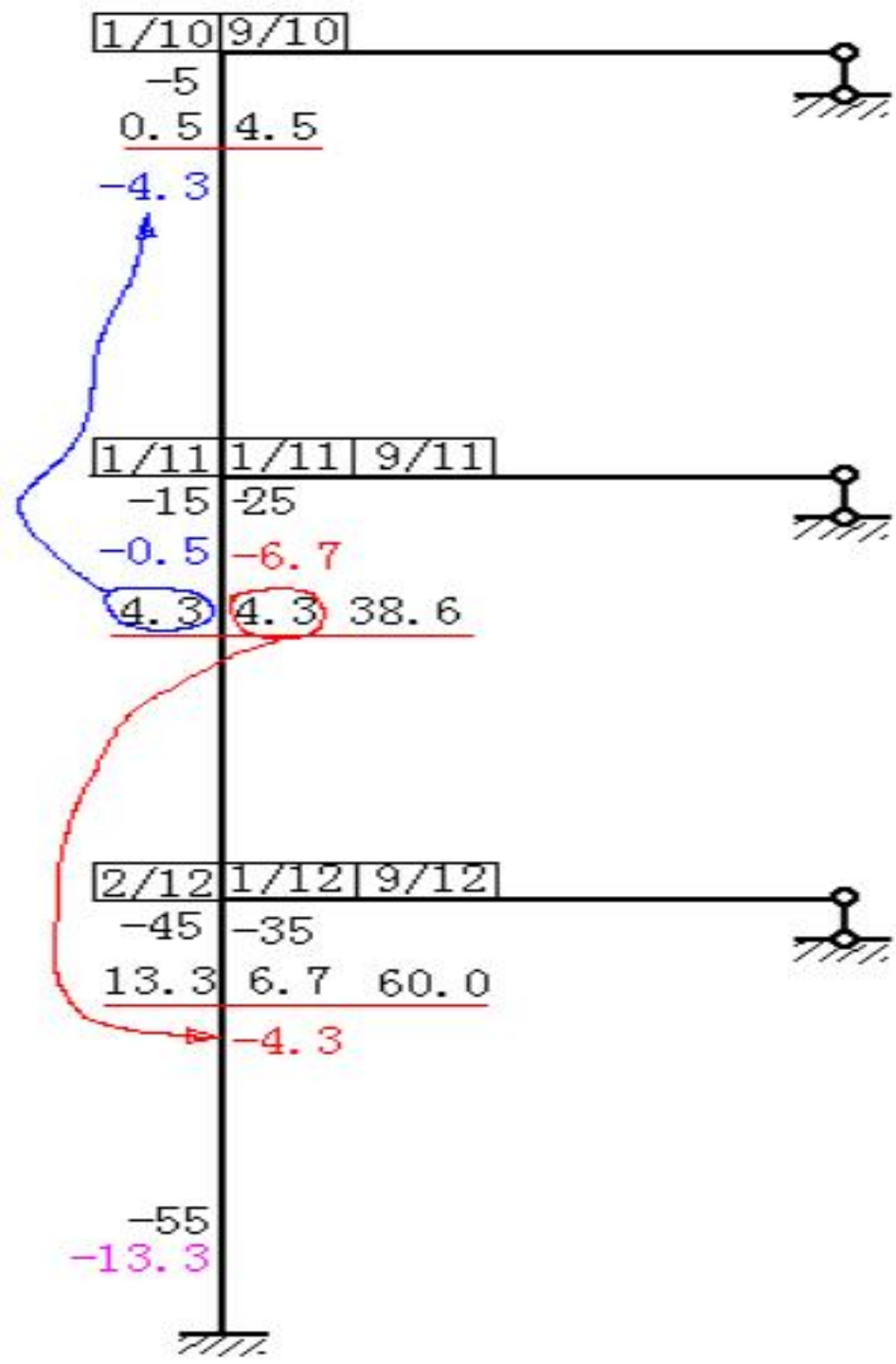


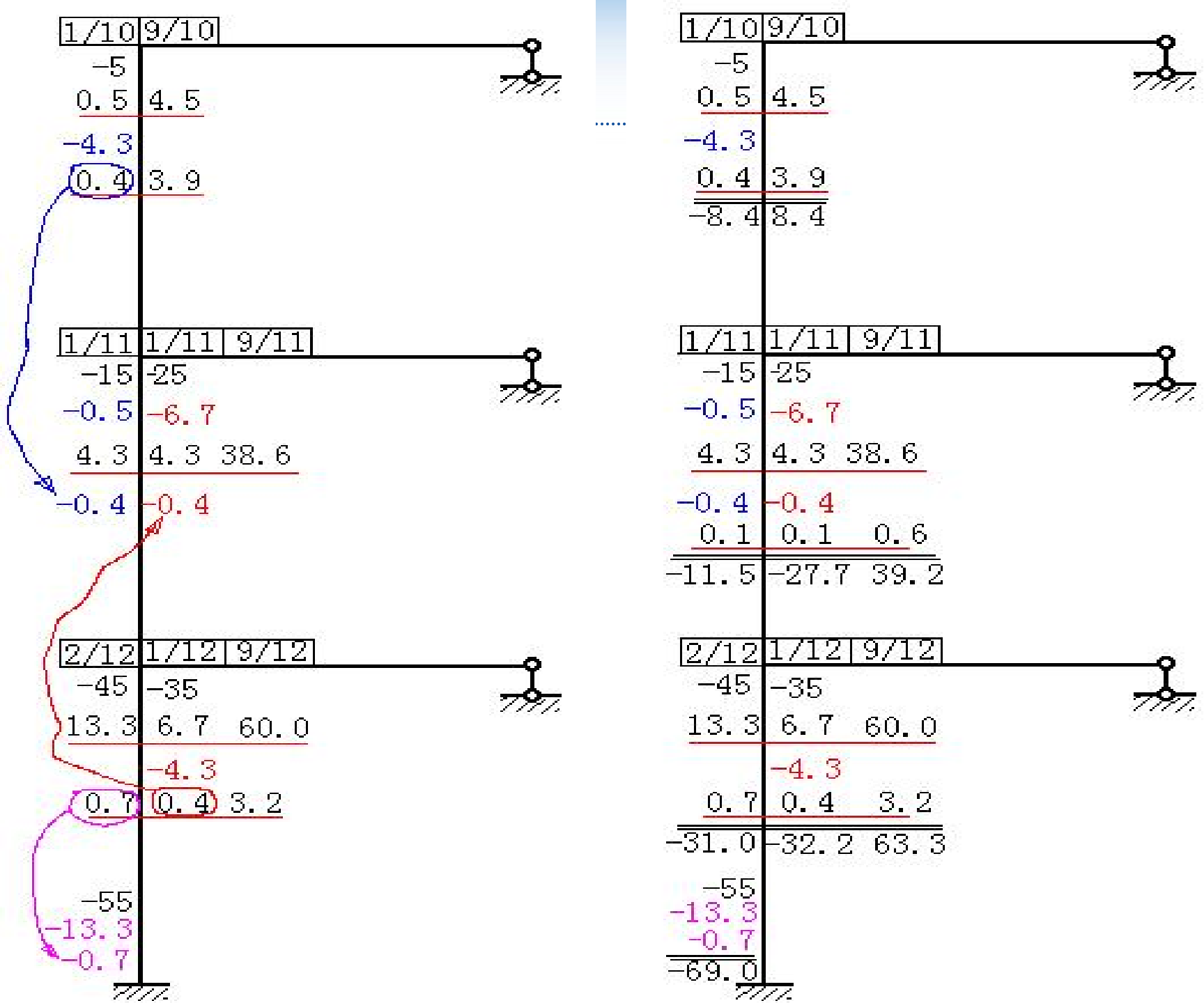
$$\mu_{AB}=\frac{S_{AB}}{S_{AB}+S_{AC}}=\frac{3\times 3i}{3\times 3i+i}=\frac{9}{10},\qquad \mu_{AC}=\frac{S_{AC}}{S_{AB}+S_{AC}}=\frac{i}{3\times 3i+i}=\frac{1}{10}$$

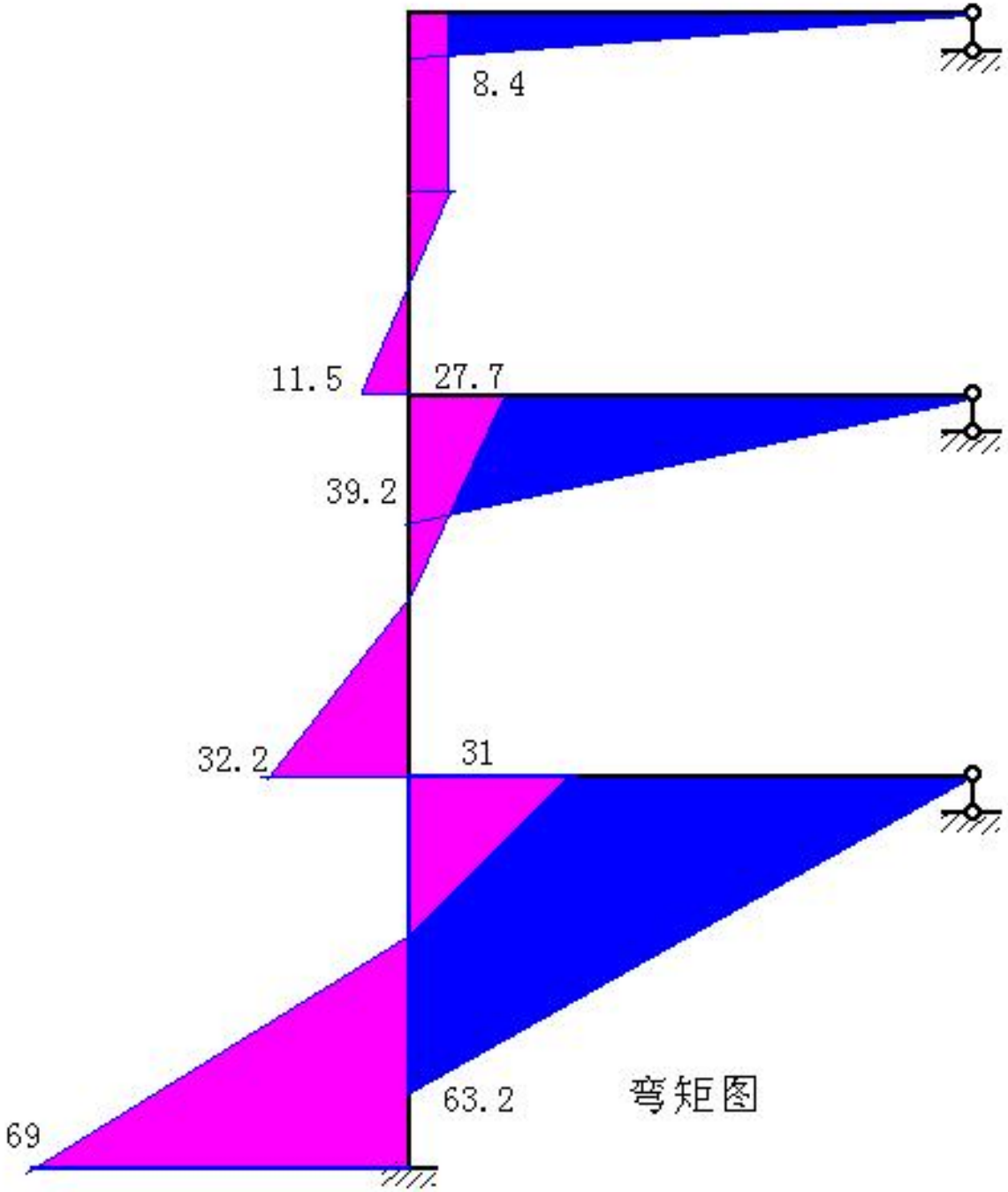
$$\mu_{CA}=\frac{S_{CA}}{S_{CA}+S_{CD}+S_{CE}}=\frac{i}{i+3\times 3i+i}=\frac{1}{11},\qquad u_{CD}=\frac{9}{11},\qquad \mu_{CE}=\frac{1}{11}$$

$$\mu_{EC}=\frac{S_{EC}}{S_{EC}+S_{EF}+S_{EG}}=\frac{i}{i+3\times 3i+2\times i}=\frac{1}{12},\qquad u_{EF}=\frac{9}{12},\qquad \mu_{EG}=\frac{2}{12}$$


1/10
9/10
-5
0.5
4.5
-1
1/11
1/11
9/11
-15
-25
-0.5
-6.7
-1
2/12
1/12
9/12
-45
-35
13.3
6.7
60.0
-55
-13.3

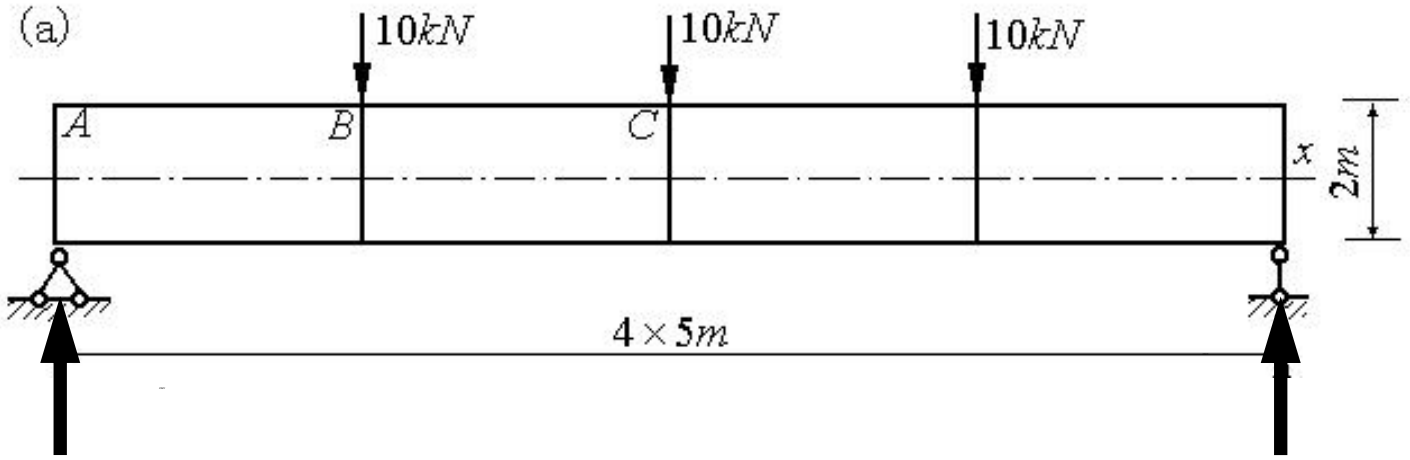

1/10
9/10
-5
0.5
4.5
-4.3
1/11
1/11
9/11
-15
-25
-0.5
-6.7
4.3
4.3
38.6
2/12
1/12
9/12
-45
-35
13.3
6.7
60.0
-4.3
-55
-13.3


1/10 9/10
-5
0.5 4.5
-4.3
0.4 3.9
1/11 1/11 9/11
-15 -25
-0.5 -6.7
4.3 4.3 38.6
-0.4 -0.4
2/12 1/12 9/12
-45 -35
13.3 6.7 60.0
-4.3
0.7 0.4 3.2
-55
-13.3
-0.7
1/10 9/10
-5
0.5 4.5
-4.3
0.4 3.9
-8.4 8.4
1/11 1/11 9/11
-15 -25
-0.5 -6.7
4.3 4.3 38.6
-0.4 -0.4
0.1 0.1 0.6
-11.5 -27.7 39.2
2/12 1/12 9/12
-45 -35
13.3 6.7 60.0
-4.3
0.7 0.4 3.2
-31.0 -32.2 63.3
-55
-13.3
-0.7
-69.0

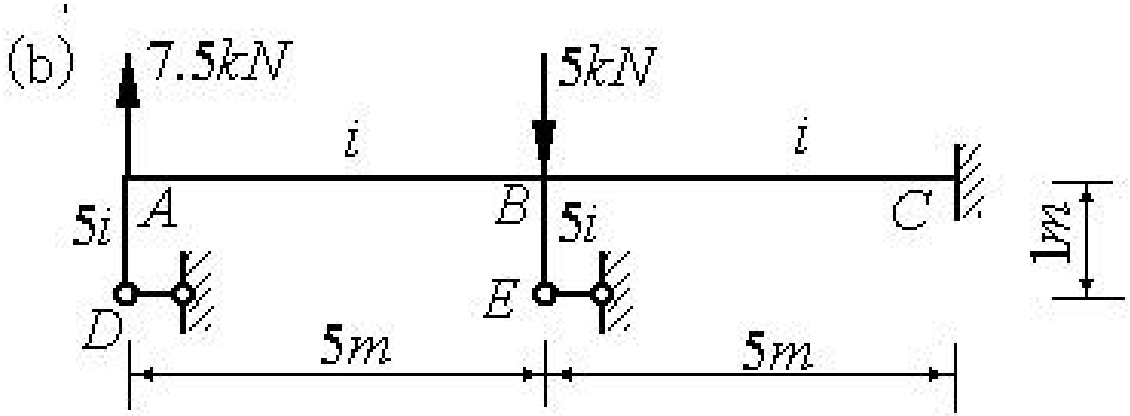


弯矩图

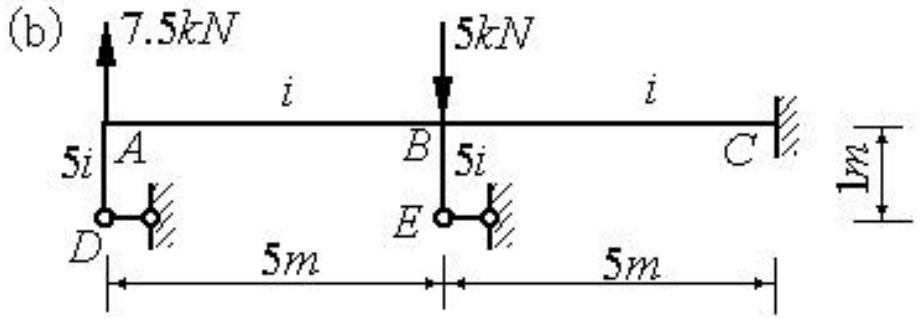
例2 用无剪力分配法计算图*a*所示空腹梁并绘弯矩图，*EI*=*C*。



解：取1/4结构，可用无剪力分配法计算。



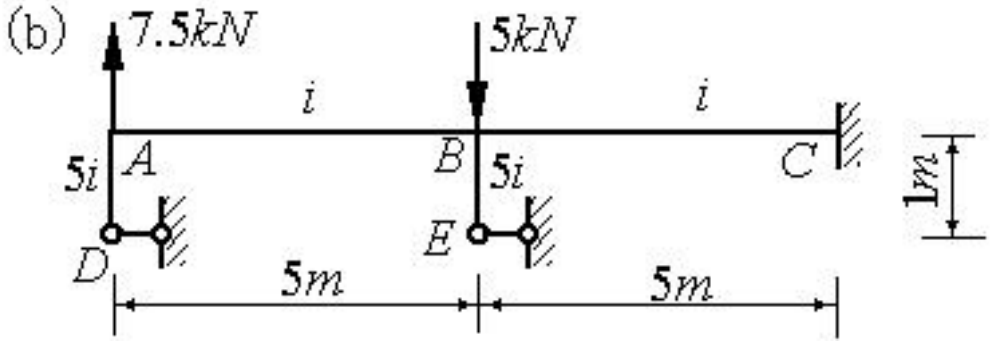
计算固端弯矩



$$M_{AB}^{F} = M_{BA}^{F} = -\frac{Pl}{2} = -\frac{7.5 \times 5}{2} = -18.75kN.m$$

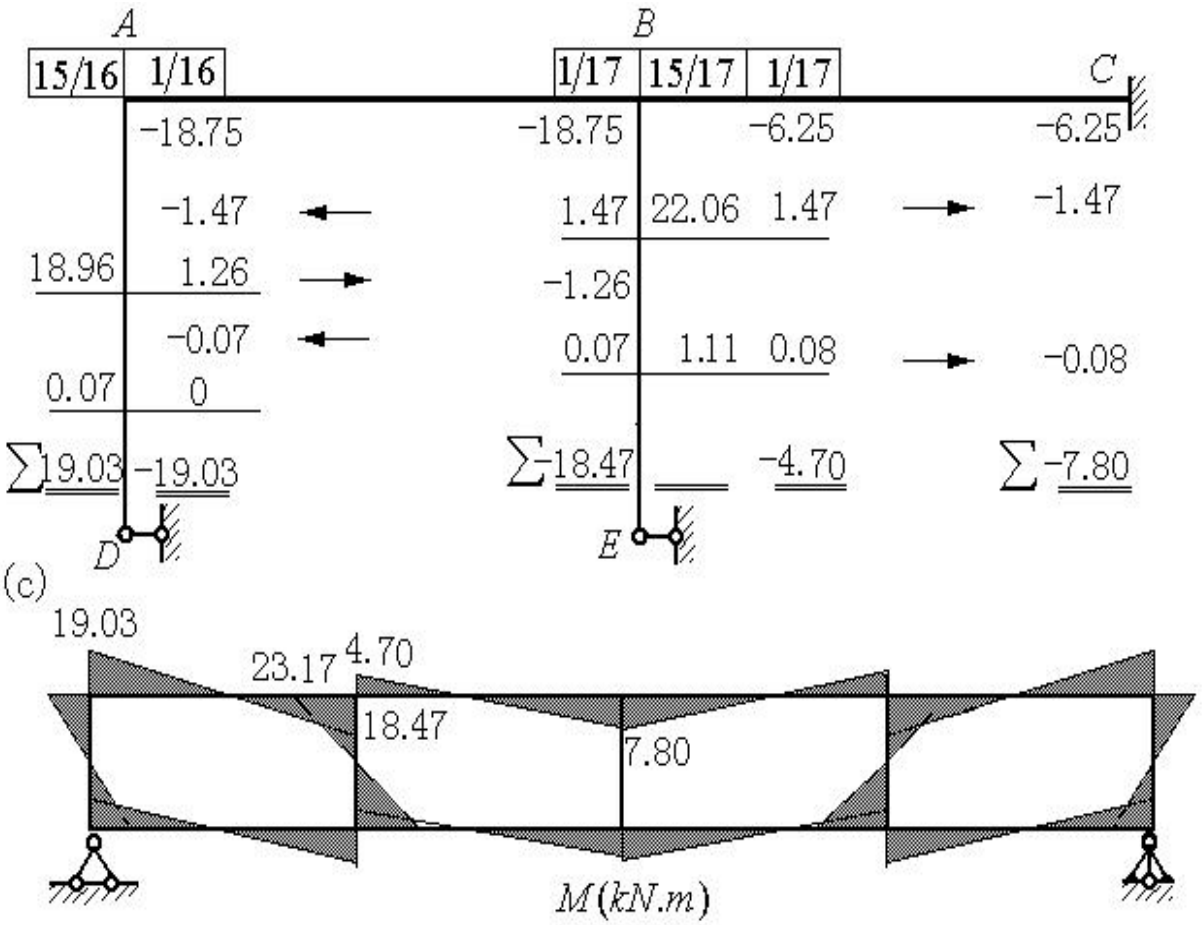
$$M_{BC}^{F} = M_{CB}^{F} = -\frac{Pl}{2} = -\frac{(7.5-5) \times 5}{2} = -6.25kN.m$$

计算分配系数



$$u_{AD}=\frac{S_{AD}}{\sum_{A}S}=\frac{3(5i)}{3(5i)+i}=\frac{15}{16}\qquad u_{AB}=\frac{S_{AB}}{\sum_{A}S}=\frac{i}{3(5i)+i}=\frac{1}{16}$$

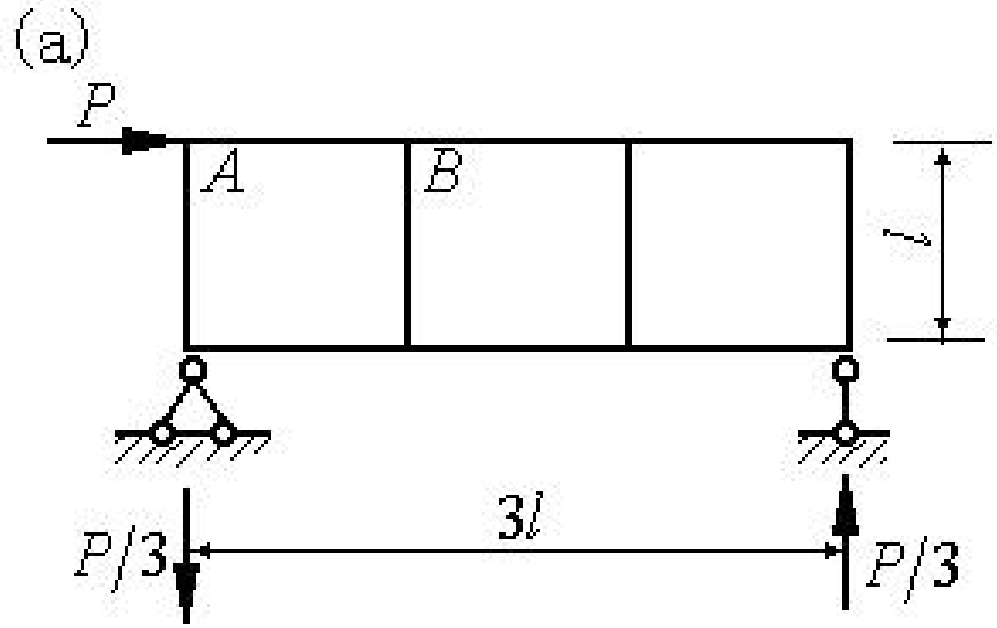
$$u_{BA}=\frac{S_{BA}}{\sum_{B}S}=\frac{3(5i)}{3(5i)+i+i}=\frac{15}{17}\qquad u_{BC}=u_{BE}=\frac{1}{17}$$


A
B
C
15/16
1/16
1/17
15/17
1/17
-18.75
-18.75
-6.25
-6.25
-1.47
1.47
22.06
1.47
-1.47
18.96
1.26
-1.26
-0.07
0.07
1.11
0.08
-0.08
0.07
0
∑19.03
-19.03
∑-18.47
-4.70
∑-7.80
D
E
(c)
19.03
23.17
4.70
18.47
7.80
M(kN.m)

例3

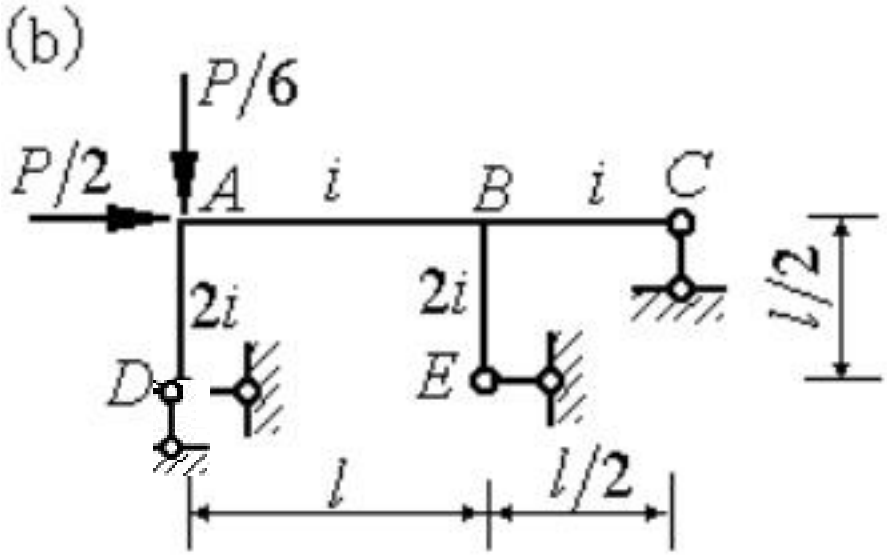
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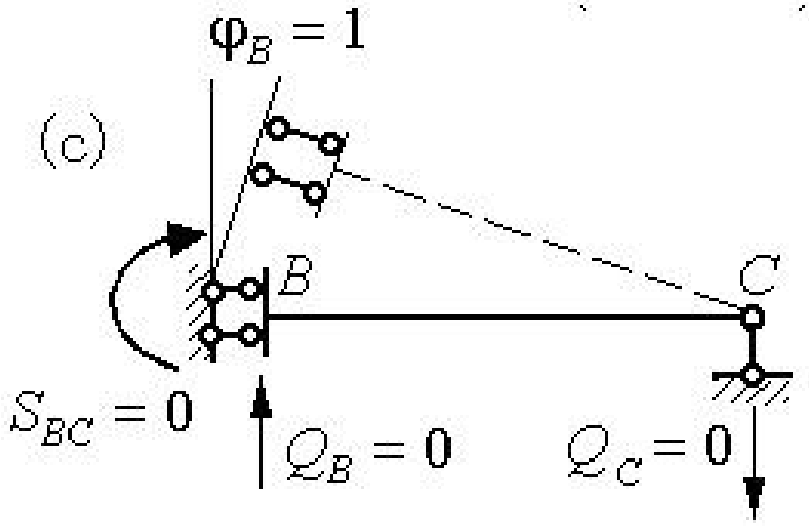
用无剪力分配法计算图*a*所示空腹梁并绘弯矩图，*EI*=*C*。



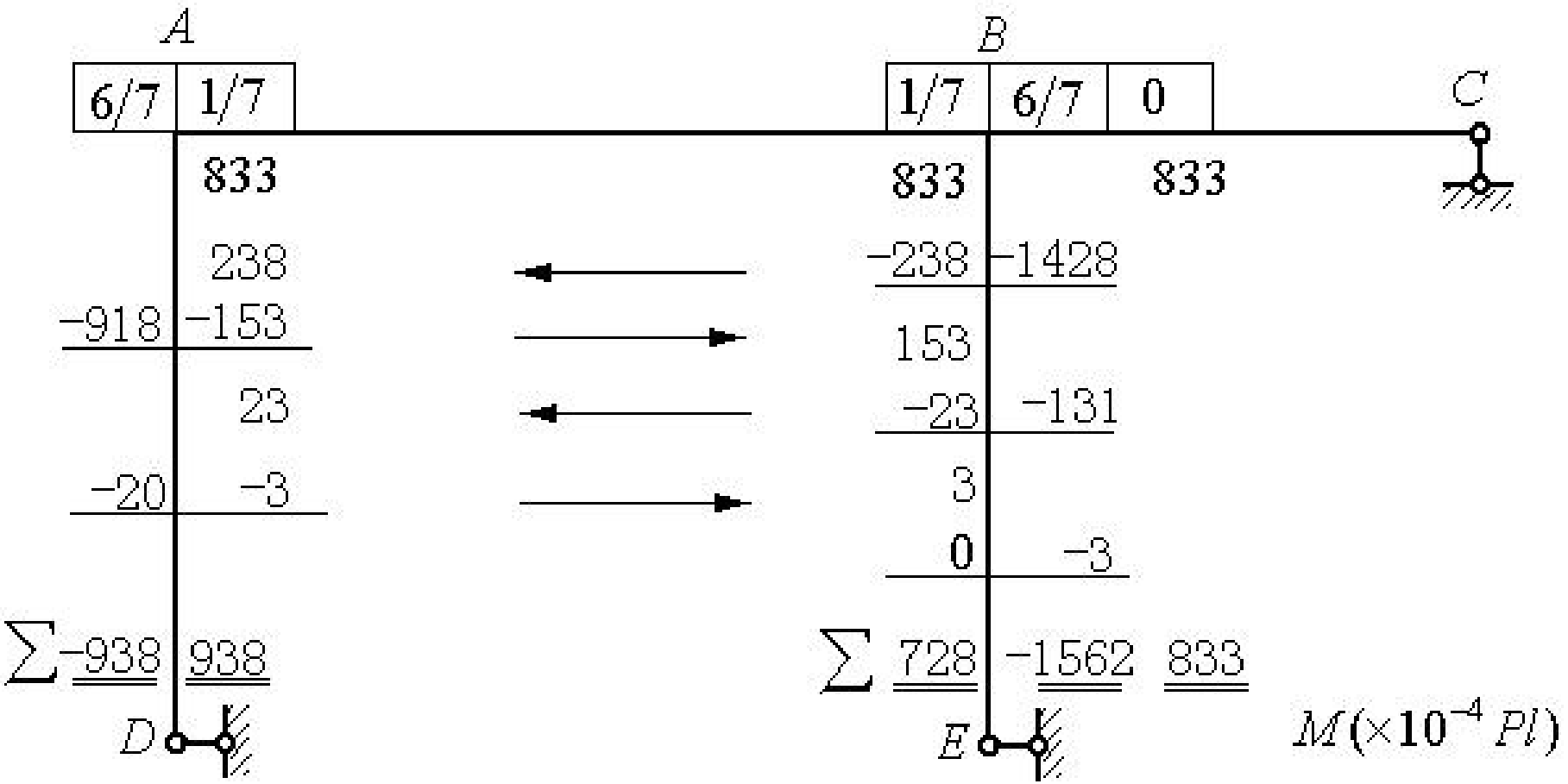
取1/4结构计算

解：上下左右均反对称，取1/4结构计算。

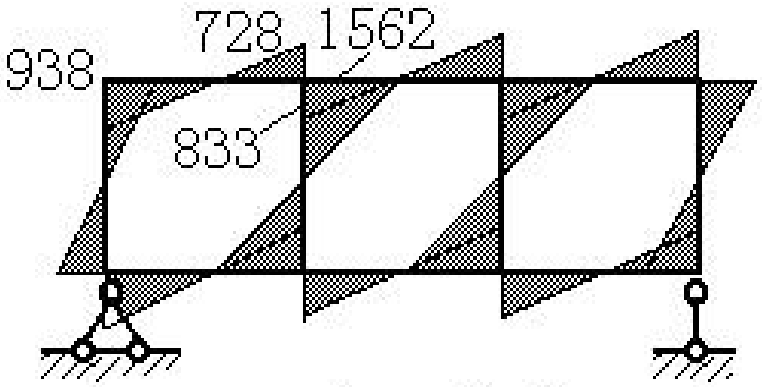




注意的是*BC*杆的分配系数为零，其推导过程如图*c*所示。



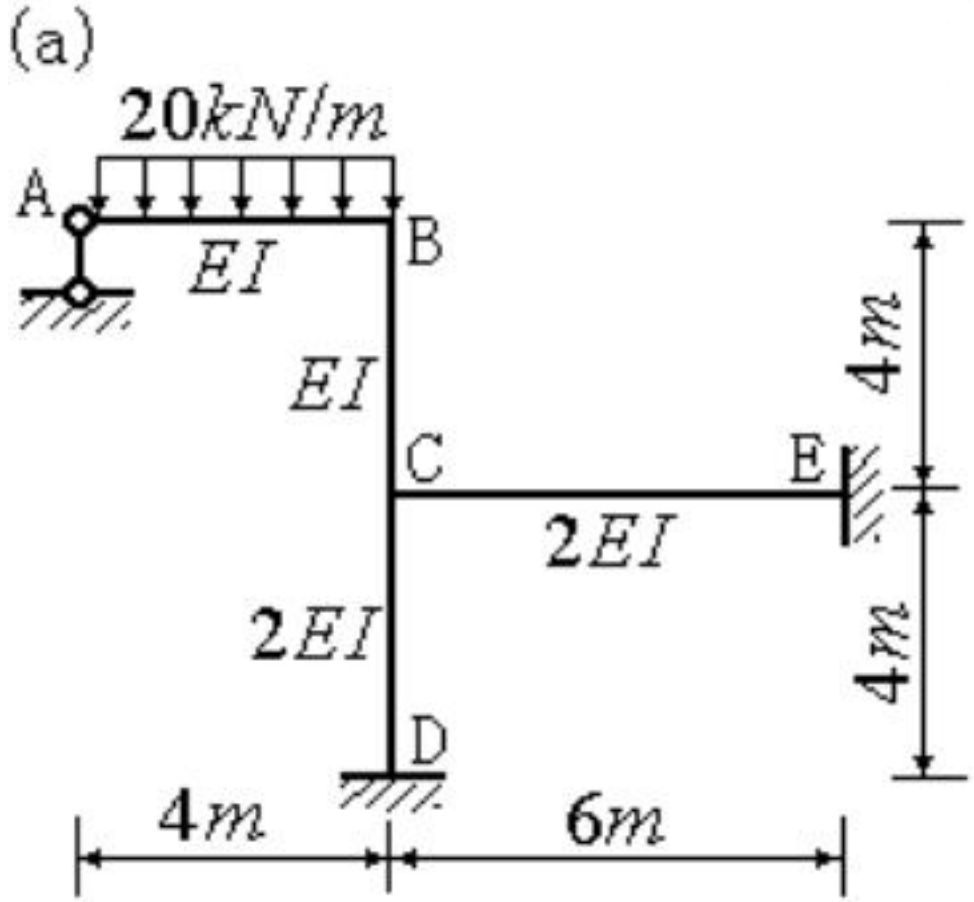
(d)



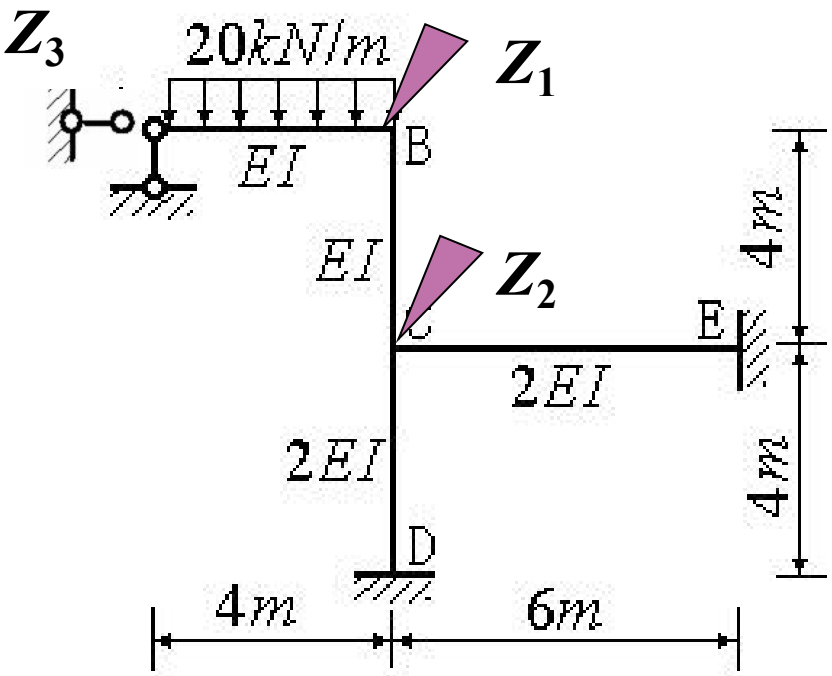
$M(\times 10^{-4}\,Pl)$

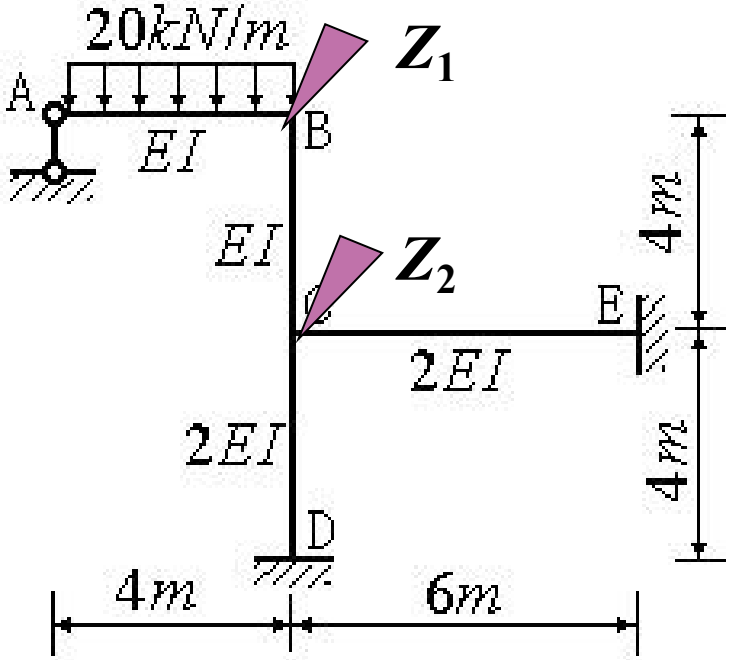
例4

选择适当方法计算图*a*所示刚架，绘制弯矩图。



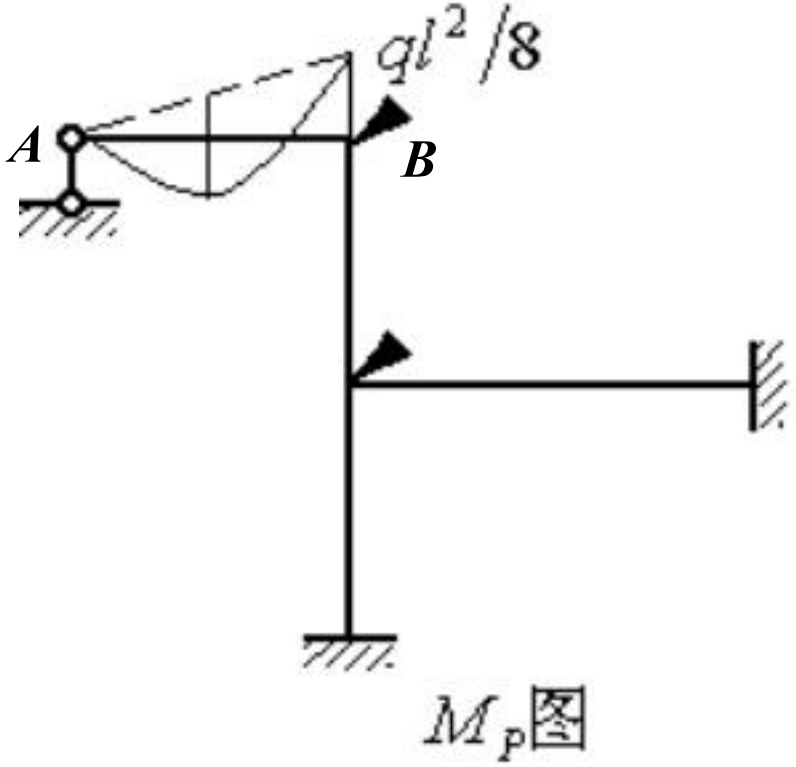
方法选择


Z3
20kN/m
Z1
B
EI
EI
Z2
C
E
2EI
2EI
D
4m
4m
4m
6m


20kN/m
Z1
A
B
EI
EI
Z2
C
E
2EI
2EI
D
4m
4m
4m
6m

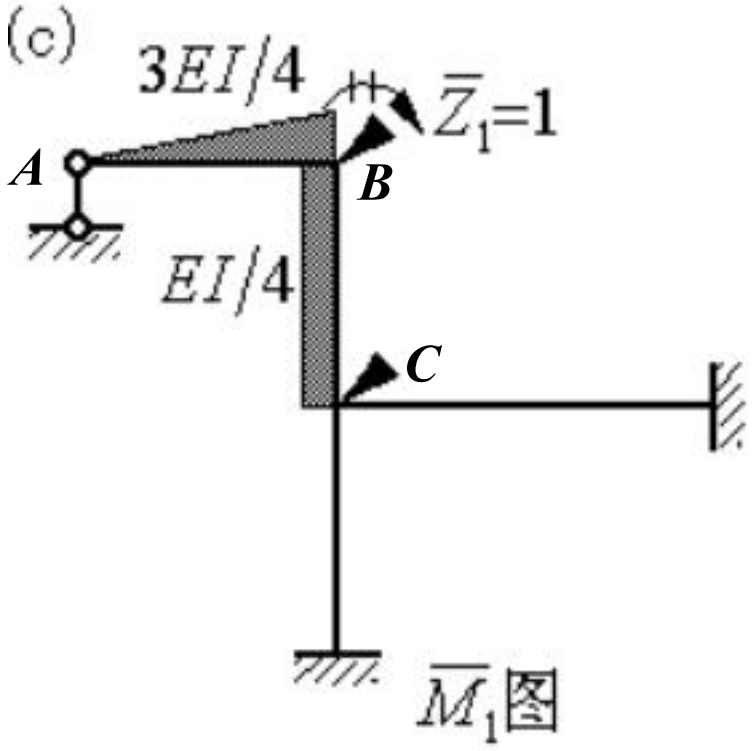
一般位移法

绘出荷载弯矩图、计算固端弯矩



$$M_{BA}^{F}=-\frac{ql^2}{8}=\frac{20\times16}{8}=40kN.m$$

绘出单位弯矩图、计算分配系数



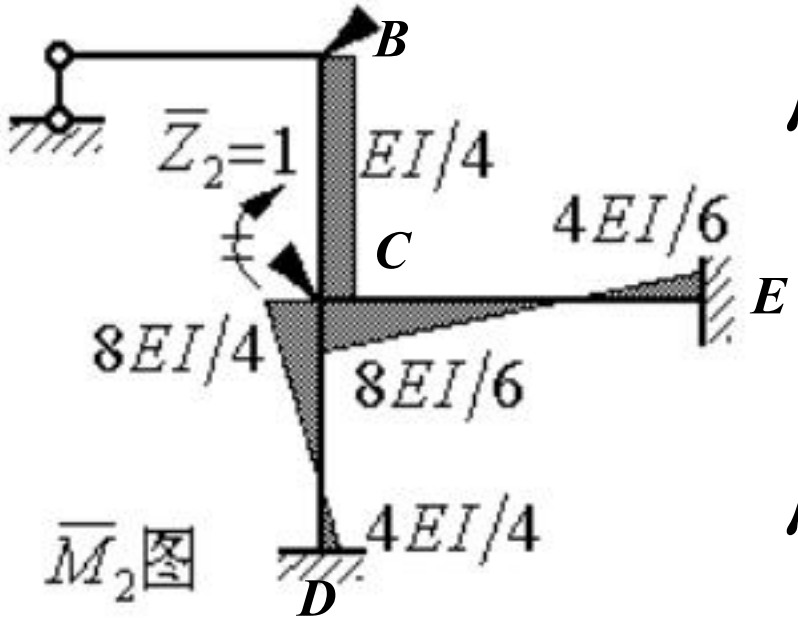
$$\mu_{BA}=\frac{S_{BA}}{S_{BA}+S_{BC}}=\frac{\frac{3}{4}}{\frac{3}{4}+\frac{1}{4}}=\frac{3}{4},$$

$$\mu_{BC}=\frac{S_{BC}}{S_{BA}+S_{BC}}=\frac{\frac{1}{4}}{\frac{3}{4}+\frac{1}{4}}=\frac{1}{4}$$

绘出单位弯矩图、计算分配系数

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(d)



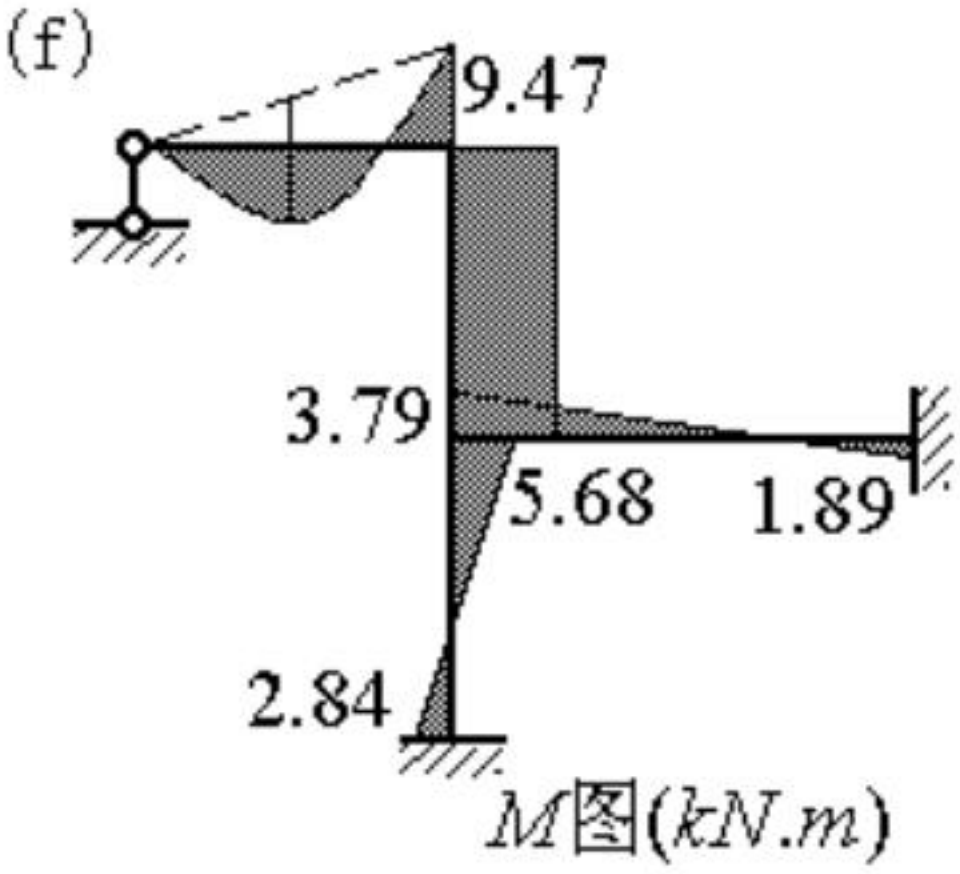
$$\mu_{CB}=\frac{S_{CB}}{S_{CB}+S_{CE}+S_{CD}}=\frac{\frac{1}{4}}{\frac{1}{4}+\frac{8}{6}+\frac{8}{4}}=\frac{3}{43}$$

$$\mu_{CE}=\frac{S_{CE}}{S_{CB}+S_{CE}+S_{CD}}=\frac{\frac{8}{6}}{\frac{1}{4}+\frac{8}{6}+\frac{8}{4}}=\frac{16}{43}$$

$$\mu_{CD}=\frac{S_{CD}}{S_{CB}+S_{CE}+S_{CD}}=\frac{\frac{8}{4}}{\frac{1}{4}+\frac{8}{6}+\frac{8}{4}}=\frac{24}{43}$$

结点	A	B		C			D	E
杆端	AB	BA	BC	CB	CE	CD	DC	EC
μ	铰支	3/4	1/4	3/43	16/43	24/43	固定	固定
M^F	0	40						
分配及传递		-30	-10	10				
			0.70	-0.70	-3.72	-5.58	-2.79	-1.86
		-0.52	-0.18	0.18				
				-0.01	-0.07	-0.10	-0.05	-0.03
$\sum M$	0	9.48	-9.48	9.47	-3.79	-5.68	-2.84	-1.89
校核		$\sum M = 0$		$\sum M = 0$				

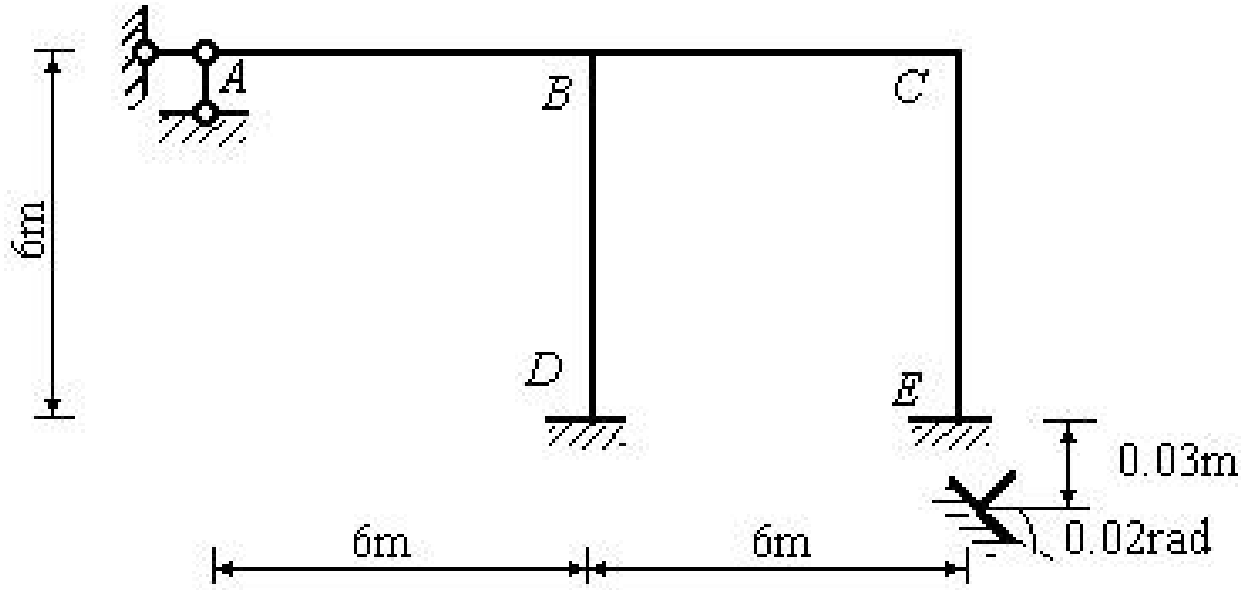
弯矩图



作业1（提示）

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用力矩分配法计算图示刚架，$EI=C$。



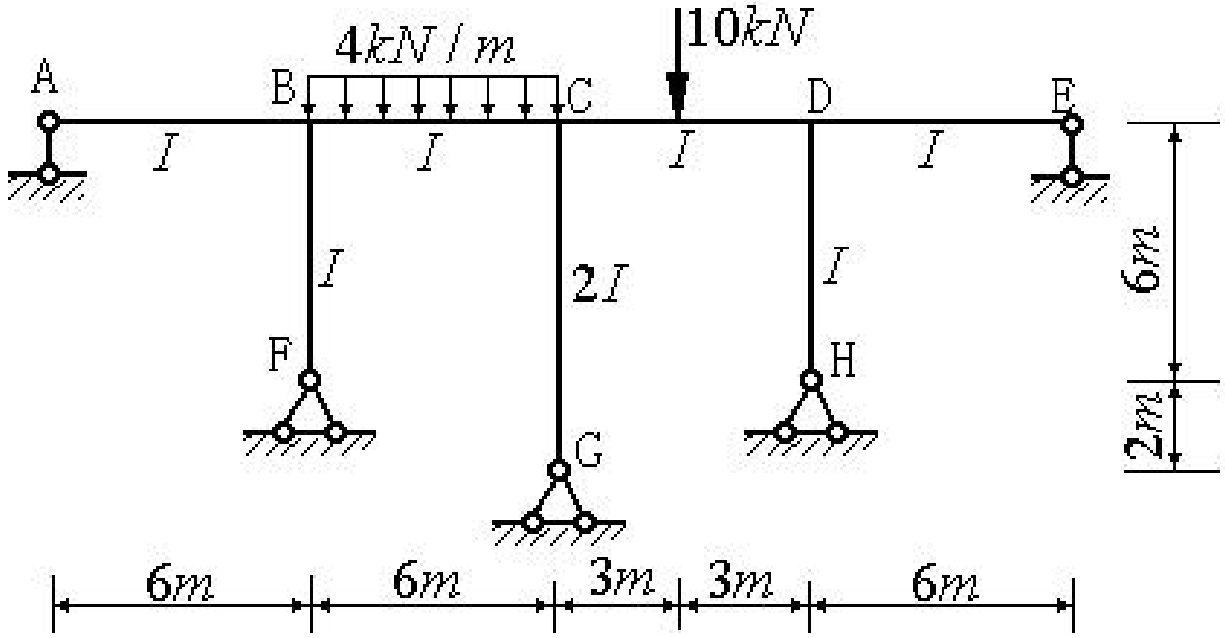
分配系数、固端弯矩及计算表格

结点	A	B			C		D	E
杆端	AB	BA	BD	BC	CB	CE	DB	EC
μ	铰支	3/11	4/11	4/11	1/2	1/2		
M^F		-400		300	300	200		400
分配及传递								
$\sum M$		-336	85	251	72	-72	43	265

作业2（提示）

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用力矩分配法计算图示刚架，$EI=C$。



分配系数、固端弯矩及计算表格

结点	A	B			C			D			E	F	G	H
杆端	AB	BA	BF	BC	CB	CG	CD	DC	DH	DE	ED	FB	GC	HD
μ		0.3	0.3	0.4	0.32	0.36	0.32	0.4	0.3	0.3				
M^F				-12	12		7.5	-7.5						
分配及传递														
$\sum M$		3.88	3.88	-7.76	12.73	-2.07	-10.66	3.94	-1.97	-1.97				

9-5 力矩分配法与位移法的联合应用

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对一般的有结点线位移的刚架，上述力矩分配法已不适用。

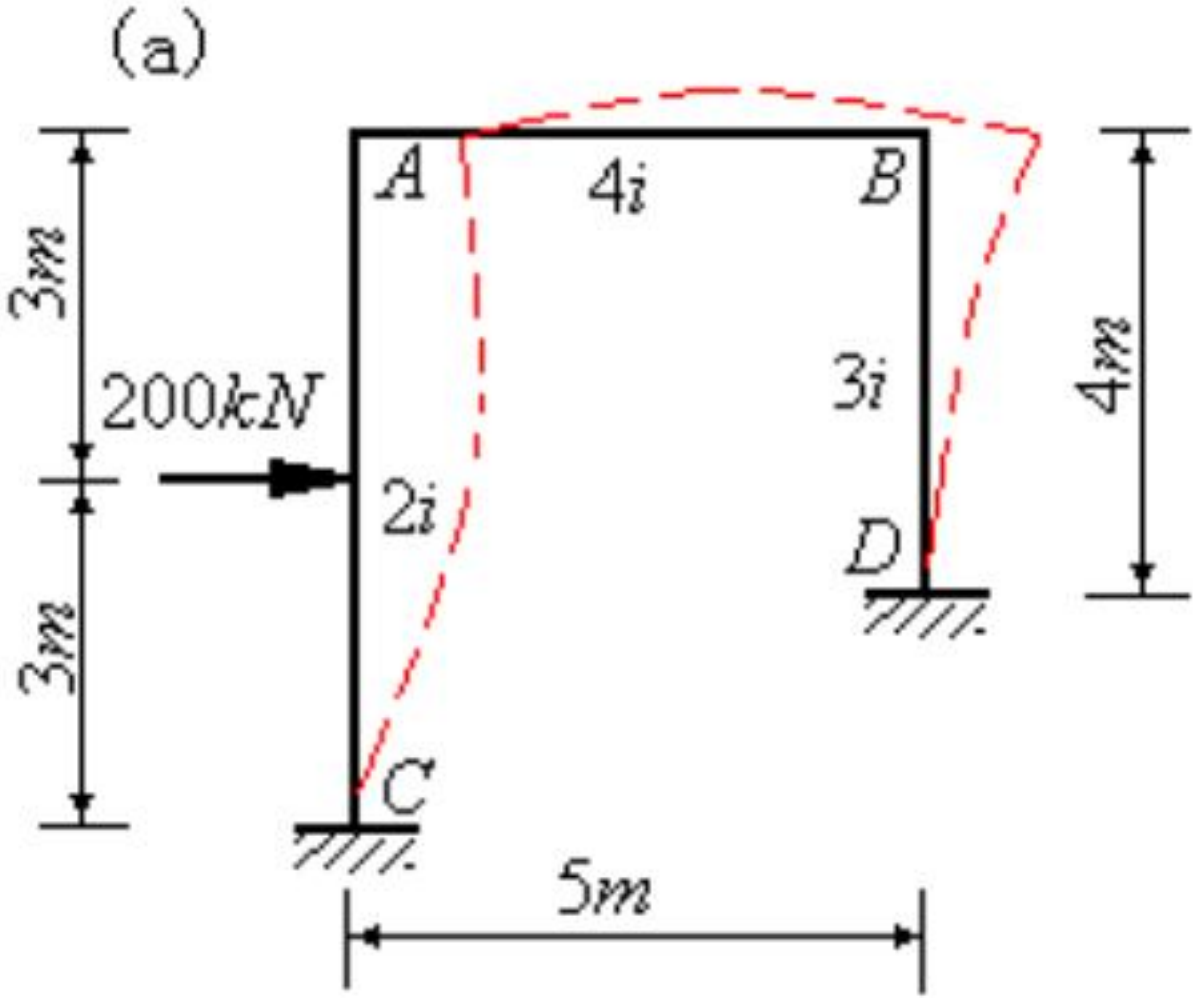
为此，可联合应用力矩分配法与位移法求解：

用力矩分配法考虑角位移的影响；

用位移法考虑线位移的影响。

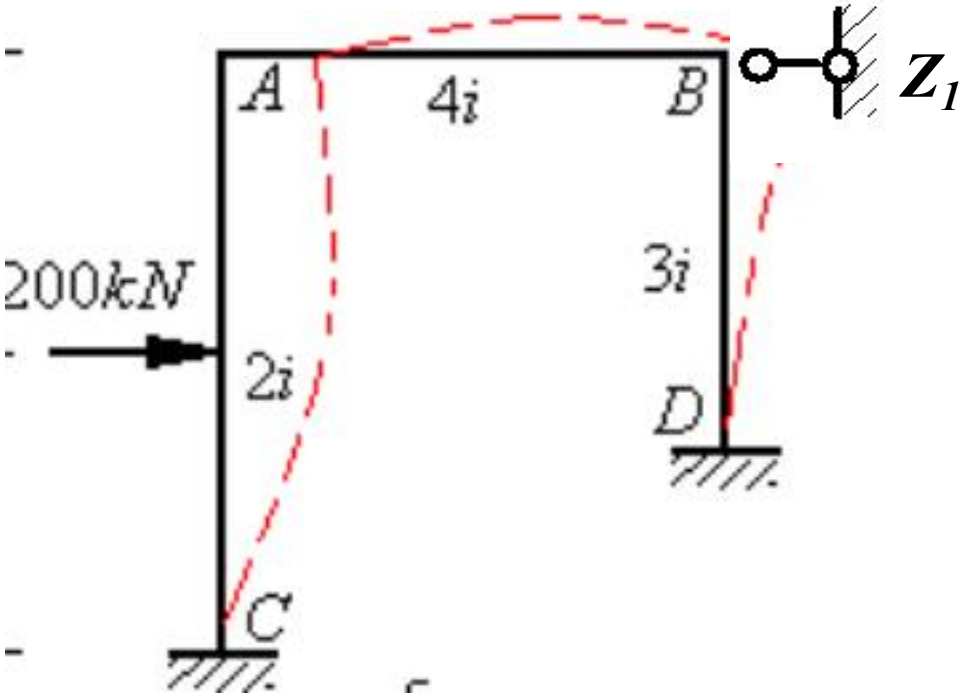
现以图示结构为例来说明及算原理

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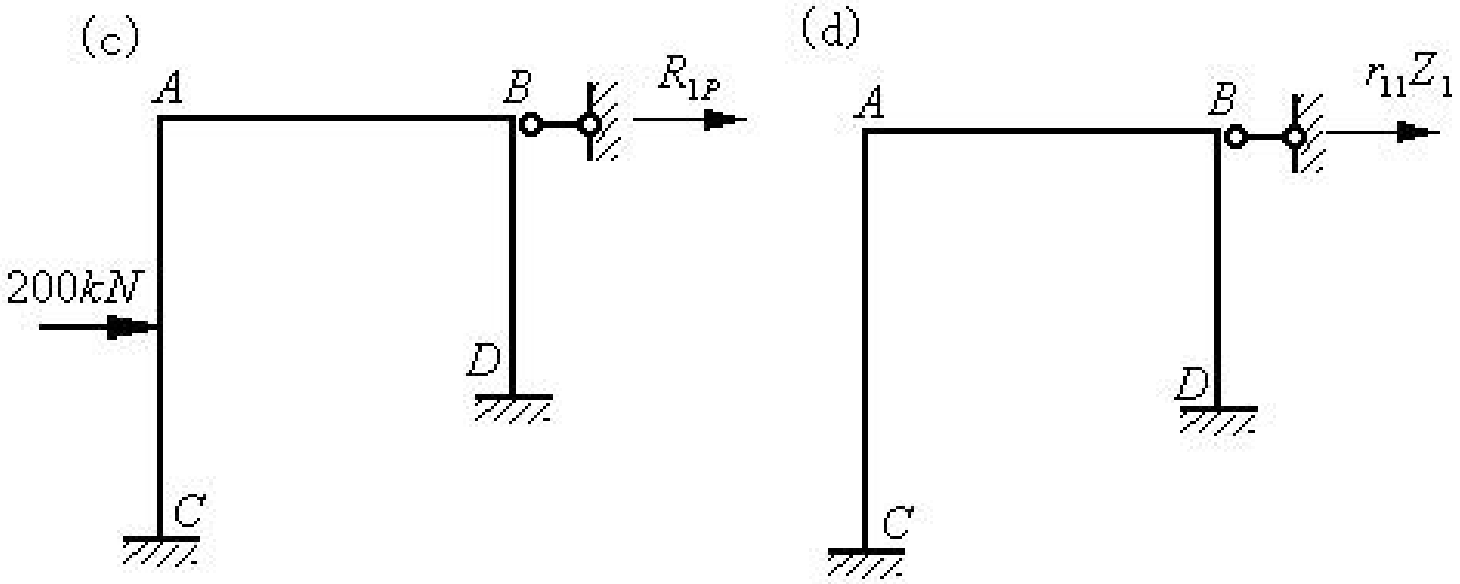
首先用位移法求解，但所取的基本体系只控制线位移，而不控制角位移。

位移法方程为： $r_{11}Z_1 + R_{1P} = 0$



基本体系的两个分解状态如下：

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而弯矩可表示为：　$M = \overline{M}_1 Z_1 + M_P$

R_{1P}, M_P　是基本结构在荷载作用下的附加反力和弯矩；

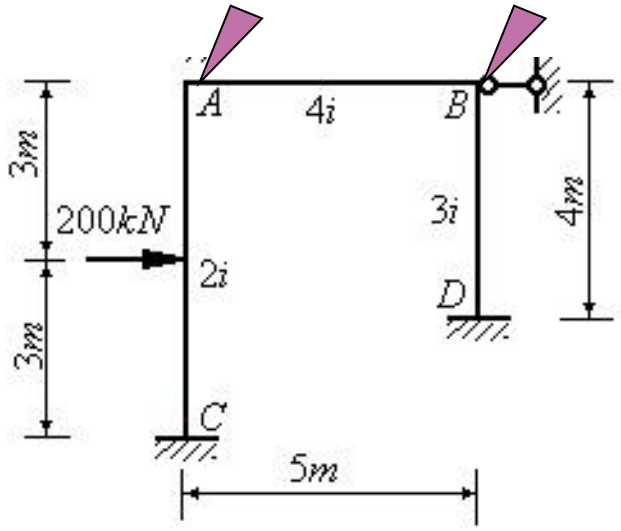
$r_{11}, \overline{M}_1$　是基本结构由于单位位移而产生的反力和弯矩。

用力矩分配法计算基本结构求$\overline{M}_1, M_P r_{11}, R_{1P}$。

由于在基本结构中，结点线位移为给定的（0和1），只有结点角位移是未知量，用力矩分配法求解是方便的。

然后由位移法方程求出线位移，按叠加法求得弯矩图。

用力矩分配法计算基本结构



计算分配系数

结点A

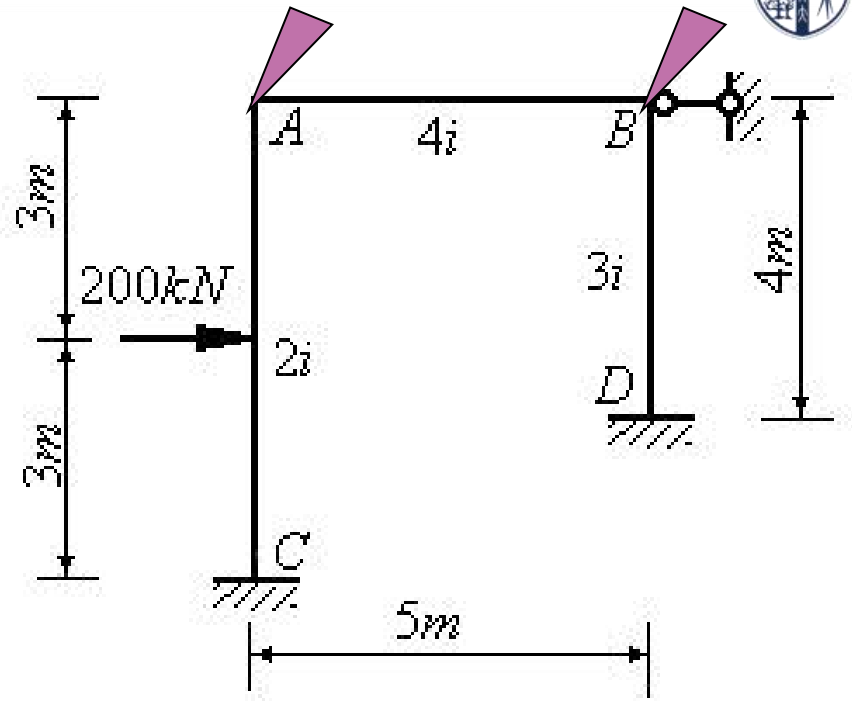
$$\mu_{AB}=\frac{S_{AB}}{S_{AB}+S_{AC}}=\frac{4\times 4i}{4\times 4i+4\times 2i}=\frac{2}{3},$$

$$\mu_{AC}=\frac{1}{3}$$

结点B:

$$\mu_{BA}=\frac{S_{BA}}{S_{BA}+S_{BD}}=\frac{4\times 4i}{4\times 4i+4\times 3i}=\frac{4}{7},\qquad \mu_{BD}=\frac{3}{7}$$

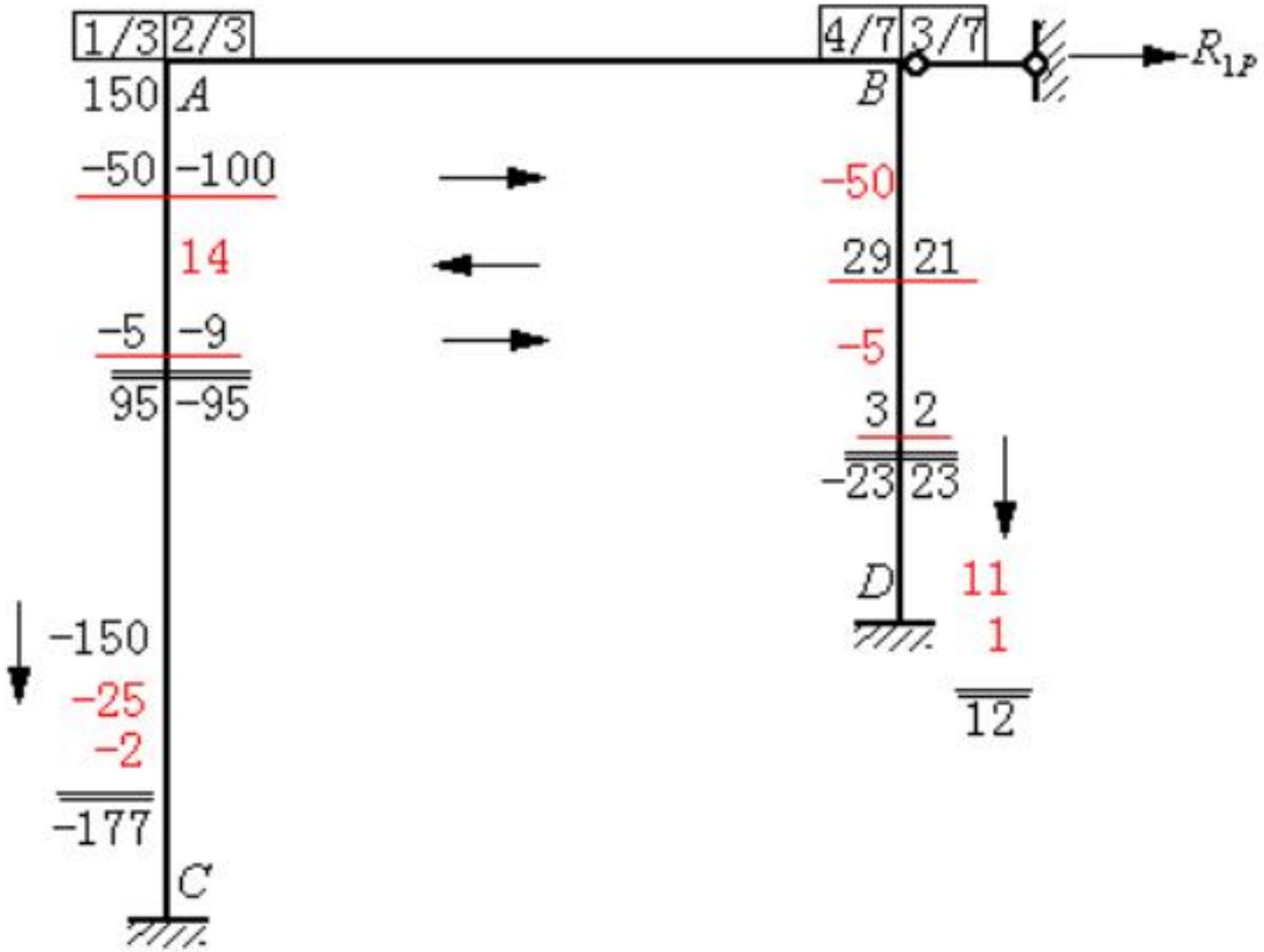
计算固端弯矩



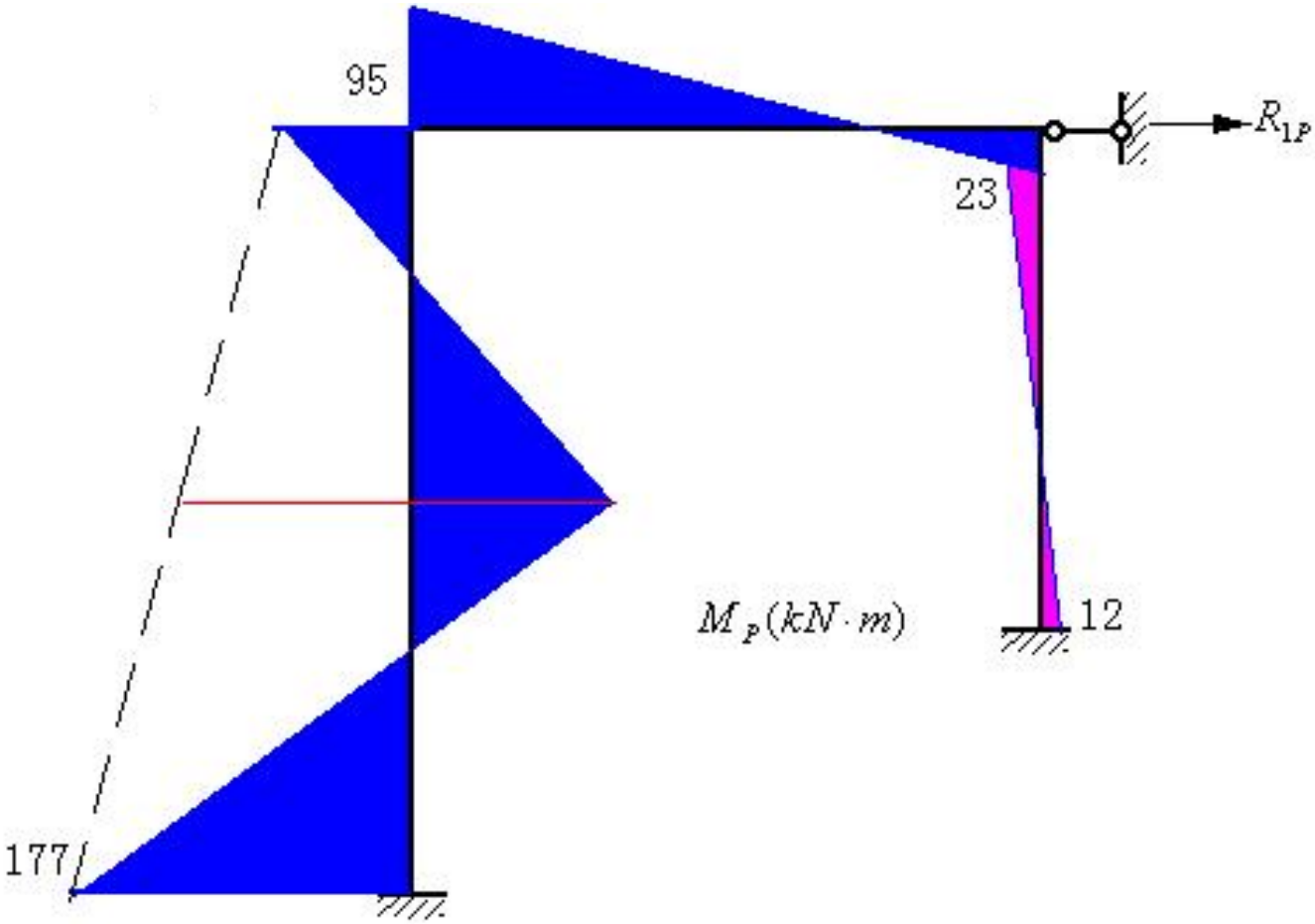
$$M_{CA}^{F} = -\frac{Pl}{8} = -\frac{200kN \times 6m}{8} = -150kN \cdot m,$$

$$M_{AC}^{F} = 150kN \cdot m$$

用力矩分配法计算基本结构

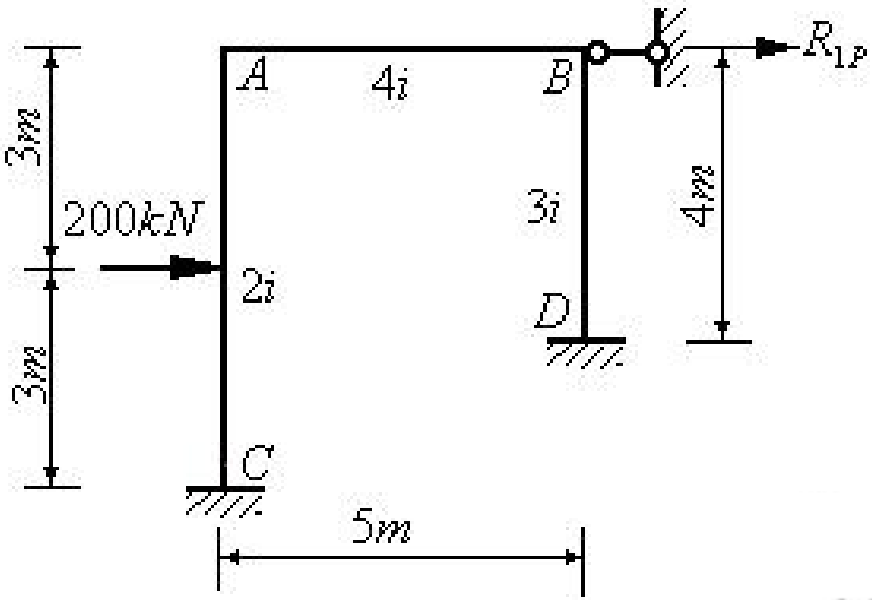


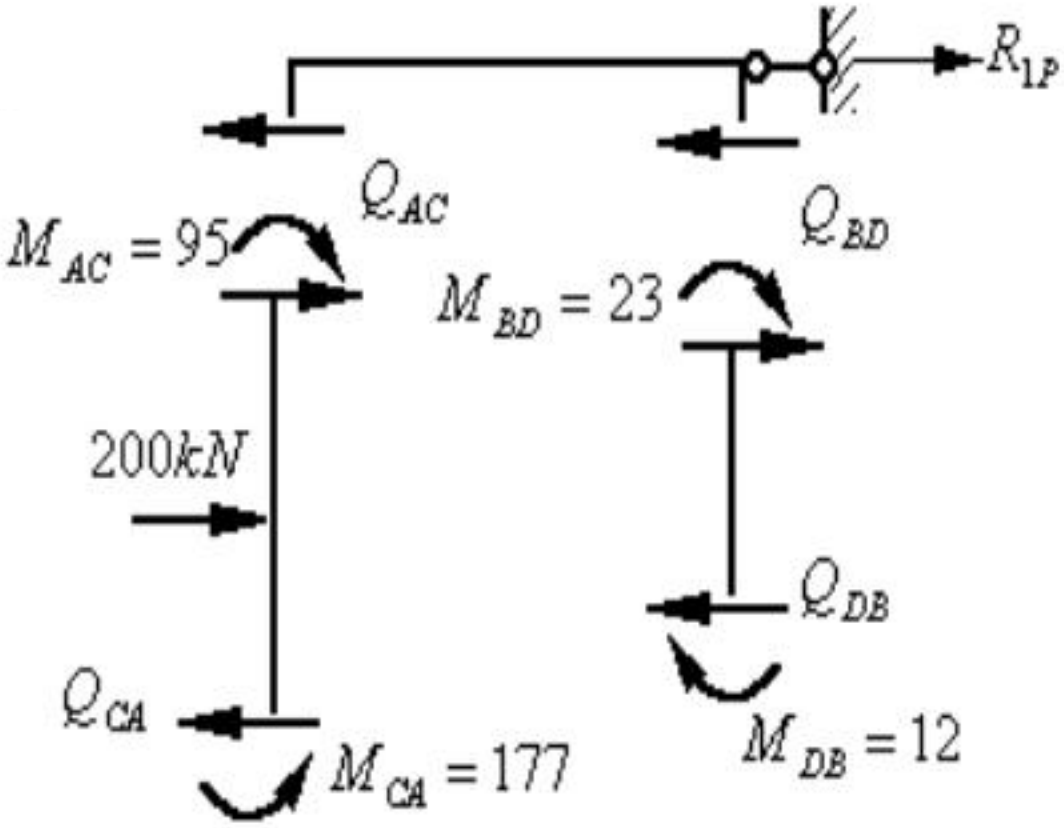
基本结构 M_P 图



计算自由项R_{1P}

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计算自由项R_{1P}

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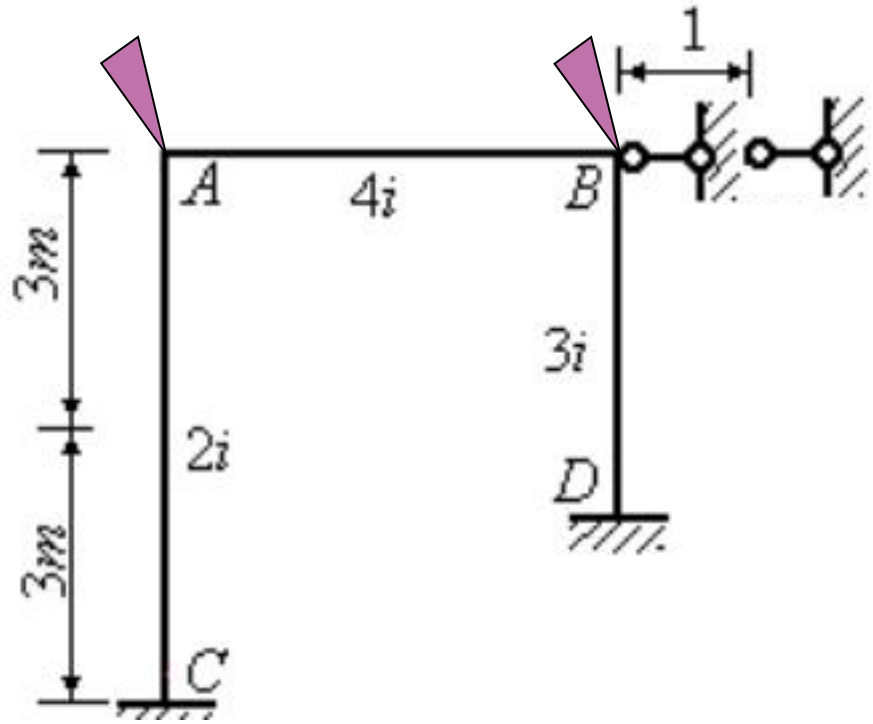
$$\sum X=0,\quad R_{1P}=Q_{AC}+Q_{BD}$$

$$Q_{AC}=-\frac{P}{2}+\frac{M_{CA}-M_{AC}}{6}=-100+\frac{177-95}{6}=-86.333$$

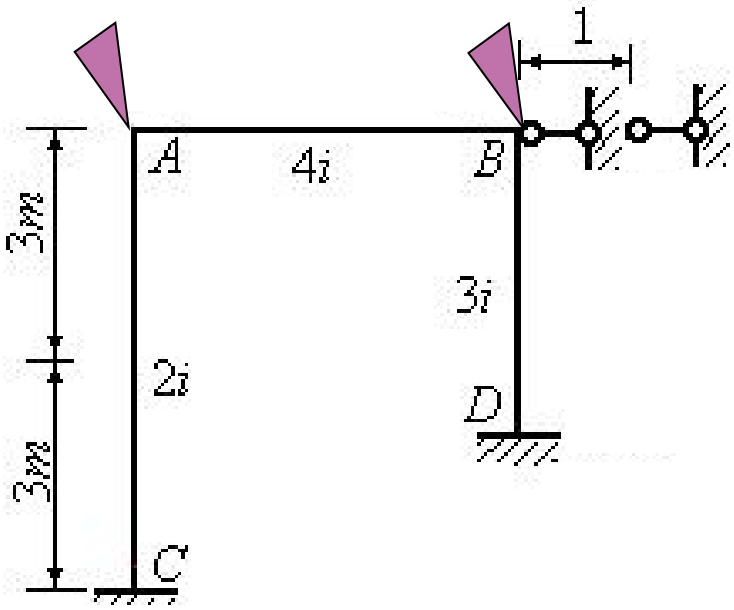
$$Q_{BD}=-\frac{M_{BD}+M_{DB}}{4}=-\frac{23+12}{4}=-8.75$$

$$R_{1P}=Q_{AC}+Q_{BD}=-86.333-8.75=-95.1kN\leftarrow$$

用力矩分配法计算基本结构 $\overline{M}_1$



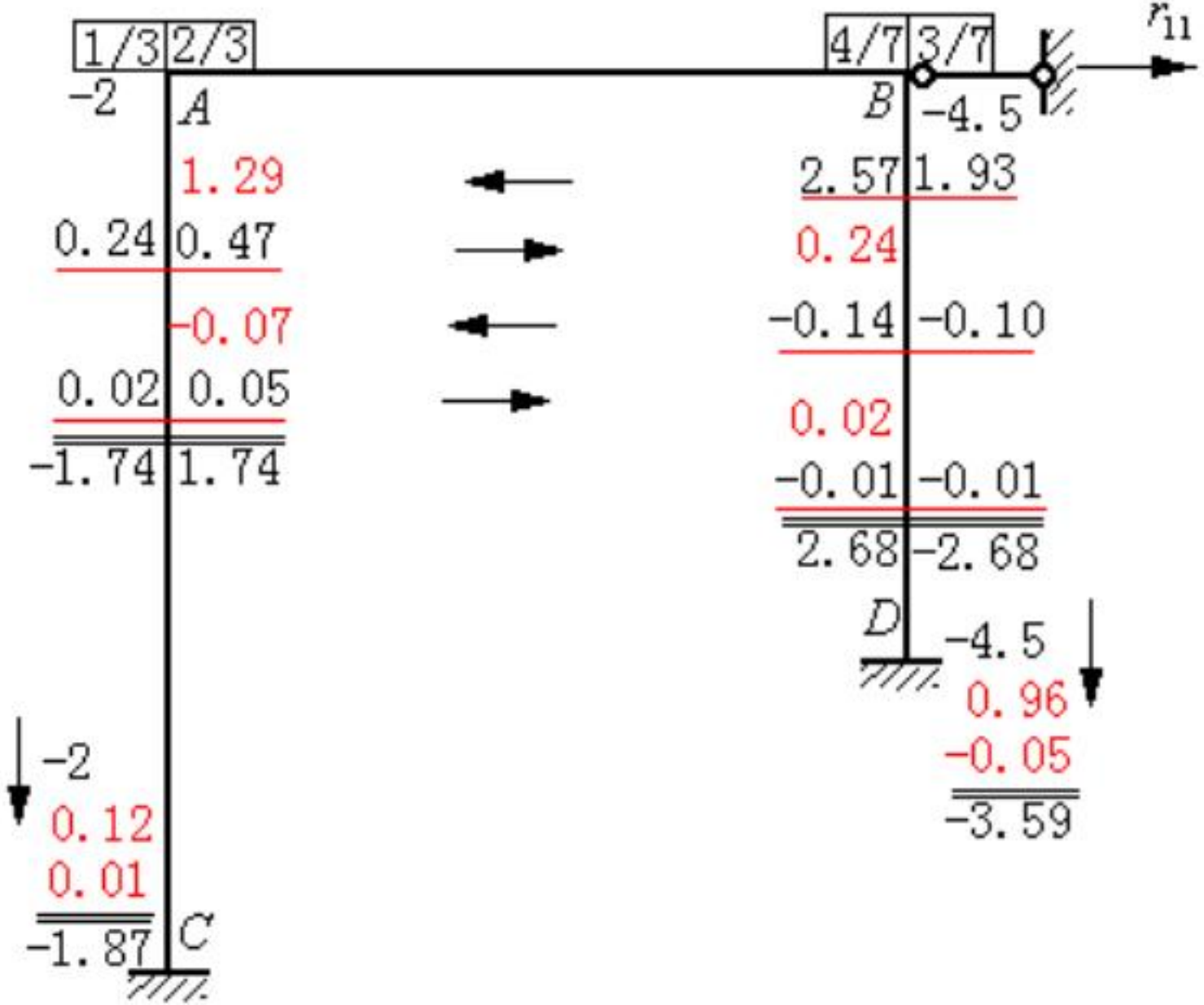
计算固端弯矩



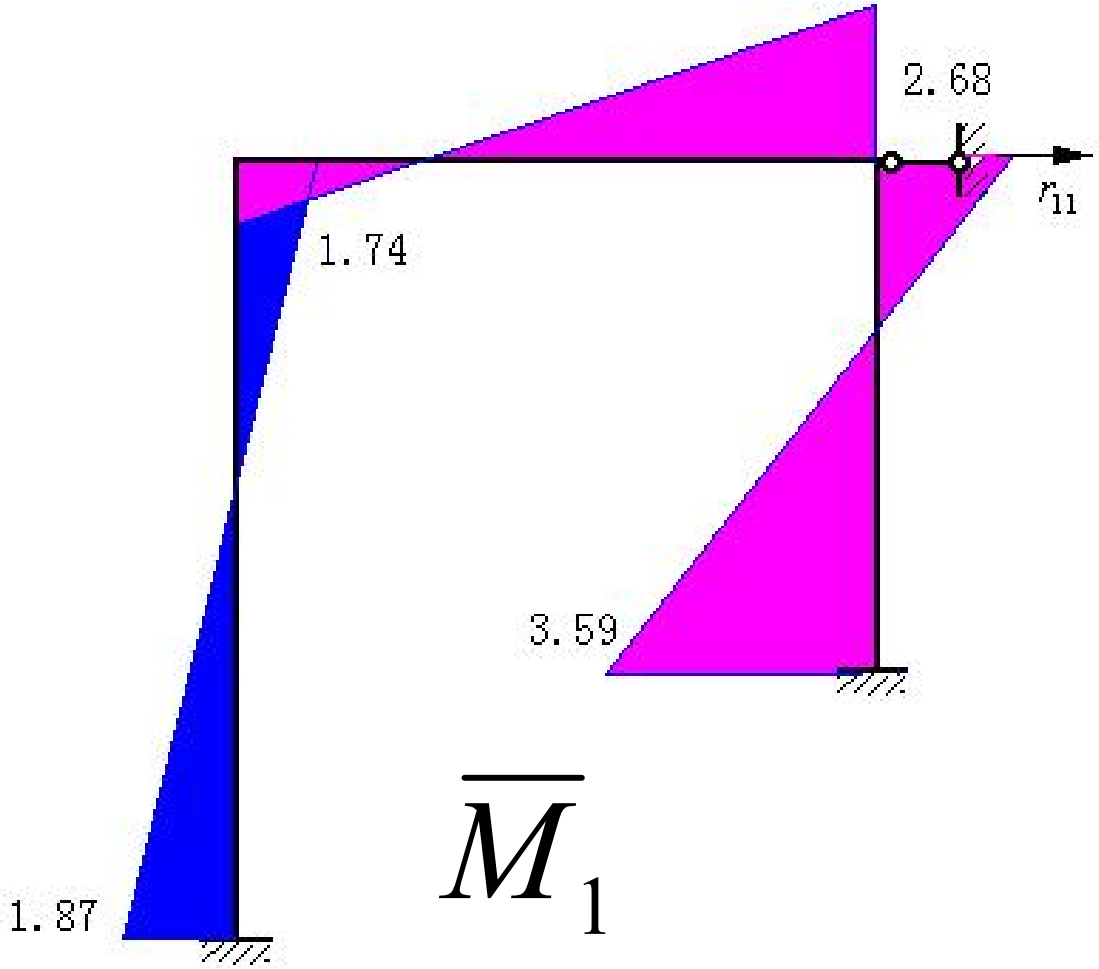
$$M_{AC}^{F} = M_{CA}^{F} = -\frac{6\times 2i}{6} = -2i,$$

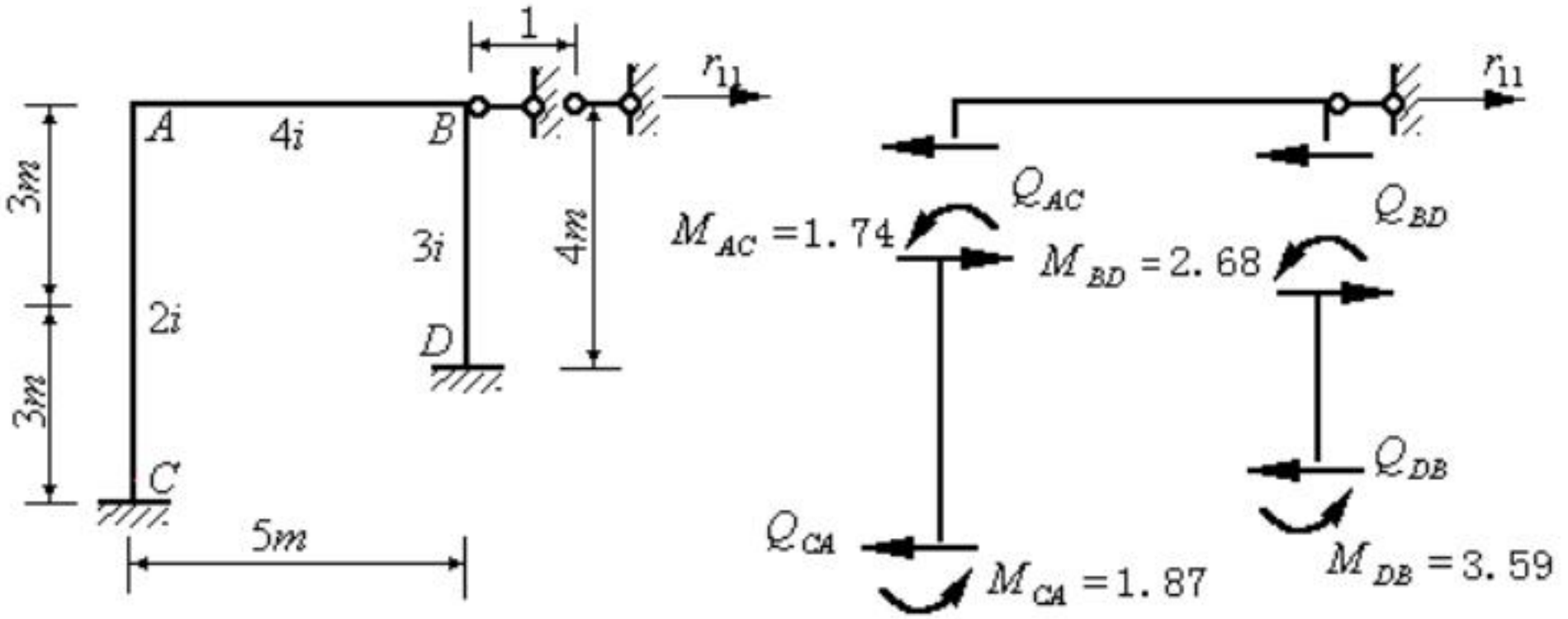
$$M_{BD}^{F} = M_{DB}^{F} = -\frac{6\times 3i}{4} = -4.5i$$

用力矩分配法计算基本结构



基本结构 $\overline{M}_1$ 图





$$\sum X = 0, \qquad r_{11} = Q_{AC} + Q_{BD}$$

$$Q_{AC} = \frac{M_{CA} + M_{AC}}{6} = \frac{1.74 + 1.87}{6} = 0.60i$$

$$Q_{BD} = \frac{M_{BD} + M_{DB}}{4} = \frac{2.68 + 3.59}{4} = 1.567i$$

$$r_{11} = Q_{AC} + Q_{BD} = 0.60i + 1.567i = 2.17i \rightarrow$$

由叠加法得最后弯矩图

将系数项和自由项代入位移法方程求得水平位移Z_1。

$$Z_1 = -\frac{R_{1P}}{r_{11}} = -\frac{-95.1}{2.17i} = \frac{43.8}{i}$$

由叠加法得弯矩图

$$M = M_P + \overline{M}_1 Z_1$$

